

We continue our discussion of the genetics of the evolutionary process. This chapter is devoted to a discussion of some of the effects of violating, or relaxing, the assumptions of the Hardy-Weinberg equilibrium other than random mating, which we discussed in chapter 19. Here we consider the effects of mutation, migration, small population size, and natural selection on the Hardy-Weinberg equilibrium. These processes usually change allelic frequencies.

MODELS FOR POPULATION GENETICS

The steps we need to take to solve for equilibrium in population genetics models follow the same general pattern regardless of what model we are analyzing. We emphasize that these models were developed to help us understand the genetic changes taking place in a population. The models shed light on nonintuitive processes and help quantify intuitive processes. The steps in the models can be outlined as follows:

1. Set up an algebraic model.
2. Calculate allelic frequency in the next generation, q_{n+1} .
3. Calculate change in allelic frequency between generations, Δq .
4. Calculate the equilibrium condition, $\hat{q}$ (q -hat), at $\Delta q = 0$.
5. Determine, when feasible, if the equilibrium is stable.

MUTATION



Mutational Equilibrium

Mutation affects the Hardy-Weinberg equilibrium by changing one allele to another and thus changing allelic and genotypic frequencies. Consider a simple model in which two alleles, A and a , exist. A mutates to a at a rate of μ (mu), and a mutates back to A at a rate of ν (nu):



If p_n is the frequency of A in generation n and q_n is the frequency of a in generation n , then the new frequency of a , q_{n+1} , is the old frequency of a plus the addition of a alleles from forward mutation and the loss of a alleles by back mutation. That is,

$$q_{n+1} = q_n + \mu p_n - \nu q_n \quad (20.1)$$

in which μp_n is the increment of a alleles added by forward mutation, and νq_n is the loss of a alleles due to back mutation. Equation 20.1 takes into account not only the rate of forward mutation, μ , but also p_n , the frequency of A alleles available to mutate. Similarly, the loss of a to A alleles is the product of both the rate of back mutation, ν , and the frequency of the a allele, q_n . Equation 20.1 completes the second modeling step, derivation of an expression for q_{n+1} , allelic frequency after one generation of mutation pressure. The third step is to derive an expression for the change in allelic frequency between two generations. This change (Δq) is simply the difference between the allelic frequency at generation $n + 1$ and the allelic frequency at generation n . Thus, for the a allele

$$\Delta q = q_{n+1} - q_n = (q_n + \mu p_n - \nu q_n) - q_n \quad (20.2)$$

which simplifies to

$$\Delta q = \mu p_n - \nu q_n \quad (20.3)$$

The next step in the model is to calculate the equilibrium condition $\hat{q}$, or the allelic frequency when there is no change in allelic frequency from one generation to the next—that is, when Δq (equation 20.3) is equal to zero:

$$\Delta q = \mu p_n - \nu q_n = 0 \quad (20.4)$$

Thus,

$$\mu p_n = \nu q_n \quad (20.5)$$

Then, substituting $(1 - q_n)$ for p_n (since $p = 1 - q$), gives

$$\mu(1 - q_n) = \nu q_n$$

or, by rearranging:

$$\hat{q} = \frac{\mu}{\mu + \nu} \quad (20.6)$$

And, since $p + q = 1$,

$$\hat{p} = \frac{\nu}{\mu + \nu} \quad (20.7)$$

We can see from equations 20.6 and 20.7 that an equilibrium of allelic frequencies does exist. Also, the equilibrium value of allele a ($\hat{q}$) is directly proportional to the relative size of μ , the rate of forward mutation toward a . If $\mu = \nu$, the equilibrium frequency of the a allele ($\hat{q}$) will be 0.5. As μ gets larger, the equilibrium value shifts toward higher frequencies of the a allele.

Stability of Mutational Equilibrium

Having demonstrated that allelic frequencies can reach an equilibrium due to mutation, we can ask whether the mutational equilibrium is stable. A stable equilibrium is

one that returns to the original equilibrium point after being perturbed. An unstable equilibrium is one that will not return after being perturbed but, rather, continues to move away from the equilibrium point. As we mentioned in the last chapter, the Hardy-Weinberg equilibrium is a neutral equilibrium: It remains at the allelic frequency it moved to when perturbed.

Stable, unstable, and neutral equilibrium points can be visualized as marbles in the bottom of a concave surface (stable), on the top of a convex surface (unstable), or on a level plane (neutral; fig. 20.1). Although more sophisticated mathematical formulas exist for determining whether an equilibrium is stable, unstable, or neutral, we will use graphical analysis for this purpose.

Figure 20.2 introduces the process of graphical analysis, which provides an understanding of the dynamics of an event or process by representing the event in graphical form. In figure 20.2, we have graphed equation 20.3, the Δq equation of mutational dynamics. The ordinate, or y-axis, is Δq , the change in allelic frequency. The abscissa, or x-axis, is q , or allelic frequency. The diagonal line is the Δq equation, the relationship between Δq and q . Note that Δq can be positive (q is increasing) or negative (q is decreasing), whereas q is always positive (0–1.0). Graphical analysis can provide insights into the dynamics of many processes in population genetics.

The diagonal line in figure 20.2 crosses the $\Delta q = 0$ line at the equilibrium value ($\hat{q}$) of 0.167. This line also shows us the changes in allelic frequency that occur in a population not at the equilibrium point. We will look

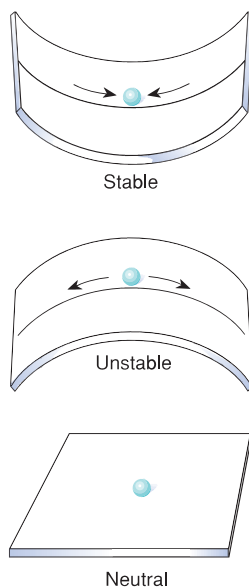


Figure 20.1 Types of equilibria: stable, unstable, and neutral.

at two examples of populations under the influence of mutation pressure, but not at equilibrium: one at $q = 0.1$ (below equilibrium) and one at $q = 0.9$ (above equilibrium).

If we substitute $q = 0.1$ into equation 20.3, we get a Δq value of 4×10^{-6} . If we substitute $q = 0.9$ into the equation, we get a Δq value of -4.4×10^{-5} . In other words, when the population is below equilibrium, q increases ($\Delta q = +4 \times 10^{-6}$); if the population is above equilibrium, q decreases ($\Delta q = -4.4 \times 10^{-5}$). We can read these same conclusions directly from the graph in figure 20.2.

We can see that the mutational equilibrium is a stable one. Any population whose allelic frequency is not at the equilibrium value tends to return to that equilibrium value. A shortcoming of this model is that it provides no obvious information revealing the time frame for reaching equilibrium. To derive the equations needed to determine this parameter is beyond our scope. (We could use computer simulation or integrate equation 20.3 with respect to time.) In a large population, any great change in allelic frequency caused by mutation pressure alone takes an extremely long time. Most mutation rates are on

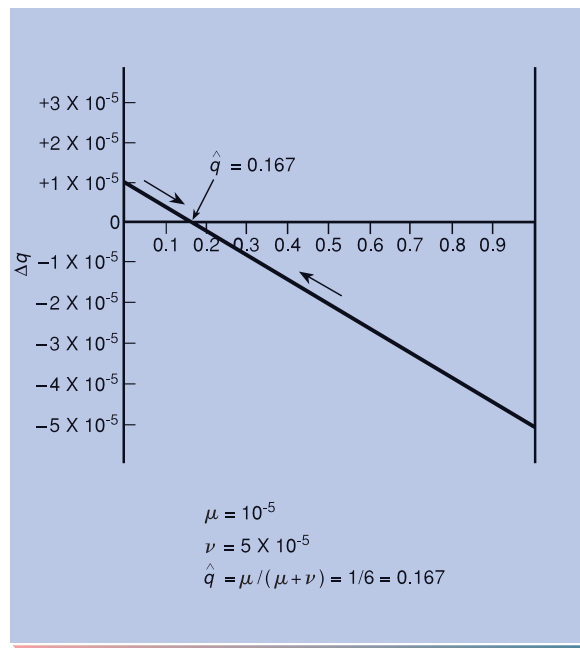


Figure 20.2 Graphical analysis of mutational equilibrium. The graph of the mutational Δq equation shows that when the population is perturbed from the equilibrium point ($q = 0.167$), it returns to that equilibrium point. At q values above equilibrium, change is negative, tending to return the population to equilibrium. At q values below equilibrium, change is positive, also tending to return the population to equilibrium.

the order of 10^{-5} , and equation 20.3 shows that change will be very slow with values of this magnitude. For example, if $\mu = 10^{-5}$, $\nu = 10^{-6}$, and $p = q = 0.5$, $\Delta q = (0.5 \times 10^{-5}) - (0.5 \times 10^{-6}) = 4.5 \times 10^{-6}$, or 0.0000045. It usually takes thousands of generations to get near equilibrium, which is approached asymptotically.

As you can see from the low values of mutation rates, it would usually be nearly impossible to detect perturbations to the Hardy-Weinberg equilibrium by mutation in any one generation. The mutation rate can, however, determine the eventual allelic frequencies at equilibrium if no other factors act to perturb the gradual changes that mutation rates cause. Mutation can also affect final allelic frequencies when it restores alleles that natural selection is removing, a situation we will discuss at the end of the chapter. More important, mutation provides the alternative alleles that natural selection acts upon.

MIGRATION

Migration is similar to mutation in the sense that it adds or removes alleles and thereby changes allelic frequencies. Human populations are frequently affected by migration.

Assume two populations, natives and migrants, both containing alleles A and a at the A locus, but at different frequencies (p_N and q_N versus p_M and q_M), as shown in figure 20.3. Assume that a group of migrants joins the native population and that this group of migrants makes up a fraction m (e.g., 0.2) of the new conglomerate population. Thus, the old residents, or natives, will make up a proportionate fraction $(1 - m)$ (e.g., 0.8) of the combined population. The conglomerate a -allele frequency, q_c , will be the weighted average of the allelic frequencies of the natives and migrants (the allelic frequencies weighted—multiplied—by their proportions):

$$q_c = mq_M + (1 - m)q_N \quad (20.8)$$

$$q_c = q_N + m(q_M - q_N) \quad (20.9)$$

The change in allelic frequency, a , from before to after the migration event is

$$\Delta q = q_c - q_N = [q_N + m(q_M - q_N)] - q_N \quad (20.10)$$

$$\Delta q = m(q_M - q_N) \quad (20.11)$$

We then find the equilibrium value, $\hat{q}$ (at $\Delta q = 0$). Remembering that, in a product series, any multiplier with the value of zero makes the whole expression zero, Δq will be zero when either

$$m = 0 \text{ or } q_M - q_N = 0; q_M = q_N$$

The conclusions we can draw from this model are intuitive. Migration can upset the Hardy-Weinberg equilib-

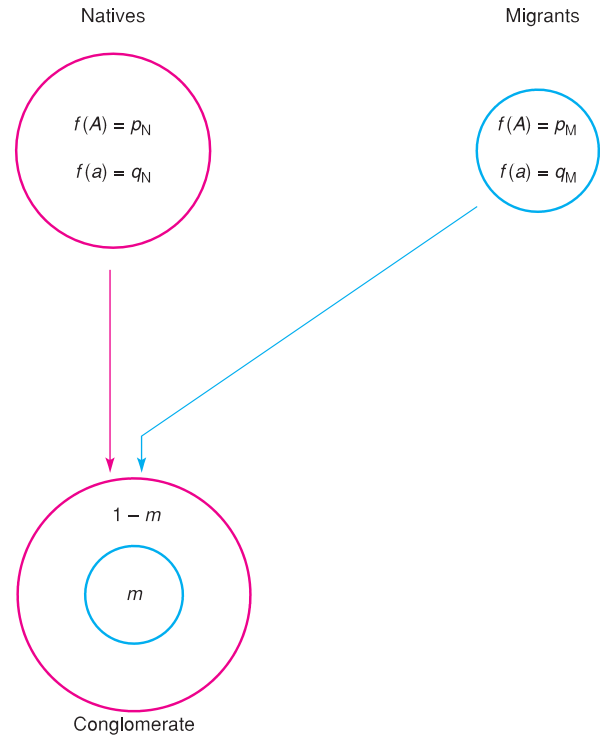


Figure 20.3 Diagrammatic view of migration. A group of migrants enters a native population, making up a proportion, m , of the final conglomerate population.

rium. Allelic frequencies in a population under the influence of migration will not change if either the size of the migrant group drops to zero (m , the proportion of the conglomerate made up of migrants, drops to zero) or the allelic frequencies in the migrant and resident groups become identical.

This migration model can be used to determine the degree to which alleles from one population have entered another population. It can analyze the allele interactions in any two populations. We can, for example, analyze the amount of admixture of alleles from Mongol populations with eastern European populations to explain the relatively high levels of blood type B in eastern European populations (if we make the relatively unrealistic assumption that each of these groups is homogeneous). The calculations are also based on a change happening all in one generation, which did not happen. Blood type and other loci can be used to determine allelic frequencies in western European, eastern European, and Mongol populations. We can rearrange equation 20.9 to solve for m , the proportion of migrants:

$$m = \frac{q_c - q_N}{q_M - q_N} \quad (20.12)$$

From one sample, we find that the B allele is 0.10 in western Europe, taken as the resident or native population (q_N); 0.12 in eastern Europe, the conglomerate population (q_C); and 0.21 in Mongols, the migrants (q_M). Substituting these values into equation 20.12 gives a value for m of 0.18. That is, given the stated assumptions, 18% of the alleles in the eastern European population were brought in by genetic mixture with Mongols.

When a migrant group first joins a native group, before genetic mixing (mating) takes place, the Hardy-Weinberg equilibrium of the conglomerate population is perturbed, even though both subgroups are themselves in Hardy-Weinberg proportions. A decrease will occur in heterozygotes in the conglomerate population as compared to what we would predict from the allelic frequencies of that population (the average allelic frequencies of the two groups). This is a phenomenon of subdivision referred to as the **Wahlund effect**. The reason this happens is because the relative proportions of heterozygotes increase at intermediate allelic frequencies. As allelic frequencies rise above or fall below 0.5, the relative proportion of heterozygotes decreases.

In a conglomerate population, the allelic frequencies will be intermediate between the values of the two subgroups because of averaging. This generally means the predicted proportion of heterozygotes will be higher than the actual average proportion of heterozygotes in the two subgroups. An example is worked out in table 20.1. Assume that the two subgroups each make up 50% of the conglomerate population. In subgroup 1, $p = 0.1$ and $q = 0.9$; in subgroup 2, $p = 0.9$ and $q = 0.1$. Each subgroup will have 18% heterozygotes. The average, $(0.18 + 0.18)/2 = 0.18$, is the proportion of heterozygotes actually in the population. However, the conglomerate allelic frequencies are $p = 0.5$ and $q = 0.5$, leading to the expectation that 50% of the population will be heterozygotes. Hence, the observed frequency of het-

erozygotes is lower than the expected frequency (i.e., the Wahlund effect).

We should note that the same logic holds even if both populations have allelic frequencies above or below 0.5. Also, this effect happens when an observer samples what he or she thinks is a single population but is actually a population subdivided into several demes. When most population geneticists sample a population and find a deficiency of heterozygotes, they first think of inbreeding and then of subdivision, the Wahlund effect. (A further complication is that inbreeding leads to subdivision, and subdivision leads to inbreeding. Statistics have been developed to try to separate the effects of these two phenomena.) As soon as random mating occurs in a subdivided population, Hardy-Weinberg equilibrium is established in one generation. We refer to a population in which the individuals are mating at random as unstructured or **panmictic**.

SMALL POPULATION SIZE



Another variable that can upset the Hardy-Weinberg equilibrium is small population size. The Hardy-Weinberg equilibrium assumes an infinitely large population because, as defined, it is **deterministic**, not stochastic. That is, the Hardy-Weinberg equilibrium predicts exactly what the allelic and genotypic frequencies should be after one generation; it ignores variation due to sampling error. Obviously, every population of organisms on earth violates the Hardy-Weinberg assumption of infinite population size.

Sampling Error



The zygotes of every generation are a sample of gametes from the parent generation. Sampling errors are the changes in allelic frequencies from one generation to the next that are due to inexact sampling of the alleles of the parent generation. Toss a coin one hundred times, and chances are, it will not land heads exactly fifty times. However, as the number of coin tosses increases, the percentage of heads will approach 50%, a percentage reached with certainty only after an infinite number of tosses. The same applies to any sampling problem, from drawing cards from a deck to drawing gametes from a gene pool.

If small population size is the only factor causing deviation from Hardy-Weinberg equilibrium, it will cause the allelic frequencies of a population to fluctuate from generation to generation in the process known as random genetic drift. In other words, an Aa heterozygote will sometimes produce several offspring that have only the A allele, or sometimes random mortality will kill a disproportionate number of aa homozygotes. In either case, the next generation may not have the same allelic frequencies as the

Table 20.1 The Wahlund Effect: Heterozygote Frequencies Are Below Expected in a Conglomerate Population

	Subgroup 1	Subgroup 2	Conglomerate	
p	0.1	0.9	0.5	
q	0.9	0.1	0.5	
			Expected	Observed
p^2	0.01	0.81	0.25	0.41
$2pq$	0.18	0.18	0.50	0.18
q^2	0.81	0.01	0.25	0.41

Note: In this example, the subgroups are of equal sizes.

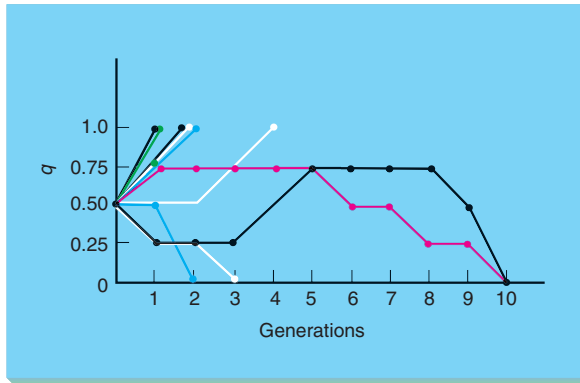


Figure 20.4 Random genetic drift. Ten populations, each consisting of two individuals with initial $q = 0.5$, all go to fixation or loss of the a allele (four or zero copies) within ten generations due to the sampling error of gametes. Once the a allele has been fixed or lost, no further change in allelic frequency will occur (barring mutation or migration). We show a population of only two individuals to exaggerate the effects of random genetic drift.

present generation. The end result will be either fixation or loss of any given allele ($q = 1$ or $q = 0$; fig. 20.4), although which will be fixed or lost depends on the original allelic frequencies. The rate of approach to reach the fixation-loss endpoint depends on the size of the population.

Simulation of Random Genetic Drift



We can investigate the process of random genetic drift mathematically by starting with a large number of populations of the same finite size and observing how the distribution of allelic frequencies among the populations changes in time due only to random genetic drift. For example, we can start with one thousand hypothetical populations, each containing one hundred individuals, with the frequency of the a allele, q , 0.5 in each (fig. 20.5). We measure time in generations, t , as a function of the population size, N (one hundred in this example). For instance, $t = N$ is generation one hundred, $t = N/5$ is generation twenty, and $t = 3N$ is generation three hundred. Then, by using computer simulation (or the **Fokker-Planck equation**, which physicists use to describe diffusion processes such as Brownian motion), we generate the series of curves shown in figure 20.6. These curves show that as the number of generations increases, the populations begin to diverge from $q = 0.5$. Approximately the same number of populations go to q values above 0.5 as go to q values below 0.5. Therefore, the distribution spreads symmetrically. When the distribution of allelic frequencies reaches the sides of the graph, some populations become fixed for the a allele and some lose it. In a sense, the sides act as sinks:

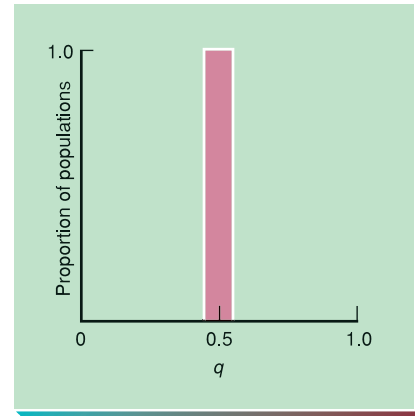


Figure 20.5 Initial conditions of random drift model. One thousand populations, each of size one hundred, and each with an allelic frequency (q) of 0.5.

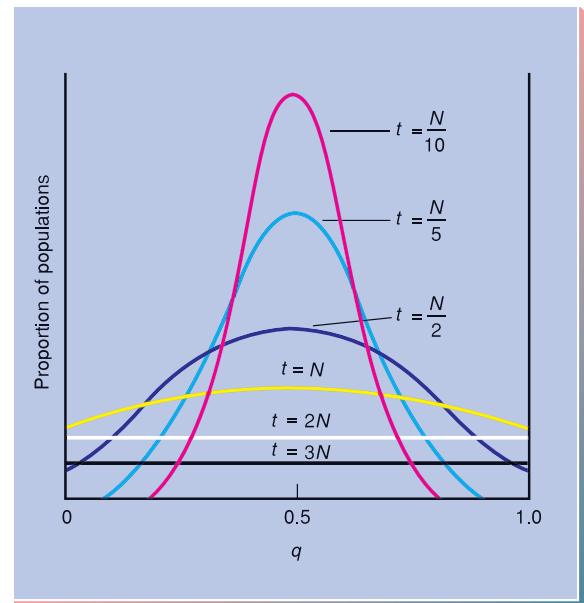


Figure 20.6 Genetic drift in small populations: $q = 0.5$. After time passes, the populations of figure 20.5 begin to diverge in their allelic frequencies. Time is measured in population size (N), showing that the effects of random genetic drift are qualitatively similar in populations of all sizes; the only difference is the timescale. (From M. Kimura, "Solution of a process of random genetic drift with a continuous model," *Proceedings of the National Academy of Sciences, USA*, 41:144-50, 1955. Reprinted by permission.)

Any population that has the a allele lost or fixed will be permanently removed from the process of random genetic drift. Without mutation to bring one or the other allele

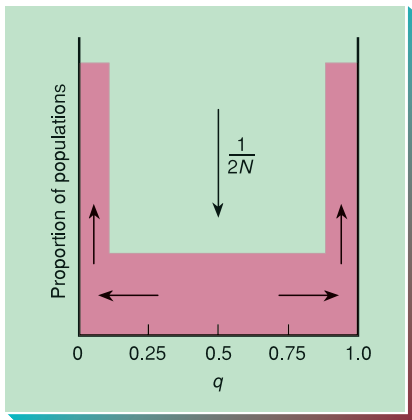


Figure 20.7 Continued genetic drift in the one thousand populations, each numbering one hundred in size, shown in figures 20.5 and 20.6. After approximately $2N$ generations, the distribution is flat, and populations are going to loss or fixation of the a allele at a rate of $1/2N$ populations per generation. (From S. Wright, "Evolution in Mendelian Populations," *Genetics*, 97:114. Copyright © 1931 Genetics Society of America.)

back into the gene pool, these populations maintain a constant allelic frequency of zero or 1.0.

At a point between N (one hundred) and $2N$ (two hundred) generations, the distribution of allelic frequencies flattens out and begins to lose populations to the edges (fixation or loss) at a constant rate, as shown in figure 20.7. The rate of loss is about $1/2N$ ($1/200$), or 0.5% of the populations per generation. If the initial allelic frequency was not 0.5, everything is shifted in the distribution (fig. 20.8), but the basic process is the same—in all populations, sampling error causes allelic frequencies to drift toward fixation or elimination. If no other factor counteracts this drift, every population is destined to eventually be either fixed for or deficient in any given allele.

The amount of time the process takes depends on population size. The example used here was based on small populations of one hundred. If we substitute one million for one hundred in figure 20.6, a flat distribution of populations would not be reached for two million generations, rather than two hundred generations. Thus, a population experiences the effect of random genetic drift in inverse proportion to its size: Small populations rapidly fix or lose a given allele, whereas large populations take longer to show the same effects. Genetic drift also shows itself in several other ways.

Founder Effects and Bottlenecks

Several well-known genetic phenomena are caused by populations starting at or proceeding through small num-

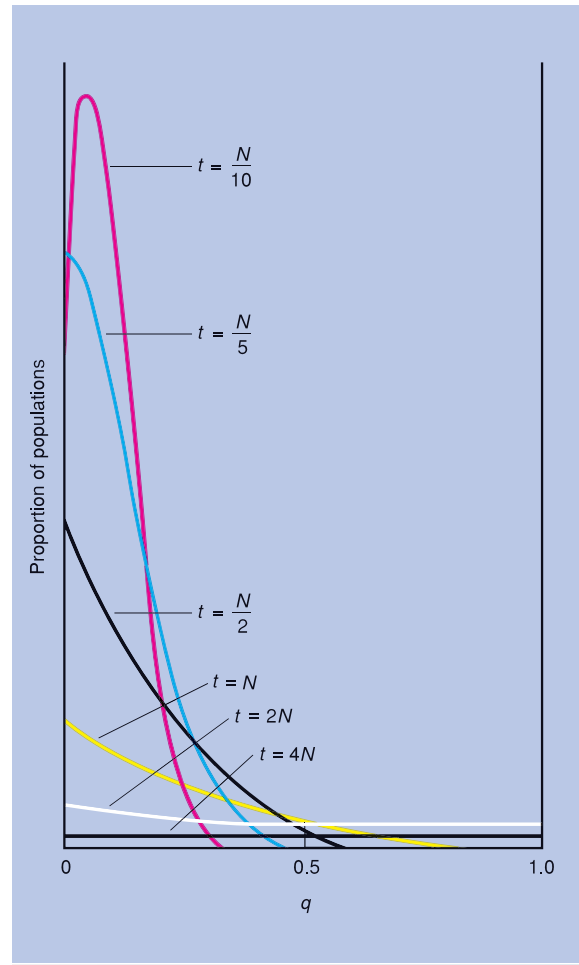


Figure 20.8 Random genetic drift in small populations with $q = 0.1$. Compare this figure with figure 20.6. In this case, the probability of fixation of the a allele is 0.1, and the probability of its loss is 0.9. (From M. Kimura, "Solution of a process of random genetic drift with a continuous model," *Proceedings of the National Academy of Sciences, USA*, 41:144–50, 1955. Reprinted by permission.)

bers. When a population is initiated by a small, and therefore genetically unrepresentative, sample of the parent population, the genetic drift observed in the subpopulation is referred to as a **founder effect**. A classic human example is the population founded on Pitcairn Island by several of the *Bounty* mutineers and some Polynesians. The unique combination of Caucasian and Polynesian traits that characterizes today's Pitcairn Island population resulted from the small number of founders for the population.

Sometimes populations go through **bottlenecks**, periods of very small population size, with predictable ge-

netic results. After the bottleneck, the parents of the next generation have been reduced to a small number and may not be genetically representative of the original population. The field mice on Muskeget Island, Massachusetts, have a white forehead blaze of hair not commonly found in nearby mainland populations. Presumably, the island population went through a bottleneck at the turn of the century, when cats on the island reduced the number of mice to near zero. The population was reestablished by a small group of mice that happened by chance to contain several animals with this forehead blaze.

NATURAL SELECTION

Although mutation, migration, and random genetic drift all influence allelic frequencies, they do not necessarily produce populations of individuals that are better adapted to their environments. Natural selection, however, tends to that end. The consequence of natural selection, Darwinian evolution, is considered in detail in the next chapter. We discuss here the algebra behind the process of natural selection. Artificial selection, as practiced by animal and plant breeders, follows the same rules.

How Natural Selection Acts

Selection, or **natural selection**, is a process whereby one phenotype and, therefore, one genotype leaves relatively more offspring than another genotype, measured by both reproduction and survival. Selection is thus a matter of **reproductive success**, the relative contribution of that genotype to the next generation. It is important to remember that selection acts on whole organisms and thus on phenotypes. However, we analyze the process by looking directly at the genotype, usually only at one locus.

Fitness

A measure of reproductive success is the **fitness**, or **adaptive value**, of a genotype. A genotype that, compared with other genotypes, leaves relatively more offspring that survive to reproduce has the higher fitness. (Note that this use of the word *fitness* differs from our common notion of physical fitness.)

Fitness is usually computed to vary from zero to one (0–1) and is always related to a given population at a given time. For example, in a normal environment, fruit flies with long wings may be more fit than fruit flies with short wings. But in a very windy environment, a fruit fly with limited flying ability may survive better than one with the long-winged genotype, which will be blown

around by the wind. Thus, fitness (usually assigned the letter W) is relative to a given circumstance. In a given environment, the genotype that leaves the most offspring is usually assigned a fitness of $W = 1$, and a lethal genotype has a fitness of $W = 0$. Any other genotype has a fitness value between zero and one. A number of factors can decrease this fitness value, W , below one. A **selection coefficient** measures the sum of forces acting to prevent reproductive success. It is usually represented by the letter s or t and is defined by the fitness equation

$$W = 1 - s \quad (20.13)$$

and

$$s = 1 - W \quad (20.14)$$

Thus, as the selection coefficient increases, fitness decreases, and vice versa.

Components of Fitness

Natural selection can act at any stage of the life cycle of an organism. It usually acts in one of four ways. (1) The reproductive success of a genotype can be affected by prenatal, juvenile, or adult survival. Differential survival of genotypes is referred to as viability selection or **zygotic selection**. (2) A heterozygote can produce gametes with differential success when one of its alleles fertilizes more often than the other. This is termed **gametic selection**. A well-studied case is the t -allele (tailless) locus in house mice; the gametes of as many as 95% of the heterozygous males of the Tt genotype carry the t allele. (This phenomenon is also referred to as **segregation distortion** or **meiotic drive**.) Selection can also take place in two areas of the reproductive segment of an organism's life cycle. (3) Some genotypes may mate more often than others (have greater mating success), resulting in **sexual selection**. Sexual selection usually occurs when members of the same sex compete for mates or when females have some form of choice. Adaptations for fighting, such as antlers in male elk, or displaying, such as the peacock's tail, are the results of sexual selection. (4) Finally, some genotypes may be more fertile than other genotypes, resulting in **fecundity selection**. The particular variable of the life cycle that selection acts upon is termed a **component of fitness**.

Effects of Selection

Figure 20.9 shows the three main ways that the sum total of selection can act. **Directional selection** works by continuously removing individuals from one end of the phenotypic (and therefore, presumably, genotypic) distribution (e.g., short-necked giraffes are removed). Removal means disappearance through death or failure to reproduce (genetic death). Thus, the mean is constantly shifted toward

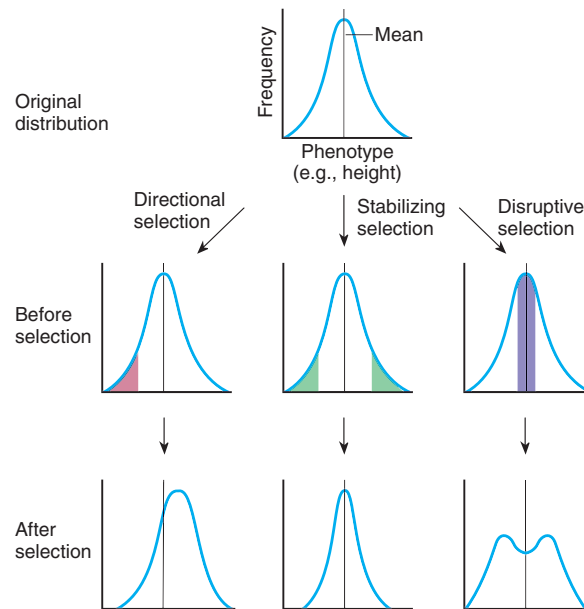


Figure 20.9 Directional, stabilizing, and disruptive selection. Colored areas show the groups being selected against. At the top is the original distribution of individuals. The final distributions after selection appear in the bottom row.

the other end of the phenotypic distribution; in our example, the mean shifts toward long-necked giraffes. The evolution of neck length in giraffes, presumably by directional selection, has been documented from the geologic record.

Stabilizing selection (fig. 20.9) works by constantly removing individuals from both ends of a phenotypic distribution, thus maintaining the same mean over time. Stabilizing selection now works on giraffe neck length—it is neither increasing nor decreasing. **Disruptive selection** works by favoring individuals at both ends of a phenotypic distribution at the expense of individuals in the middle. It, like stabilizing selection, should maintain the same mean value for the phenotypic distribution. Disruptive selection has been carried out successfully in the laboratory for bristle number in *Drosophila*. Starting with a population with a mean number of sternopleural chaeta (bristles on one of the body plates) of about eighteen, investigators succeeded after twelve generations of getting a fly population with one peak of bristle numbers at about sixteen and another at about twenty-three (fig. 20.10).

Selection Against the Recessive Homozygote



We can analyze selection by using our standard model-building protocol of population genetics—namely, de-

fine the initial conditions; allow selection to act; calculate the allelic frequency after selection (q_{n+1}); calculate Δq (change in allelic frequency from one generation to the next); then calculate equilibrium frequency, $\hat{q}$, when Δq becomes zero; and examine the stability of the equilibrium. In the analysis that follows, we consider a single autosomal locus in a diploid, sexually reproducing species with two alleles and assume that selection acts directly on the phenotypes in a simple fashion (i.e., it occurs at a single stage in the life of the organism, such as larval mortality in *Drosophila*). After selection, the individuals remaining within the population mate at random to form a new generation in Hardy-Weinberg proportions.

Selection Model

In table 20.2, we outline the model for selection against the homozygous recessive genotype. The initial population is in Hardy-Weinberg equilibrium. Even with selection acting during the life cycle of the organism, Hardy-Weinberg proportions will be reestablished anew after each round of random mating, although presumably at new allelic frequencies. All selection models start out the same way. They diverge at the point of assigning fitness, which depends on the way natural selection is acting. In the model in table 20.2, the dominant homozygote and the heterozygote have

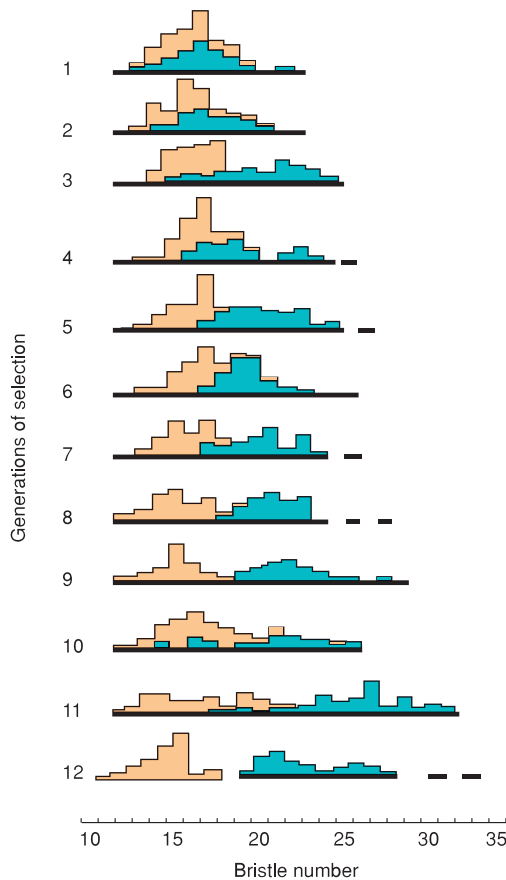


Figure 20.10 Disruptive selection in *Drosophila melanogaster*. After twelve generations of selection for flies with either many or few bristles (chaetae) on the sternopleural plate, the population was bimodal. In other words, many flies in the population had either few or many bristles, but few flies had an intermediate bristle number. (Reprinted with permission from *Nature*, Vol. 193, J. M. Thoday and J. B. Gibson, "Isolation by Disruptive Selection." Copyright © 1962 Macmillan Magazines Limited.)

the same fitness ($W = 1$). Natural selection cannot differentiate between the two genotypes because they both have the same phenotype. The recessive homozygote, however, is being selected against, which means that it has a lower fitness than the two other genotypes ($W = 1 - s$).

After selection, the ratio of the different genotypes is determined by multiplying their frequencies (Hardy-Weinberg proportions) by their fitnesses. The procedure follows from the definition of fitness, which in this case is a relative survival value. Thus, only $1 - s$ of the aa genotype survives for every one of the other two genotypes. For example, if s were 0.4, then the fitness of the aa type would be $1 - s$, or 0.6. For every ten AA and Aa

Table 20.2 Selection Against the Recessive Homozygote: One Locus with Two Alleles, A and a

	Genotype			Total
	AA	Aa	aa	
Initial genotypic frequencies	p^2	$2pq$	q^2	1
Fitness (W)	1	1	$1 - s$	
Ratio after selection	p^2	$2pq$	$q^2(1 - s)$	$1 - sq^2 = \bar{W}$
Genotypic frequencies after selection	$\frac{p^2}{\bar{W}}$	$\frac{2pq}{\bar{W}}$	$\frac{q^2(1 - s)}{\bar{W}}$	1

individuals that survive to reproduce, only six aa individuals would survive to reproduce. The total of the three genotypes after selection is $1 - sq^2$. That is,

$$p^2 + 2pq + q^2(1 - s) = p^2 + 2pq + q^2 - sq^2 = 1 - sq^2$$

Mean Fitness of a Population

The value $(1 - sq^2)$ is referred to as the **mean fitness of the population**, $\bar{W}$, because it is the sum of the fitnesses of the genotypes multiplied (weighted) by the frequencies at which they occur. Thus, it is a weighted mean of the fitnesses, weighted by their frequencies. The new ratios of the three genotypes can be returned to genotypic frequencies by simply dividing by the mean fitness of the population, $\bar{W}$, as in the last line of table 20.2. (Remember that a set of numbers can be converted to proportions of unity by dividing them by their sum.) The new genotypic frequencies are thus the products of their original frequencies times their fitnesses, divided by the mean fitness of the population.

After selection, the new allelic frequency (q_{n+1}) is the proportion of aa homozygotes plus half the proportion of heterozygotes, or

$$\begin{aligned} q_{n+1} &= \frac{q^2(1 - s)}{1 - sq^2} + \frac{pq}{1 - sq^2} \\ &= \frac{q(q - sq + p)}{1 - sq^2} \\ &= \frac{q(1 - sq)}{1 - sq^2} \end{aligned} \quad (20.15)$$

This model can be simplified somewhat if we assume that the aa genotype is lethal. Its fitness would be zero,

and s , the selection coefficient, would be one. Equation 20.15 would then change to

$$q_{n+1} = \frac{q(1 - q)}{1 - q^2} \quad (20.16)$$

Since $(1 - q^2)$ is factorable into $(1 - q)(1 + q)$, equation 20.16 becomes

$$\begin{aligned} q_{n+1} &= \frac{q(1 - q)}{(1 - q)(1 + q)} \\ &= \frac{q}{1 + q} \end{aligned} \quad (20.17)$$

The change in allelic frequency is then calculated as

$$\Delta q = q_{n+1} - q = \frac{q}{1 + q} - q$$

To solve this equation, q is multiplied by $(1 + q)/(1 + q)$ so that both parts of the expression are over a common denominator:

$$\begin{aligned} \Delta q &= \frac{q - q(1 + q)}{1 + q} \\ &= \frac{-q^2}{1 + q} \end{aligned} \quad (20.18)$$

This is the expression for the change in allelic frequency caused by selection. Since selection will not act again until the same stage in the life cycle during the next generation, equation 20.18 is also an expression for the change in allelic frequency between generations.

Two facts should be apparent from equation 20.18. First, the frequency of the recessive allele (q) is declining, as indicated by the negative sign of the fraction. This fact should be intuitive because of the way selection was defined in the model (eliminating aa homozygotes). Second, the change in allelic frequency is proportional to q^2 , which appears in the numerator of the expression. In other words, allelic frequency is declining as a relative function of the number of homozygous recessive individuals in the population. This fact is consistent with the premise of the selection model (with selection against the homozygous recessive genotype). This final formula supports the methodology of the model.

Equilibrium Conditions

Next we calculate the equilibrium q by setting the Δq equation equal to zero, since a population in equilibrium will show no change in allelic frequencies from one generation to the next:

$$\frac{-q^2}{1 + q} = 0 \quad (20.19)$$

For a fraction to be zero, the numerator must equal zero. Thus, $q^2 = 0$, and $\hat{q} = 0$. At equilibrium, the a allele should be entirely removed from the population. If the aa homozygotes are being removed, and if there is no mutation to return a alleles to the population, then eventually the a allele disappears from the population.

Time Frame for Equilibrium

One shortcoming of this selection model is that it is not immediately apparent how many generations will be required to remove the a allele. The deficiency can be compensated for by using a computer simulation or by introducing a calculus differential into the model. Either method would produce the frequency-time graph of figure 20.11. This figure clearly shows that the a allele is removed more quickly when selection is stronger (when s is larger) and that the curves appear to be asymptotic—the a allele is not immediately eliminated and would not be entirely removed until an infinitely large number of generations had passed. There is a reason for the asymptotic behavior of the graph: As the a allele becomes rarer and rarer, it tends to be found in heterozygotes (table 20.3). Since selection can remove only aa homozygotes, an a allele hidden in an Aa heterozygote will not be selected against. When $q = 0.5$, there are two heterozygotes for every aa homozygote. When

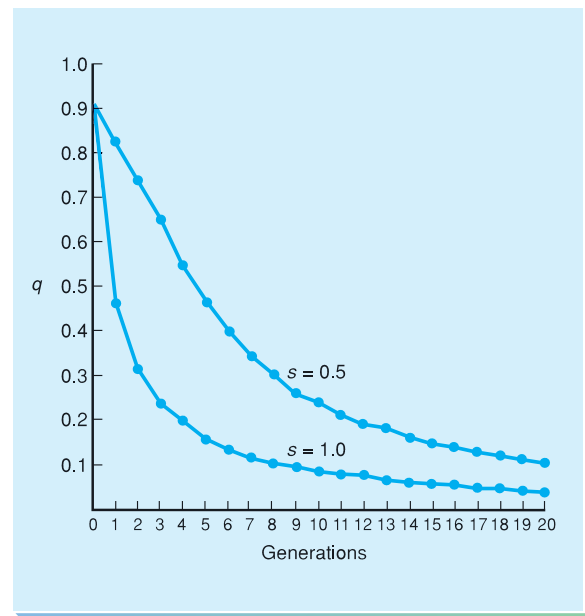


Figure 20.11 Decline in q (the frequency of the a allele) under different intensities of selection against the aa homozygote. Note that the loss of the a allele is asymptotic in both cases, but the drop in allelic frequency is more rapid with the larger selection coefficient.

Table 20.3 Relative Occurrence of Heterozygotes and Homozygotes as Allelic Frequency Declines: $q = f(a)$; $p = f(A)$

q	$f(Aa)$ ($2pq$)	$f(aa)$ (q^2)	$f(Aa)/f(aa)$
0.5	0.50	0.25	2
0.2	0.32	0.04	8
0.1	0.18	0.01	18
0.01	0.0198	0.0001	198
0.001	0.001998	0.000001	1,998

$q = 0.001$, there are almost two thousand heterozygotes per aa homozygote. Remember, only the recessive homozygote is selected against. Natural selection cannot distinguish the dominant homozygote from the heterozygote.

Selection-Mutation Equilibrium

Although a deleterious allele is eliminated slowly from a population, the time frame is so great that there is opportunity for mutation to bring the allele back. Given a population in which alleles are removed by selection and added by mutation, the point at which no change in allelic frequency occurs, the **selection-mutation equilibrium**, may be determined as follows. The new frequency (q_{n+1}) of the recessive a allele after nonlethal selection ($s < 1$) against the recessive homozygote is obtained by equation 20.15:

$$q_{n+1} = \frac{q(1 - sq)}{1 - sq^2}$$

Change in allelic frequency under this circumstance will thus be

$$\begin{aligned} \Delta q &= q_{n+1} - q = \frac{q(1 - sq)}{(1 - sq^2)} - \frac{q(1 - sq^2)}{(1 - sq^2)} \\ &= \frac{q - sq^2 - q + sq^3}{(1 - sq^2)} \\ &= \frac{-sq^2(1 - q)}{1 - sq^2} \end{aligned} \quad (20.20)$$

Equation 20.20 is the general form of equation 20.18 for any value of s . The change in allelic frequency due to mutation can be found by using equation 20.4:

$$\Delta q = \mu p - \nu q$$

where μ and ν are the rate of forward and back mutation, respectively. When equilibrium exists, the change

from selection will just balance the change from mutation. Thus,

$$\mu p - \nu q + \frac{-sq^2(1 - q)}{1 - sq^2} = 0$$

and

$$\mu p - \nu q = \frac{sq^2(1 - q)}{1 - sq^2} \quad (20.21)$$

Now, some judicious simplifying is justified, because in a real situation, q will be very small because the a allele is being selected against. Thus, νq will be close to zero, and $1 - sq^2$ will be close to unity. Equation 20.21, therefore, becomes:

$$\begin{aligned} \mu p &\approx sq^2(1 - q) \\ \mu(1 - q) &\approx sq^2(1 - q) \\ q^2 &\approx \mu/s \\ \hat{q} &\approx \sqrt{\mu/s} \end{aligned} \quad (20.22)$$

In the case of a recessive lethal, s would be unity, so

$$q^2 \approx \mu \text{ and } \hat{q} \approx \sqrt{\mu}$$

If a recessive homozygote has a fitness of 0.5 ($s = 0.5$) and a mutation rate, μ , of 1×10^{-5} , the allelic frequency at selection-mutation equilibrium will be

$$\begin{aligned} \hat{q} &\approx \sqrt{\mu/s} \approx \sqrt{1 \times 10^{-5}/0.5} \approx \sqrt{2 \times 10^{-5}} \\ &\approx 0.004 \end{aligned}$$

If the recessive phenotype were lethal, then

$$\begin{aligned} \hat{q} &\approx \sqrt{\mu/s} \approx \sqrt{1 \times 10^{-5}/1} \\ &\approx 0.003 \end{aligned}$$

These are very low equilibrium values for the a allele.

Types of Selection Models

In view of the limited ways that fitnesses can be assigned, only a limited number of selection models are possible. Table 20.4 lists all possible selection models if we assume that fitnesses are constants and the highest fitness is one. (You might now go through the list of models and determine the equilibrium conditions for each.) Note that two possible fitness distributions are missing. There is no model in which fitnesses are $1 - s$, 1, and 1 for the A_1A_1 , A_1A_2 , and A_2A_2 genotypes, respectively (remembering that $p = f[A_1]$ and $q = f[A_2]$). That model is for selection against the A_1A_1 homozygote. Some reflection should show that this is the same model as model 1 of table 20.4, except that the A_1 allele is acting like a recessive allele. In other words, natural selection acts against A_1A_1 homozygotes, but not against the A_1A_2 and A_2A_2 genotypes. Thus, the model reduces to model 1 if we treat A_1 as the recessive allele and A_2 as the

Table 20.4 All Possible One-Locus, Two-Allele Selection Models (Assuming All Selection Coefficients Are Constants)

Type of Selection	Genotypic Fitness		
	$A_1 A_1$	$A_1 A_2$	$A_2 A_2$
1. Against recessive homozygotes	1	1	$1 - s$
2. Against heterozygotes	1	$1 - s$	1
3. Against one allele	1	$1 - s_1$	$1 - s_2$
4. Against homozygotes	$1 - s_1$	1	$1 - s_2$

dominant allele. Similarly, the $(1 - s_1, 1 - s_2, 1)$ model is eliminated for the same reason (allele A_2 is acting like the dominant allele and A_1 like the recessive allele). We now describe the outcome of each of the models in the table.

In both models 1 and 3 (table 20.4), selection is against genotypes containing the A_2 allele. Model 1, which we just derived in detail, is the model for a deleterious recessive allele. Almost any enzyme defect in a metabolic pathway fits this model, such as PKU, alkaptonuria, Tay-Sachs disease, and so on. In model 3, however, natural selection can detect the heterozygote, as is the case with deleterious alleles that are not completely recessive. An example would be the hemoglobin anomaly called thalassemia, a disorder common in some European and Asian populations, that produces a severe anemia in homozygotes and a milder anemia in heterozygotes. It should be clear that selection can more quickly eliminate a partially recessive allele than a completely recessive allele because the allele can no longer “hide” in the heterozygote.

Dominant or semidominant alleles (model 3) are usually more quickly removed from a population because they are completely open to selection. It takes an infinite number of generations to remove a recessive lethal allele, but only one generation for natural selection to remove a completely dominant lethal allele (see model 3, where $s_1 = s_2 = 1$). Examples of dominant deleterious traits in

people are Huntington disease, facioscapular muscular dystrophy, and chondrodystrophy.

Model 2 is interesting because selection against the heterozygote leads to an unstable equilibrium at $q = 0.5$. If one heterozygote is removed by selection, one each of the two alleles is eliminated. However, if p and q are not equal (and thus not equal to 0.5), then one A_1 allele is not the same proportion of the A_1 alleles as one A_2 allele is of all the A_2 alleles. In other words, in a population of fifty individuals with $q = 0.1$ and $p = 0.9$, one A_2 allele is 10% (1/10) of the A_2 alleles, whereas one A_1 allele is only 1.1% (1/90) of the A_1 alleles. Removing one each of the two alleles causes a decrease in q . Therefore, a population following model 2 is at equilibrium at $p = q = 0.5$. However, this is an unstable equilibrium. Any perturbation that changes the allelic frequencies causes the rarer allele to be selected against and eventually removed from the population. An example is the maternal-fetal incompatibility at the Rh locus in human beings. The disease erythroblastosis occurs only in heterozygous fetuses ($Rh^+ Rh^-$) in Rh-negative ($Rh^- Rh^-$) mothers. Heterozygotes are, therefore, selected against.

In model 4, selection is against homozygotes. This model is called the **heterozygote advantage**, and we will derive its equilibrium condition because the results are important to evolutionary theory (table 20.5). At equilibrium

$$\Delta q = \frac{pq(s_1 p - s_2 q)}{\bar{W}} \quad (20.23)$$

For this expression to be zero, either

$$p = 0, q = 0, \text{ or } (s_1 p - s_2 q) = 0$$

If $p = 0$ or $q = 0$, the result is trivial; the equilibrium exists only because of the absence of one of the alleles. The more meaningful equilibrium occurs when $s_1 p - s_2 q = 0$. In that case

$$s_1 p = s_2 q \text{ or } s_1(1 - q) = s_2 q$$

and

$$\hat{q} = \frac{s_1}{s_1 + s_2} \quad (20.24)$$

Table 20.5 Selection Model of Heterozygote Advantage: The A Locus with A_1 and A_2 Alleles

	Genotype			Total
	$A_1 A_1$	$A_1 A_2$	$A_2 A_2$	
Initial genotypic frequencies	p^2	$2pq$	q^2	1
Fitness (W)	$1 - s_1$	1	$1 - s_2$	
Ratio after selection	$p^2(1 - s_1)$	$2pq$	$q^2(1 - s_2)$	$1 - s_1 p^2 - s_2 q^2 = \bar{W}$
Genotypic frequencies after selection	$\frac{p^2(1 - s_1)}{\bar{W}}$	$\frac{2pq}{\bar{W}}$	$\frac{q^2(1 - s_2)}{\bar{W}}$ ¹	

BOX 20.1

It is surprising how much insight we can gain into the processes of population genetics by modeling them on a computer. The simple computer program presented here calculates changing allelic frequencies due to random mating when alleles at a locus are under a heterozygote-advantage selection regime. The program is written in the Microsoft® Visual Basic language. You can simulate any of the selection models described in chapter 20 by simply changing the variables. Also, this program can model many of the other processes discussed in this and the

Experimental
Methods

*A General Computer
Program to Simulate the
Approach to Allelic
Equilibrium Under
Heterozygote Advantage*

last chapter; usually, only a few lines need to be changed to look at an entirely different process. Other com-

puter programs can substitute. Output should be graphed. The program should be rerun several times with various sets of values for the allelic frequencies and fitnesses. If the outcome isn't clear by twenty-five generations, the number of generations can be increased with a few small changes in the program.

In the computer program (fig. 1), p is set to 0.9, q is $1 - p$ (0.1), and the three fitnesses are named w_{11} , w_{12} , and w_{22} for the AA , Aa , and aa genotypes, respectively. In this case, w_{11} is set to 0.4, w_{12} to 1, and w_{22} to 0.6, a model of heterozygote advantage;

continued

```
Sub Command1_Click ()
    Static q(25)
    Static p(25)
    Picture1.Cls

    'Set variables

    p(1) = .9
    w11 = .4
    w12 = 1
    w22 = .6
    q(1) = 1 - p(1)

    'Calculate p and q values

    For i = 2 To 25
        wbar = p(i - 1) ^ 2 * w11 + 2 * p(i - 1) * q(i - 1) * w12 + q(i - 1) ^ 2 * w22
        q(i) = (q(i - 1) ^ 2 * w22 + p(i - 1) * q(i - 1) * w12) / wbar
        p(i) = 1 - q(i)
    Next i

    'Draw axes and grid

    Picture1.Scale (-1, 1.1)-(26, -.1)
    Picture1.Line (0, 0)-(0, 1)
    For i = 0 To 10
        Picture1.Line (0, .1 * i)-(25, .1 * i)
    Next i
    For i = 5 To 25 Step 5
        Picture1.Line (i, 0)-(i, 1)
    Next i

    'Draw q values

    Picture1.DrawWidth = 5
    For i = 1 To 25
        Picture1.PSet (i, q(i))
    Next i
End Sub
```

Figure 1 A Microsoft® Visual Basic computer program for the simulation of heterozygote advantage. The first statement indicates that the program is run by clicking a command button. Twenty-five values of q and p are calculated and stored for printing. The program also prints a grid of lines at increments of $q = 0.1$ and generations = 5.

BOX 20.1 CONTINUED

the number of generations is twenty-five. The program calculates the mean fitness of the population, $\bar{w}$, as $p^2(w_{11}) + 2pq(w_{12}) + q^2(w_{22})$; it then calculates the new allelic frequencies after one generation of se-

lection (the new proportion of aa homozygotes plus half the proportion of heterozygotes). The program then repeats this process twenty-five times, storing each new q in the array $q(i)$. The graphic output shown in fig-

ure 2 results. As you can see, q is approaching 0.6. If you would like to see the values generated by the program, appropriate print statements can be added.

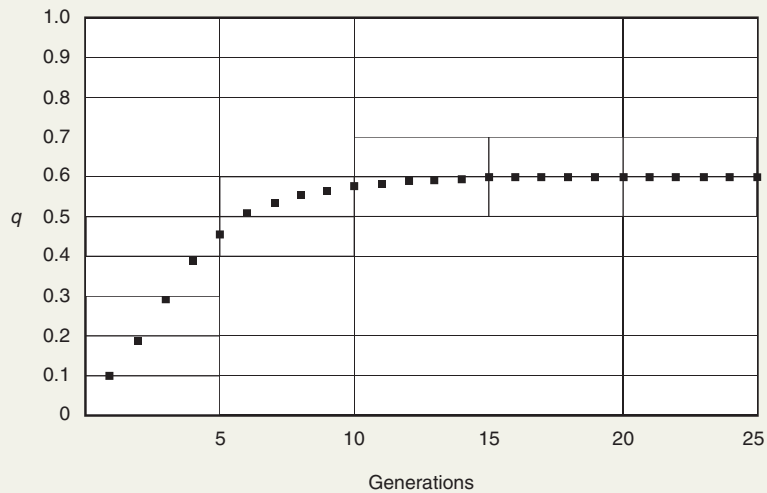


Figure 2 Graphical output of the computer program from figure 1, with axis labels drawn in. The frequency of the a allele, q , begins at 0.1 and asymptotes toward 0.6.

Since $p + q = 1$,

$$\hat{p} = \frac{s_2}{s_1 + s_2} \quad (20.25)$$

Several interesting conclusions follow. First, unlike the other models of selection, this model allows a population to maintain both alleles. We can demonstrate that this equilibrium is stable by graphing the Δq value against q . Such a graph appears in figure 20.12, in which q is the frequency of allele A_2 and the fitnesses of genotypes A_1A_1 , A_1A_2 , and A_2A_2 are assumed to be 0.8, 1, and 0.7, respectively. Note that if the equilibrium is perturbed by an increase or decrease in q , the population returns to the point of equilibrium. Second, the equilibrium is independent of the original allelic frequencies since it involves only the selection coefficients, s_1 and s_2 . Last, the equilibrium for each allele (equations 20.24 and 20.25) is directly proportional to the selection coefficient against the other allele. As the selection against A_1 increases (s_1 increases), the equilibrium shifts toward a higher value of q (more A_2 alleles; box 20.1).

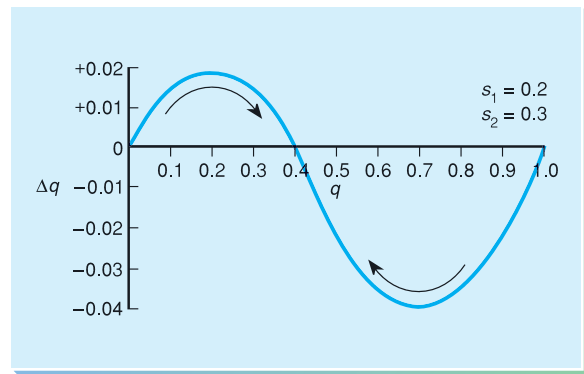


Figure 20.12 Plot of allelic frequency (q) versus change in allelic frequency (Δq) for a polymorphism maintained by heterozygote advantage. In this case, $s_1 = 0.2$ and $s_2 = 0.3$; the equilibrium value, $\hat{q}$, is 0.4. When perturbed, the population tends to return to this value unless the perturbation brings q to either 1.0 or 0.0, in which case the population is either fixed for the a allele or has lost it. In both cases, no further change in allelic frequency will take place, barring mutation or migration.

S U M M A R Y

STUDY OBJECTIVE 1: To develop ways to analyze population genetics problems 571

A five-step protocol is presented to determine equilibrium allelic frequencies.

STUDY OBJECTIVE 2: To analyze the effects of mutation, migration, and population size on the Hardy-Weinberg equilibrium 571–577

The effects of relaxing some of the assumptions of the Hardy-Weinberg equilibrium are analyzed. Both mutation and migration transport alleles in and out of a population. Mutation provides the variability on which natural selection acts, but it usually does not directly affect the equilibrium because mutation rates are usually very low. If two randomly mating populations merge, or if two randomly mating demes are mistakenly treated as a single deme, the conglomerate will be deficient in heterozygotes. This deviation is called the Wahlund effect.

Finite population size is a source of sampling error. It results in changes in allelic frequencies known as random genetic drift. The smaller the population, the more rapidly allelic frequencies change. The dynamics of random genetic drift were studied graphically.

STUDY OBJECTIVE 3: To study the ways in which natural selection results in organisms adapted to their environments 577–584

Natural selection is defined by differential reproductive success. Depending upon which phenotypes are most fit, natural selection can act in several ways to change allelic and genotypic frequencies. Selection against the recessive homozygote acts to remove the allele from the population. Mutation brings the allele back into the population. Thus, a selection-mutation equilibrium maintains the unfavorable allele at a relatively low frequency. Heterozygote advantage maintains both alleles in a population.

S O L V E D P R O B L E M S

PROBLEM 1: At a particular locus, there are two alleles, B and b . The mutation rate of B to b is 3.5×10^{-4} , whereas the mutation rate of b to B is 6×10^{-8} . What is the equilibrium frequency of the b allele, assuming no other factor is operating in this population to disturb the Hardy-Weinberg equilibrium?

Answer: We let $q = f(b)$, $\mu = 3.5 \times 10^{-4}$, and $\nu = 6 \times 10^{-8}$. We then simply substitute μ and ν into equation 20.6:

$$\begin{aligned}\hat{q} &= \mu/(\mu + \nu) = 3.5 \times 10^{-4} / \\ &\quad (3.5 \times 10^{-4} + 6 \times 10^{-8}) \\ &= 0.9998\end{aligned}$$

PROBLEM 2: Given a population of about one million cicadas with a frequency of the a allele at the A locus of 0.75, what is the probability that the a allele will be lost due to random genetic drift? How much longer will the possible loss of the allele take than the loss of the allele would take in a population of one thousand?

Answer: Regardless of the size of a finite population, random genetic drift takes place. The probability of the loss of an allele with a frequency of 0.75 is 0.25; the prob-

ability of its fixation is 0.75 (see fig. 20.8). Since it is convenient to measure time (number of generations) within populations of finite size in units of population size, we can see that an event that takes N generations will be one thousand generations in the small population, but one million generations in the large population. Thus, random genetic drift occurs in the larger population at about one-thousandth the rate of the small population.

PROBLEM 3: In a laboratory colony of fruit flies, the fitnesses of the genotypes of an electrophoretic locus (malate dehydrogenase) are determined. Three genotypes, FF , FS , and SS , have fitnesses of 0.85, 1.0, and 0.6, respectively. What is the equilibrium frequency of the slow allele (S)?

Answer: If the fitnesses of the three genotypes FF , FS , and SS are as given, then the locus is exhibiting heterozygote advantage with selection coefficients of the two homozygotes of $s_1 = 0.15$ ($1 - 0.85$) and $s_2 = 0.4$ ($1 - 0.6$). If q is the frequency of the slow allele, then, using equation 20.24,

$$\hat{q} = s_1/(s_1 + s_2) = 0.15/(0.15 + 0.4) = 0.27$$

EXERCISES AND PROBLEMS *

MUTATION

- Consider a locus with alleles A and a in a large, randomly mating population under the influence of mutation.
 - If the mutation rate of A to a is 6×10^{-5} , and the back-mutation rate to A is 7×10^{-7} , what is the equilibrium frequency of a ?
 - If $q = 0.9$ in generation n , what would it be one generation later, under only the influence of mutation?
- Derive an expression for mutation equilibrium when no back mutation is occurring.
- Consider a population in which $p = 0.9$ and $q = 0.1$. If the forward mutation rate, $A \rightarrow a$, is 5×10^{-5} and the reverse mutation rate, $a \rightarrow A$, is 2×10^{-5} , calculate the equilibrium frequency, $\hat{q}$, of the a allele.
- If the forward mutation rate, $A \rightarrow a$, is five times the reverse mutation rate, what is the equilibrium frequency of the a allele?

MIGRATION

- The following data refer to the R^o allele in the Rh blood system:

frequency in western Europeans = 0.62
frequency in eastern Europeans = 0.45
frequency in Mongols = 0.03

What is the total proportion of alleles that have entered the eastern European population?
- Given the data from problem 1 of chapter 19, what factors could have caused the population to leave Hardy-Weinberg equilibrium? (See also SMALL POPULATION SIZE and NATURAL SELECTION)
- In a population of nine hundred butterflies, the frequency (p) of the fast allele of the enzyme phosphoenol pyruvate is 0.6, and the frequency of the slow form (q) is 0.4. Ninety butterflies migrate to this population, and the migrants have a slow-allele frequency of 0.8. Calculate the allelic frequencies of the new population.
- If the frequency of the N allele is 0.25 in a native population, 0.32 in a conglomerate population, and 0.4 in a migrant population, what percentage of the N alleles in the conglomerate population were derived from the migrant population?
- In a particular population, the frequency of allele t was 0.25 in a migrant population and 0.45 in the

conglomerate population. If the migration rate was 0.1, calculate the frequency of t in the original, native population.

SMALL POPULATION SIZE

- In a population of five hundred individuals with a frequency of allele A of 0.7, what is the ultimate fate of the A allele? What is the probability that the population will eventually lose the A allele? How many are $N/5$ generations? $4N$ generations?

NATURAL SELECTION

- Differentiate among stabilizing, directional, and disruptive selection.
- Derive a model of selection in which the fitness of the heterozygote is half the fitness of one of the homozygotes and twice the fitness of the other. Give expressions for the following:
 - Mean population fitness
 - Equilibrium allelic frequency (stable?)
- Derive an expression for the equilibrium allelic frequencies under a model in which selection acts against heterozygotes. Is the equilibrium stable?
- Table 20.6 describes selection at the A locus in a given diploid species in which $p = f(A)$ and $q = f(a)$.
 - Describe the type of selection occurring here. Why does the total equal one before selection but $\bar{W}$, after?
 - Derive an equation for q after one generation of selection (q_{n+1}).
 - This system will reach equilibrium, with $\hat{p} = s_2/(s_1 + s_2)$. If selection is twice as strong against aa as against AA , what are the equilibrium allelic frequencies? If $s_1 = 0.1$ and $s_2 = 0.3$, what percentage of heterozygotes is at equilibrium?
- Given a locus with alleles A and a in a sexually reproducing, diploid population in Hardy-Weinberg equilibrium, set up a model and the initial formula for the frequency of the dominant allele after one generation ($p_n + 1$) if selection acts against the dominant phenotype. What are the equilibrium conditions?
- There is a locus with alleles A and a in a large, randomly mating, diploid, sexually reproducing population. Allele A mutates to a at a rate of μ , and no back mutation takes place. However, the aa homozygote is selected against with a fitness of $1 - s$. Give a formula for the equilibrium condition. If $\mu = 5 \times 10^{-5}$

* Answers to selected Exercises and Problems are on page A-22.

Table 20.6

	Genotypes			Total
	<i>AA</i>	<i>Aa</i>	<i>aa</i>	
Before selection	p^2	$2pq$	q^2	1
Fitness (<i>W</i>)	$1 - s_1$	1	$1 - s_2$	
After selection	$p^2(1 - s_1)$	$2pq$	$q^2(1 - s_2)$	$\bar{W} = 1 - s_1p^2 - s_2q^2$

and $s = 0.15$, what are the equilibrium allelic frequencies?

- If a locus has alleles A_1 and A_2 , what is the equilibrium frequency of A_1 if both homozygotes are lethal?
- The following data were collected from a population of *Drosophila* segregating sepia (s) and wild-type (s^+) eye colors. One sample was taken when the eggs were deposited, and another was taken later among adults. Reconstruct the mode of selection.

	s^+s^+	s^+s	ss
Egg	25	50	25
Adult	30	60	10

- The data in table 20.7 come from T. Dobzhansky's work with chromosomal inversions in *Drosophila pseudoobscura*. They represent four samples from various altitudes in the Sierra Nevada Mountains in California. What would you say about, and what would you do in the lab to determine, the fitnesses of the inversions? What factors could cause the changes in fitness?
- In a particular population with two alleles at a locus, the frequency of AA individuals = 0.25, Aa = 0.5, and aa = 0.25. If the AA genotype fitness = 1, Aa = 0.8, and aa = 0.6, what will the frequencies of A and a be in the next generation? Assume mutations do not occur.

Table 20.7 Data from Dobzhansky's Work

Elevation of Sample	Inversion			Others
	<i>ST</i>	<i>AR</i>	<i>CH</i>	
6,800 ft	26	44	16	14
4,600 ft	32	37	19	12
3,000 ft	41	35	14	10
800 ft	46	25	16	13

Note: *ST* = standard; *AR* = Arrowhead; *CH* = Chiricahua

- Calculate the frequency of the recessive b allele in a population one generation after selection if in the original population $q = f(b) = 0.7$ and the relative fitness of bb homozygotes is 0.4.
- A type of dwarfism in dogs is caused by a recessive allele. The mutation rate from the normal to the mutant allele has been estimated at 5×10^{-5} , and the fitness of the dwarf is 0.2 when compared with normal individuals. Calculate the equilibrium frequency of the dwarf allele.
- A recessive allele ($q = 0.5$) was initially neutral, but suddenly the environment changed and the recessive homozygote became lethal. What is q one generation after selection begins? What is the expected frequency of the recessive allele two generations after selection?

CRITICAL THINKING QUESTIONS

- If the selection model of heterozygous disadvantage leads to the elimination of the rarer allele, why would such systems (e.g., the Rh blood system) still exist.
- A scientist studied the distribution of electrophoretic genotypes in a sample of an insect species and found a deficiency of heterozygotes. How could this come about?

21

EVOLUTION AND SPECIATION

STUDY OBJECTIVES

1. To analyze the mechanisms of evolution and speciation 589
2. To investigate the mechanisms of the maintenance of genetic variation in natural populations, both selective and neutral 596
3. To discuss sociobiology, the evolution of social behavior 603

STUDY OUTLINE

Darwinian Evolution 589

Evolution and Speciation 589

Mechanisms of Cladogenesis 592

Phyletic Gradualism Versus Punctuated Equilibrium 594

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The cactus ground-finch (*Geospiza scandens*) from
Santa Cruz Island, Galápagos. (© Frans Lanting/
Photo Researchers, Inc.)

Populations change, or evolve, through natural selection and the other forces that perturb the Hardy-Weinberg equilibrium. The merger of population genetics theory with the classical Darwinian view of evolution is known as **neo-Darwinism**, or the “new synthesis.” In the two previous chapters, we laid the theoretical groundwork for an understanding of the process of evolution in natural populations. In this chapter, we concern ourselves with long-term evolution and speciation.

DARWINIAN EVOLUTION

Charles Darwin (fig. 21.1) was a British naturalist who published his theory of evolution in 1859 in a book entitled *The Origin of Species by Means of Natural Selection, or the Preservation of Favored Races in the Struggle for Life*. This book provided overwhelming support for evolution as well as a mechanism for the process. Darwin had been greatly influenced by the writings of the Reverend Thomas Malthus, who is best known for his theory that populations increase exponentially, whereas their food supplies increase arithmetically. Malthus, who proposed his theory in *An Essay on the Principle of Population* in 1798, was referring specifically to human populations and was trying to encourage people to reduce their birthrate



Figure 21.1 Charles Darwin (1809–1882). Darwin was an English naturalist who first established the theory of organic evolution by natural selection. (Painting by George Richmond, 1840. Downe House, Downe, Kent. © Archiv/Photo Researchers, Inc.)

rather than let their offspring starve to death. Malthus’s writings impressed upon Darwin the realization that under limited resources—the usual circumstance in nature—not all organisms survive. In nature, organisms compete for the resources needed to survive.

Darwin sailed aboard the HMS *Beagle*, a ship that circled the world from 1831 to 1836 with the primary purpose of charting the coast of South America. During his travels on the *Beagle*, Darwin amassed great quantities of observations (especially on South America and the Galápagos Islands) that led him to suggest a theory. Darwin proposed that organisms become adapted to their environment by the process of natural selection. In outline, the process works according to the following principles:

1. *Variation is a characteristic of virtually every group of animals and plants.* Darwin saw variation as an inherent property among individuals of all populations.
2. *Every group of organisms overproduces offspring.* Most populations maintain a relatively constant density over time. Thus, every parent, on average, just replaces itself. Therefore, most of the offspring the individuals of a population produce will die before they reproduce. Hence, in every group of organisms, there is an overabundance of young.
3. *Those that do survive and reproduce will pass on their genes in greater proportion.* This step is the cornerstone and the best-known part of Darwin’s theory. Among all the organisms competing for a limited array of resources, only the organisms best able to obtain and utilize these resources survive (**survival of the fittest**). If the favorable characteristics of these individuals are inherited, these traits pass on to the next generation. These organisms then have the greatest reproductive success (box 21.1).

Thus, over time, if advantageous mutations arise, or if the environment changes, the characteristics of a population should change through the process of natural selection (directional or disruptive selection). A particularly well-adapted population in a stable environment may maintain its numbers through the forces of stabilizing selection (see fig. 20.9). Nonrandom mating, genetic drift, and migration may also play a role in population differentiation.

EVOLUTION AND SPECIATION

The term **evolution** describes a change in genotypic frequencies, which usually results in a population of individuals better adapted to the environment than their ancestors were. **Speciation** comes in two different forms. (1) It may be the evolution of a population over time until the current population cannot be classified as

BOX 21.1

Ethics and Genetics

Attacks on Darwinism

From time to time, attacks on neo-Darwinism are mounted, usually by persons who either view evolutionary theory as antireligious or who misunderstand Darwin's theory. One attack, entitled "Darwin's Mistake," by Tom Bethell, was published in *Harper's* magazine in 1976.

Bethell began by pointing out that Darwinian theory is a tautology rather than a predictive theory. (The term *tautology* means a statement that is true by definition.) That is, evolution is the survival of the fittest. But who are the fittest? Obviously, the individuals who survive. Thus, without an independent criterion for fitness other than survival, we are left with the statement that evolution is the survival of the survivors. This, indeed, is a tautology. But it is possible to assign independent criteria for fitness. Darwin wrote extensively about artificial selection in pigeons, in which the breeders' choice was the criterion for fitness. (Many novel breeds of pigeon have been created this way.)

Plant and animal breeders have practiced artificial selection extensively. Here, survival is not the criterion for fitness; productivity is.

It is more difficult to establish *a priori* independent criteria of fitness in nature. Often, uncontrolled or unseen vagaries have major impacts on the course of events. Surely the temperature became colder before the mammoths became woolly. Is it then reasonable to predict that elephants would get woolly if the climate became colder in Africa today? The answer is no, for several reasons. First, the elephants might adapt to colder weather in any of a large number of different ways—they could get fatter, they could migrate, and so on. To some extent,

adaptation depends not only on the changing environment, but also on the reserve variation within the gene pool of the species. Second, the elephants could become extinct; they might not be able to adapt at all. And third, if the climatic changes were not severe, the elephants might not change at all.

Predicting the exact course of evolution is nearly impossible. To provide independent criteria for fitness in nature is, therefore, very difficult. Some modern evolutionary biologists, although not doubting neo-Darwinism, do worry to some extent about the difficulties in testing modern evolutionary theory. However, lower-level experiments are done to test various aspects of evolution in specific systems. For example, in 1993, B. Grant and P. Grant hypothesized that changes in bill size would occur in the finch *Geospiza fortis* (see fig. 21.6) because of changing food size due to changing weather on the Galápagos Islands. Their proposal seems to be correct. In addition, the support for Darwinism (the fossil

belonging to the same **species** as the original population. This process is known as **anagenesis**, or **phyletic evolution** (*an* is Latin for without, *genesis* is Latin for birth or creation). (2) Speciation may also be the divergence of a population into two distinct forms (species) that exist simultaneously. This branching process is known as **cladogenesis** (*clado* is Greek for branch; fig. 21.2). What do we mean by the term *species*?

Before Darwin's time, **typological thinking** prevailed, and a species was defined as a group of organisms that were morphologically similar. All variants were considered imperfections of the model or type. One of Darwin's greatest contributions to modern biological theory was to treat variation as a normal phenomenon in a group of organisms. The modern **biological species concept** groups together as members of the same species organisms that can potentially interbreed. A species, therefore, is a group of organisms that can mate among themselves to produce fertile offspring.

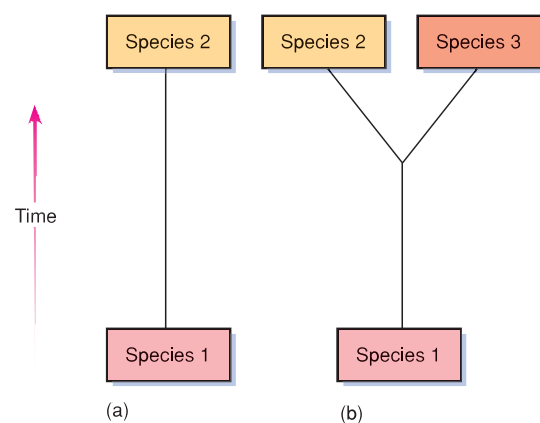


Figure 21.2 Forms of speciation. In anagenesis (a), a species changes over time until it is so different from its progenitor that it is classified as a new species. In cladogenesis (b), speciation takes place as a branching process wherein one species becomes two or more.

record, embryology, comparative anatomy, geographic distributions, etc.) is so overwhelming that the general nature of evolution is not in doubt. We can clearly trace its path, although we cannot make exact predictions for its future.

From a philosophical point of view, neo-Darwinism is the general paradigm (broad concept) defining “normal” biology. Every scientific endeavor works under the umbrella of a paradigm. When enough inconsistencies appear, a new paradigm is sought to replace the old in what Thomas Kuhn called a “scientific revolution.” In physics, relativity overthrew Newtonian principles. In biology, Darwinism overthrew the concept of a recent, biblically described origin of animals and plants. Darwinism became the paradigm because it explained many things in a consistent fashion that a recent origin of all forms of life could not. Neo-Darwinism will remain the current paradigm unless it is overthrown by a better theory that explains previous inconsistencies. To date, no major

inconsistencies suggest that neo-Darwinism is not correct.

In his article, Bethell went on to try to refute neo-Darwinism using the following argument: Survival of the fittest can be redefined to mean that some organisms have more offspring than others. Thus, natural selection cannot be a creative force because the only thing it works on is organisms alive now, some having more offspring than others. How, asks Bethell, can this possibly give us tigers and horses from ancestors that did not look like tigers and horses? The answer is that mutation produces variants in the population. The organism best able to compete will leave the most offspring. With an array of different genotypes in a population, natural selection determines which genotypes will increase in future generations. Traits that give the bearer an advantage increase in the population, and evolution takes place. Natural selection was the force behind the evolution from the small Eocene horse to the modern *Equus*.

Misinterpretation of mutation is the basis for other attacks on Darwinism. For example, Darwinian evolution has been attacked as not feasible, since most mutations are deleterious. How, the argument goes, can evolution proceed by a combination of deleterious events? The answer is that although most mutations are deleterious, some are not. This is especially true in changing environments; yesterday’s deleterious mutant may be today’s favored mutant.

The most recent attacks on Darwinism have been launched by creationists, who have attempted to pass laws in many states requiring schools to teach the biblical version of creation as an alternative to Darwinism. The courts have rejected this position because creationism is not a scientific theory. It does not follow the rules of the scientific method wherein empirical evidence can refute it.

Unfortunately, the definition of species on the basis of interbreeding cannot be used in many places, mostly due to the technical problems of applying it. Taxonomists and paleontologists, who often use nonliving specimens (preserved or fossilized), use the **morphological species concept** as a working definition. Under this concept, two organisms are classified as belonging to the same species if they are morphologically similar. They are classified as belonging to two different species if they are as different as two organisms belonging to two recognized species. Other problems arise for taxonomists since speciation is a dynamic process. For example, isolated subgroups of a population may be in various stages of becoming new species; the rate of successful interbreeding among individuals from these subgroups may range from 0 to 100%. How should the in-betweens be classified? There is no correct answer. It depends on the circumstances.

Still other problems make it necessary to turn to the morphological species concept. Haploid and asexual

species are hard to classify. Also, two organisms that will not interbreed in nature may do so in a laboratory setting. Thus, the interbreeding test carried out in the laboratory (as is done frequently) is not necessarily an adequate criterion for speciation. Other problems arise in classifying groups that are geographically isolated from each other, such as populations on islands. These individuals are physically isolated, but in many cases they can interbreed freely when brought together with their mainland counterparts. So, although there is a good theoretical definition of a species (potentially interbreeding individuals), more often than not it is necessary for biologists to apply the morphological species concept to determine whether two populations belong to the same species. In some cases, no decision can be made about the species status of a population. It is clear that a population has evolved, but it is not clear whether it has evolved enough to be called a new species. However, this is more of a problem for taxonomists and evolutionary biologists than for the organisms themselves.

Mechanisms of Cladogenesis

Reproductive Isolation

How does one species become two? Basically, **reproductive isolating mechanisms** must evolve to prevent two subpopulations from interbreeding when they come into contact. Reproductive isolating mechanisms are environmental, behavioral, mechanical, and physiological barriers that prevent individuals of two species from producing viable offspring. Following is a modification of the classification system of isolating mechanisms suggested by evolutionary biologist G. L. Stebbins:

1. *Prezygotic mechanisms* prevent fertilization and zygote formation.
 - a. Residential—The populations live in the same region, but occupy different habitats.
 - b. Seasonal or temporal—The populations exist in the same region, but are sexually mature at different times.
 - c. Ethological (in animals only)—The populations are isolated by incompatible premating behavior.
 - d. Mechanical—Cross-fertilization is prevented or restricted by incompatible differences in reproductive structures.
2. *Postzygotic mechanisms* affect the hybrid zygotes after fertilization has taken place.
 - a. F_1 hybrid breakdown— F_1 hybrids are inviable or weak.
 - b. Developmental hybrid sterility—Hybrids are sterile because gonads develop abnormally or because meiosis breaks down before it is completed.
 - c. Segregational hybrid sterility—Hybrids are sterile because of abnormal distribution to the gametes of whole chromosomes, chromosome segments, or combinations of genes.
 - d. F_2 breakdown— F_1 hybrids are normal, vigorous, and fertile, but the F_2 generation contains many weak or sterile individuals.

Allopatric, Parapatric, and Sympatric Speciation

Reproductive isolating mechanisms are barriers to **gene flow**, the spread of genes between populations. These isolating mechanisms can evolve in three different ways, each of which defines a different mechanism of speciation. Usually, the mode of speciation is dictated by both the properties of the genetic systems of the organisms and stochastic (random) or accidental events. For example, vertebrates tend to have different speciation modes than phytophagous (plant-feeding) insects.

The appearance of a geographic barrier, such as a river or mountain, through the range of a species physically isolates populations of the species. Physical isolation can also occur if migrants cross a particular barrier

and begin a new population (founder effect). The physically isolated populations can then evolve independently. If reproductive isolating mechanisms evolve, then two distinct species are formed, and if they come together in the future, they remain distinct species. Speciation that occurs because reproductive isolating mechanisms evolve during physical separation of the populations is called **allopatric speciation** (fig. 21.3). As evolutionary biologist Guy Bush pointed out, “Although examples in nature are difficult to substantiate . . . it [allopatric speciation] has been convincingly demonstrated in frogs . . . and lizards.”

Reproductive isolating mechanisms usually originate incidentally to the speciation process. That is, they arise incidentally during the process of evolution in isolated populations rather than being selected for. When isolated populations come together again, incomplete isolating mechanisms may allow hybrids to form. If the hybrids are normal and viable and can freely interbreed with individuals of each parent population, then no speciation has taken place. However, if the hybrids are at a disadvantage, natural selection may favor stronger isolating mechanisms. In this case, organisms that mate with individuals from the other population leave fewer offspring. The result is a more effective barrier to hybridization. Regions in which previously isolated populations come into contact and produce hybrids are called **hybrid zones**.

Until recently, evolutionary biologists believed that allopatric speciation was the general rule. Many now believe that two other modes of speciation may occur frequently in certain groups of organisms. **Parapatric speciation** occurs when a population of a species that occupies a large range enters a new niche or habitat (fig. 21.3). Although no physical barrier arises, the new niche acts as a barrier to gene flow between the population in the new niche and the rest of the species. Here again, reproductive isolating mechanisms evolve to produce two species where there was only one before. Parapatric speciation is believed to have occurred often in relatively nonviable animals such as snails, flightless grasshoppers, and annual plants. **Sympatric speciation** occurs when a polymorphism, which is the occurrence of alternative phenotypes in the same population, arises within an interbreeding population before a shift to a new niche. This mode of speciation may be common in parasites and phytophagous insects. For example, if a polymorphism arises within a parasitic species that allows an individual with a certain genotype to adapt to a new host, this genotype may be the forerunner of a new species. If the parasite not only feeds on the new host but also mates on the new host, a barrier to gene flow arises, although the parasite may be surrounded by other members of its species with the original genotype. Sympatric speciation can thus occur in the middle of a species range rather than at the edges (fig. 21.3).

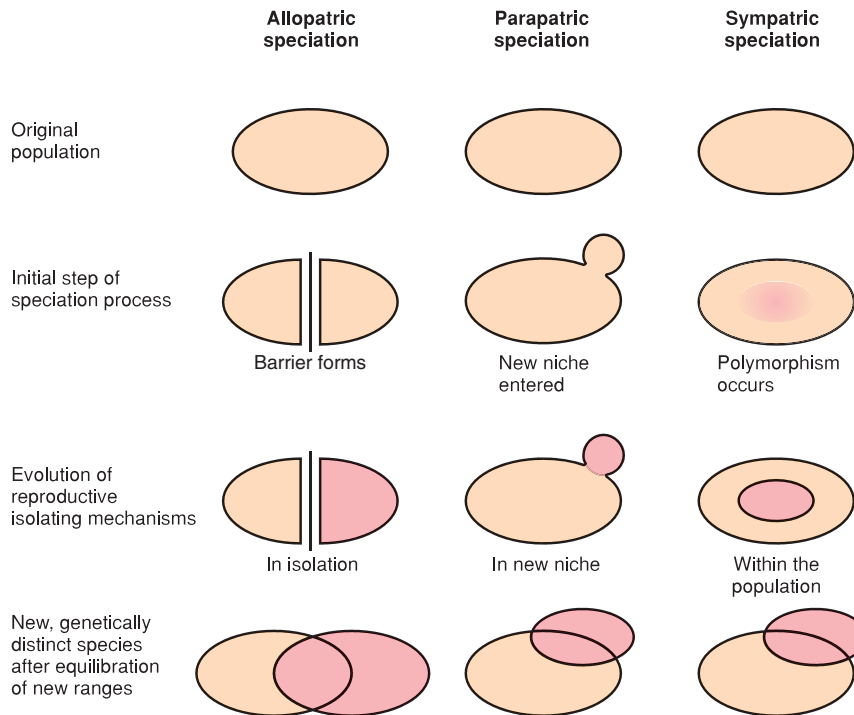


Figure 21.3 The three general mechanisms of speciation. In allopatric speciation, reproductive isolation evolves after the population has been geographically divided. In parapatric speciation, reproductive isolation evolves when a segment of the population enters a new niche. In sympatric speciation, reproductive isolation evolves while the incipient group is still in the vicinity of the parent population.

An example of incipient sympatric speciation has been seen recently in host races of the apple maggot fly (*Rhagoletis pomonella*) in North America (fig. 21.4). This fly was found originally only on hawthorn plants. However, in the nineteenth century, it spread as a pest to newly introduced apple trees. In fact, races are now known on pear and cherry trees and on rose bushes. These races have developed genetic, behavioral, and ecological differences from the original hawthorn-dwelling parent. Evolutionary biologists view this as an opportunity to observe sympatric speciation as it occurs.

Another form of sympatric speciation occurs when cytogenetic changes take place that result in “instantaneous speciation.” These cytogenetic changes include polyploidy and translocations. For example, if polyploid offspring cannot produce fertile hybrids with individuals from a parent population, then the polyploid is reproductively isolated. This mechanism is much more common in plants because they can exist vegetatively despite odd ploidy and they usually do not have chromosomal sex-determining mechanisms, which are especially vulnerable to ploidy problems (see chapter 8).

The end result of cladogenesis is the divergence of a homogeneous population into two or more species. One of the classic examples of cladogenesis appears in



Figure 21.4 The apple maggot fly, *Rhagoletis pomonella*. This species has exhibited host range expansion since the nineteenth century from hawthorn to apple, cherry, and roses. Host races are presumably the initial step in sympatric speciation. Magnification 10×. (Source: Jeffrey L. Feder and Guy L. Bush, Zoology Department, Michigan State University.)

the ground finches of the Galápagos Islands. These birds are very well studied not only because they present a striking case of speciation, but also because Darwin studied them and was strongly influenced by them in his views. Figure 21.5 is a map of the Galápagos Islands, and figure 21.6 is a diagram of the species of Darwin's finches.

An original flock of finches somehow reached the Galápagos Archipelago from South America, 700 miles away, and with time spread to the various islands of the Galápagos Archipelago. Given the limited ability of the birds to get from island to island, allopatric speciation took place. On each island, the finch population evolved reproductive isolating mechanisms while evolving to fill certain niches not already filled on the islands. For example, in South America, no finches have evolved to be like woodpeckers because many woodpecker species already live there. But the Galápagos Islands, being isolated from South America, have what is called a **depauperate fauna**, a fauna lacking many species found on the mainland. The islands lacked woodpeckers, and a very useful food resource for birds—insects beneath the bark of trees—was going unused. Finches that could

make use of this resource would be at an advantage and would thus be favored by natural selection. On one island, a finch did evolve to use this food resource. The woodpecker finch acts like a woodpecker by inserting cactus needles into holes in dead trees to extract insects. Darwin wrote: "Seeing this gradation and diversity of structure in one small, intimately related group of birds, one might really fancy that from an original paucity of birds in this archipelago, one species had been taken and modified for different ends."

Phyletic Gradualism Versus Punctuated Equilibrium

Darwin visualized cladogenesis as a gradual process, which we refer to as **phyletic gradualism**. However, an alternative view arose in 1972, when N. Eldredge and S. J. Gould suggested that speciation itself, and the morphological changes accompanying speciation, occur rapidly, separated by long periods of time when little change occurs (*stasis*). They called their model **punctuated equilibrium** (periods of stasis punctuated by rapid evolutionary change). Although figure 21.7 presents what appear to be two clear alternatives, in practice the models are very hard to tell apart. They both start with the same ancestral species and predict the same number of modern species. Allopatric, parapatric, and sympatric speciation mechanisms apply to both punctuated equilibrium and phyletic gradualism. The only major difference between the models is the rate of change, and this can only be discovered from an almost complete fossil record. The punctuated equilibrium model has brought much excitement to modern evolutionary biology. We await a time in the near future when we can decide which model has predominated in evolutionary history.

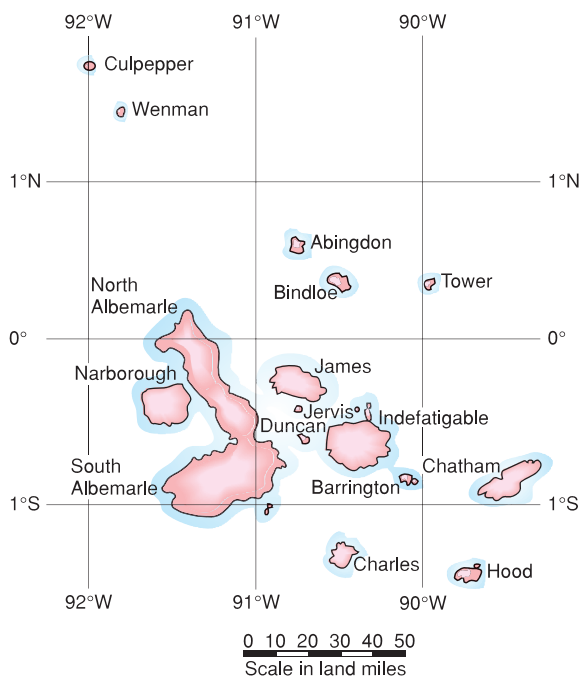
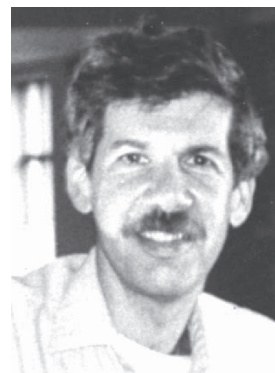


Figure 21.5 The Galápagos Archipelago is located about 700 miles west of Ecuador. This isolated chain of islands is a natural laboratory for the study of evolutionary processes. (From David Lack, *Darwin's Finches*. Copyright © 1947 by Cambridge University Press, New York, NY. Reprinted by permission.)



Stephen J. Gould (1943–).
(Courtesy of Dr. Stephen J. Gould
and the Harvard University News
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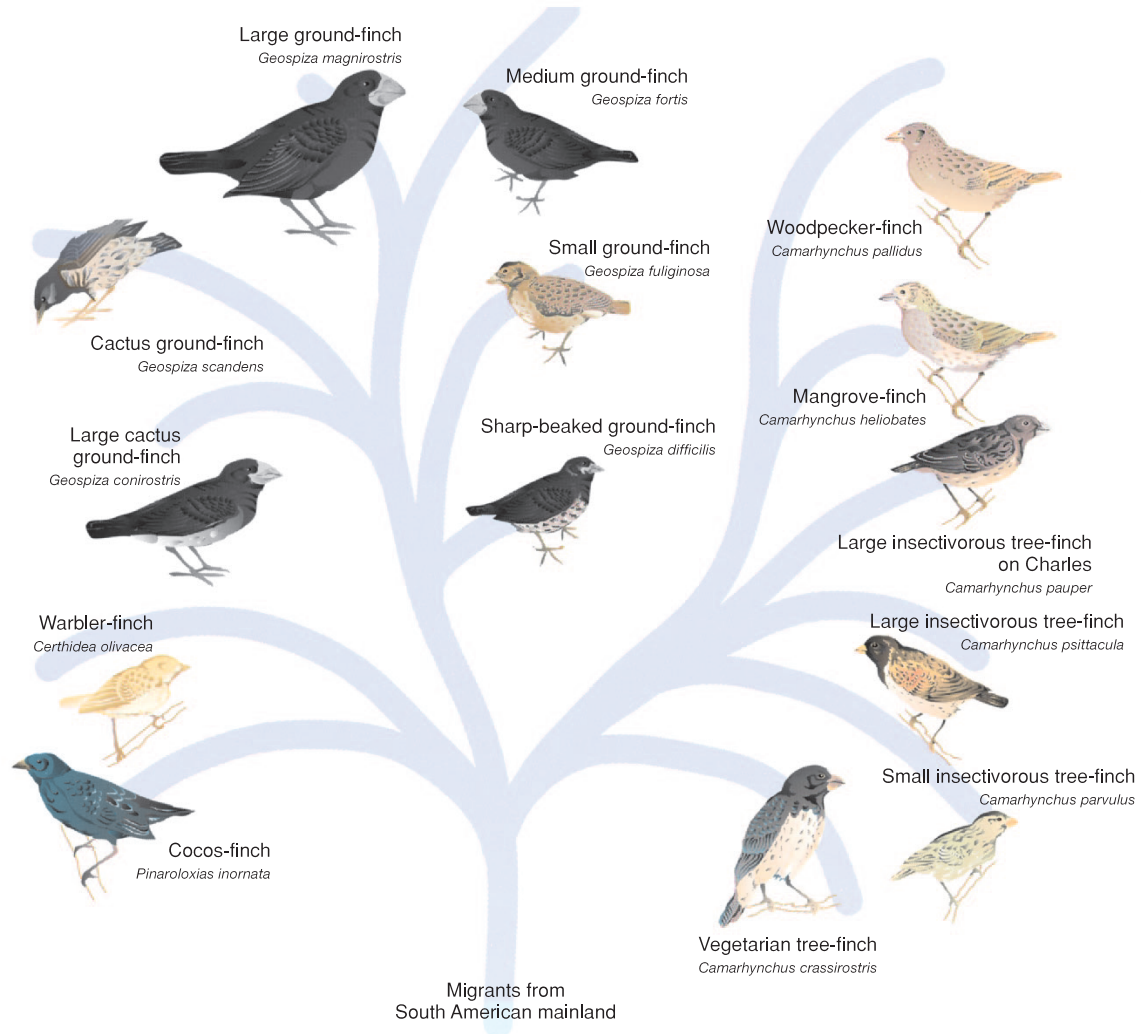


Figure 21.6 Species of Darwin's finches. These birds apparently evolved from a single group of migrants from the South American mainland. Isolated on the different islands, the birds evolved to fill many vacant niches.

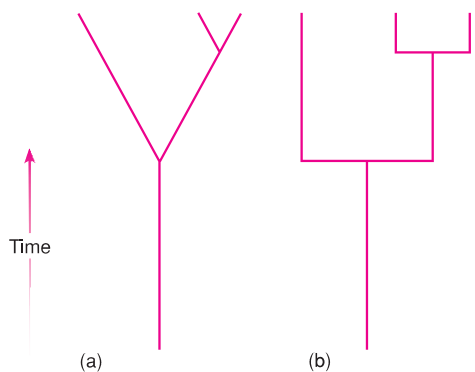


Figure 21.7 Diagrammatic interpretation of cladogenesis. (a) Phyletic gradualism is depicted as a gradual divergence over time. (b) Punctuated equilibrium is depicted as a rapid divergence of two groups after long periods of no change. The horizontal axis is some arbitrary measure of species differences.

GENETIC VARIATION

Darwinian evolution depends on the variation within a population. E. B. Ford, a British evolutionary biologist, applied the term **genetic polymorphism** to the occurrence of more than one allele at a given locus. Usually, we consider a locus polymorphic if a second allele occurs in the population at a frequency of 5% or more. Before the mid-1960s, the general belief was that only a few loci were polymorphic in any individual or any population.

In 1966, two researchers found a way to sample the genome in what they perceived was a random manner. R. C. Lewontin and J. L. Hubby used acrylamide-gel electrophoresis (see chapter 5) to investigate variability in a fruit fly species, *Drosophila pseudoobscura*. (H. Harris reported independent, similar work with human DNA.) Lewontin and Hubby reasoned that choosing enzymes and general proteins that are amenable to separation by electrophoresis, is, in fact, choosing a random sample of the genome of the fruit fly. If this is the case, then the degree of polymorphism found by electrophoretic sampling would provide an estimate of the amount of variability occurring in the individual organism and in the population. Their results were startling.

Lewontin and Hubby found that the species was polymorphic at 39% of eighteen loci examined, the average population was polymorphic at 30% of its loci, and the average individual was heterozygous at 12% of its loci. The high rate of polymorphism sparked two inter-related controversies. The first was whether electrophoresis does, in fact, randomly sample the genome. The second was whether most electrophoretic alleles are maintained in the population by natural selection. Let us return to the arguments after looking at ways in which genetic polymorphisms could be maintained in natural populations.



Edmund Brisco Ford
(1901–1988). (Courtesy of
Professor Edmund Brisco Ford.)



Richard C. Lewontin (1929–).
(Courtesy of Dr. Richard C. Lewontin.)

Maintaining Polymorphisms

Heterozygote Advantage

When selection acts against both homozygotes, an equilibrium is achieved, dependent solely on the selection coefficients, that maintains both alleles (see chapter 20). The classic example of heterozygote advantage in human beings is sickle-cell anemia. Sickle-cell hemoglobin (Hb^S) differs from normal hemoglobin (Hb^A) because it has a valine in place of a glutamic acid in position number 6 of the beta chain of the globin molecule. When the availability of oxygen is reduced, the erythrocytes containing sickle-cell hemoglobin change from round to sickle-shaped cells (see fig. 2.28). There are two unfortunate consequences: (1) sickle-shaped cells are rapidly broken down, which causes anemia as well as hypertrophy of the bone marrow, and (2) the sickle cells clump, which blocks capillaries and produces local losses of blood flow that result in tissue damage.

This condition of reduced fitness would lead one to predict that the sickle-cell allele would be selected against in all populations and, therefore, would be rare. But this is not the case. The sickle-cell allele is common in many parts of Africa, India, and southern Asia. What could possibly maintain this detrimental allele? In the search for an answer to this question, biologists discovered that the distribution of the sickle-cell allele coincided well with the distribution of malaria. The following facts have now been uncovered. The sickle-cell homozygote ($Hb^S Hb^S$) almost always dies of anemia. The sickle-cell heterozygote ($Hb^A Hb^S$) is only slightly anemic and has resistance to malaria. The normal homozygote ($Hb^A Hb^A$) is not anemic and has no resistance to malaria. Thus, in areas where malaria is common, the most fit genotype of the three appears to be the sickle-cell heterozygote, which has resistance to malaria and only a minor anemia.

This conclusion is supported by the changes in allelic frequencies that occur when a population from a malarial area moves to a nonmalarial area. Since the normal homozygote is no longer at risk for malaria, selection acts mainly on the sickle-cell homozygote and, to a slight extent, on the heterozygote. Table 21.1 shows data for African blacks versus African Americans. The African population is, of course, under malarial risk, whereas the American population is not. The sickle-cell hemoglobin allele (Hb^S) is reduced in frequency in African Americans.

Heterozygote advantage is an expensive mechanism for maintaining a polymorphism. Losses must occur in both homozygous groups in order for the polymorphism to exist. Thus, part of the reproductive output of a population is lost each generation to maintain each polymorphism under heterozygote advantage. In the case of sickle-cell anemia, this means a tragic loss of human life due to either anemia or malaria. (The loss of individuals

Table 21.1 Sickle-Cell Anemia Frequencies in African Blacks and African Americans

	Percentage of Homozygotes ($Hb^A Hb^A$)	Percentage of Heterozygotes ($Hb^A Hb^S$)	Frequency of Hb^S (q)
African Blacks (Midcentral Africa)	82	18	0.09
African Americans	92	8	0.04

to maintain genetic variation at a particular locus is called **genetic load**. In the sickle-cell case, it is due to the segregation of individuals with lowered fitness and is therefore called **segregational load**.) Very few other examples of heterozygote advantage have been documented.

Frequency-Dependent Selection

All the selection models discussed so far (chapter 20) have had selection coefficients that were constants. This is not always the case. For example, L. Ehrman has shown that when a female fruit fly has a choice between mates with different genotypes, the female fly chooses to mate with a male with a rare genotype. **Frequency-dependent selection** is selection in which the fitnesses of genotypes change according to their frequencies in the population.

The population geneticist Bruce Wallace has coined the terms *hard selection* and *soft selection* to deal with cases of frequency and density dependence. (Density-dependent selection exists when the fitness of a genotype changes as population density changes. We will not deal with that here.) Wallace defined soft selection as selection in which the selection coefficients depend on the frequency and density of genotypes. Hard selection is selection that is independent of both frequency and density. For example, the low fitness of sickle-cell anemia homozygotes involves hard selection because of the objectively deleterious effects of the anemia. Soft selection

could be envisioned as selection that might act on aggressive behavioral genotypes in some lemming and field mouse species. When population density and frequency of the genotypes are low, these animals survive and reproduce. As population density increases, there can be a selection for more aggressive genotypes because they may be more successful in obtaining resources. As density increases further and the frequencies of the aggressive genotypes increase, they may be selected against because of the preoccupation of these aggressive individuals with territory defense under crowded conditions. This has been suggested as a mechanism of wildlife's "lemming cycle," rapid declines in the density of lemming and field mouse populations every three to five years.

A model for frequency-dependent selection can be constructed by assigning fitnesses that are not constants. One way to do this is to assign fitnesses that are a function of allelic frequencies. Thus, the assigned fitnesses for one locus with two alleles could be $(1.5 - p)$, 1, and $(1.5 - q)$ for the AA , Aa , and aa genotypes, respectively (table 21.2). An interesting outcome of this model is that at $p = q = 0.5$, the system is in equilibrium, and no selection takes place because all the fitnesses are equal to 1.

Another way of looking at frequency-dependent selection is to look at the situation in which each genotype exploits a slightly different resource. As a genotype becomes rare, competition for the resource that genotype uses will likely decrease, and the genotype will thus have an advantage over the common genotypes, which are competing for resources. This type of selection is probably very common.

Transient Polymorphism

A genetic polymorphism can result when an allele is being eliminated either by random or selective mechanisms. If a population starts out homozygous for the a allele, for example, and a mutation brings in a more favored A allele, the population gradually becomes all A through directional selection. However, during the process of replacement, both alleles are present.

Other Systems

Selection at one stage in the life cycle of an organism can balance a different form of selection at another stage in the life cycle. For example, an allele can be favored in a



Lee Ehrman (1935–). (Courtesy of Dr. Lee Ehrman. Photo by Jan Robert Factor.)

Table 21.2 Selection Model of Frequency-Dependent Selection: The *A* Locus with the *A* and *a* Alleles

	Genotype			Total
	<i>AA</i>	<i>Aa</i>	<i>aa</i>	
Initial genotypic frequencies	p^2	$2pq$	q^2	1
Fitness (W)	$1.5 - p$	1	$1.5 - q$	
Ratio after selection	$p^2(1.5 - p)$	$2pq$	$q^2(1.5 - q)$	$\bar{W} = 0.5 + 2pq$
Genotypic frequencies after selection	$\frac{p^2(1.5 - p)}{\bar{W}}$	$\frac{2pq}{\bar{W}}$	$\frac{q^2(1.5 - q)}{\bar{W}}$	1

larva but selected against in an adult. There can also be a balance of selection in different parts of the habitat in a heterogeneous environment. For instance, an allele can be favored in a wet part of the habitat but selected against in a dry part.

Maintaining Many Polymorphisms

In summary, allelic polymorphisms in a population were classically accounted for by heterozygote advantage, frequency-dependent selection, or, infrequently, some other mechanism. Until Lewontin and Hubby did their work, heterozygote advantage was considered the most common method of maintaining a polymorphism at a given locus. The maintenance of an allele by heterozygote advantage costs the population a certain number of its offspring due to the mortality (or sterility) of the homozygotes. Most populations can afford the loss if polymorphisms are maintained at only a few loci. After Lewontin and Hubby reported that polymorphisms seemed to exist at a large proportion of loci, new explanations were needed to account for them. Three explanations were considered:

1. Electrophoresis (the technique used in Lewontin and Hubby's research) does not randomly sample the genome, and thus the large amount of variability they found does not really exist.
2. New population genetic models can be derived that explain how natural selection maintains this large amount of variability.
3. Electrophoretic alleles are not under selective pressure. That is, allozymic forms of an enzyme all perform the function of the enzyme equally well. This idea is called the **neutral gene hypothesis**.

Sampling the Genome

Does electrophoresis randomly sample the genome? Since, on the basis of DNA content, the genome of higher organisms has the potential to contain half a million genes, this question may be difficult to resolve. Since the

original reports of Lewontin and Hubby and Harris, numerous studies on many different organisms agree, for the most part, on the high amount of polymorphism in natural populations (table 21.3). However, several lines of evidence suggest that the results from electrophoresis are actually underestimates of the true amount of genetic variability present in a population.

The majority of amino acid substitutions, for example, do not change the charge of the protein. Thus, what appear to be single bands on an electrophoretic gel could actually be heterogeneous mixtures of the products of several alleles. Also, we now know that glycolytic enzymes are less polymorphic than other enzymes. Since glycolysis is a limited process in which most enzymes are not involved, it follows that the average heterozygosity over all loci should be slightly higher than the original estimates that included glycolytic enzymes. Recent technical advances of multidimensional electrophoresis and DNA sequencing support the hypothesis that electrophoresis does randomly sample the genome. However, DNA sequencing studies have shown that abundant variation exists, especially in the third (wobble) position of codons, and in parts of introns. Heterozygosity at the DNA sequence level seems to approach 100%.

Multilocus Selection Models

Can standard genetic models account for the high degree of variability in natural populations? If each locus is considered independently, then for each polymorphic locus, offspring in a population lost to maintain that polymorphism by heterozygote advantage are independent of offspring lost due to selection at other loci. The losses would soon outstrip the reproductive capacity of any species. Models proposed since Lewontin and Hubby's report have suggested that natural selection favors the individuals that are the most heterozygous overall. Individuals selected against because of their homozygosity would be individuals with many homozygous loci. In other words, natural selection acts on the entire genome, not on each locus separately. We can show algebraically that the large

Table 21.3 Survey of Genic Heterozygosity

Species	Number of Populations	Number of Loci	Proportion of Loci Polymorphic per Population	Heterozygosity per Locus	Standard Error of Heterozygosity
<i>Homo sapiens</i>	1	71	0.28	0.067	0.018
<i>Mus musculus musculus</i>	4	41	0.29	0.091	0.023
<i>M. m. brevisrostris</i>	1	40	0.30	0.110	—
<i>M. m. domesticus</i>	2	41	0.20	0.056	0.022
<i>Peromyscus polionotus</i>	7 (regions)	32	0.23	0.057	0.014
<i>Drosophila pseudoobscura</i>	10	24	0.43	0.128	0.041
<i>D. persimilis</i>	1	24	0.25	0.106	0.040
<i>D. obscura</i>	3 (regions)	30	0.53	0.108	0.030
<i>D. subobscura</i>	6	31	0.47	0.076	0.024
<i>D. willistoni</i>	2–21	28	0.86	0.184	0.032
	10	20	0.81	0.175	0.039
<i>D. melanogaster</i>	1	19	0.42	0.119	0.037
<i>D. simulans</i>	1	18	0.61	0.160	0.052
<i>Limulus polyphemus</i>	4	25	0.25	0.061	0.024

Source: *The Genetic Basis of Evolutionary Change* by R. C. Lewontin, (New York: Columbia University Press, 1974). Reprinted with permission of the publisher.

Note: See source (Lewontin, 1974) for individual references.

number of polymorphisms that exist in natural populations could be maintained according to these models.

Neutral Alleles

The high incidence of polymorphism that electrophoresis reveals may not be important from an evolutionary point of view. If all or most electrophoretic alleles are neutral (i.e., if no allele is more fit than its alternative) or only very slightly deleterious, there is virtually no selection at these loci, and the variation observed in the population is merely a chance accumulation of a combination of mutation and genetic drift. This model, proposed by M. Kimura of Japan, is an alternative to the natural selection model.



Motoo Kimura
(1924–1994). (Courtesy of
Dr. Motoo Kimura.)

Which Hypothesis Is Correct?

Researchers who favor the concept that most electrophoretic alleles are neutral do not deny that selection exists. They do not hold that evolution is non-adaptive, but say merely that most of the molecular variation (electrophoretic) found in nature is not related to fitness—it is neutral. Thus, the demonstration that selection actually exists, in electrophoretic systems or otherwise, is not proof against the neutralist view. No one denies the explanation for the maintenance of sickle-cell anemia. Selection at several other electrophoretic systems is also known.

For example, R. Koehn showed that different alleles of an esterase locus in a freshwater fish in Colorado produced proteins with different enzyme activities at different water temperatures. Koehn then showed that the alleles were distributed as one would predict on the basis of the water temperature. In other words, the distribution of alleles correlated with the distribution of water temperature. The enzyme produced by the *ES-1^a* allele functioned best at warm temperatures, whereas the enzyme produced by the *ES-1^b* allele functioned best at cold temperatures. The cold-adapted enzyme was prevalent in the fish in colder waters (higher latitudes), and the warm-adapted enzyme was prevalent in the fish in warmer waters (lower latitudes; fig. 21.8).

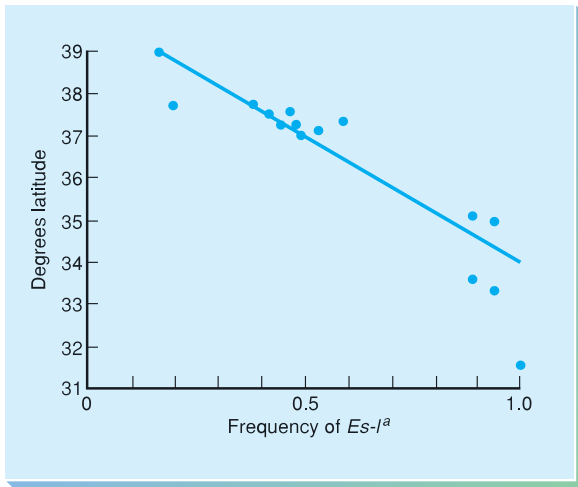


Figure 21.8 Relation of latitude and frequency of the warm-adapted esterase allele *Es-1^a* in populations of the fish *Catostomus clarki*. Note how the frequency of the allele increases as latitude decreases (warmer water). (From Richard Koehn, "Functional and evolutionary dynamics of polymorphic esterases in catostomid fishes," *Transactions of the American Fisheries Society*, 99:223. Copyright © 1970 American Fisheries Society, Bethesda, MD.)

Isolated instances of selection, however, do not adequately prove the case for maintaining variation by means of natural selection or disprove the case for maintaining variation of neutral alleles. Both theories recognize natural selection as the guiding force in producing adapted organisms. What is needed is proof that the majority of poly-

morphic loci are either being selected or are neutral. For this proof, many loci must be examined independently—a very difficult undertaking—or some grand pattern must emerge supporting one hypothesis or the other.

Grand Patterns of Variation

Clinal Selection

Data on the geographic distribution of alleles fail to adequately support either theory. Often, a single allele predominates over the range of a species (fig. 21.9). Changes in allelic frequency from one geographic area to another can often be attributed to **clinal selection**, selection along a geographic gradient, in which allelic frequencies change as altitude, latitude, or some other geographic attribute changes. Note in figure 21.9 the general increase in the *Es-5^b* frequency from west to east in the southern United States. But, in line with the neutralist view, geographic patterns similar to those in figure 21.9 can also be produced by neutral alleles with a very low level of migration, as little as one individual per one thousand per generation.

Molecular Evolutionary Clock

The advancing technology that made it possible to detect the sequence of amino acids in a protein also made it possible to discover how much the proteins and DNA of various species differ. In chapter 17, we discussed the use of mitochondrial DNA (mtDNA) to determine evolutionary relationships. Currently, protein, nuclear DNA, and mtDNA clocks are being studied.

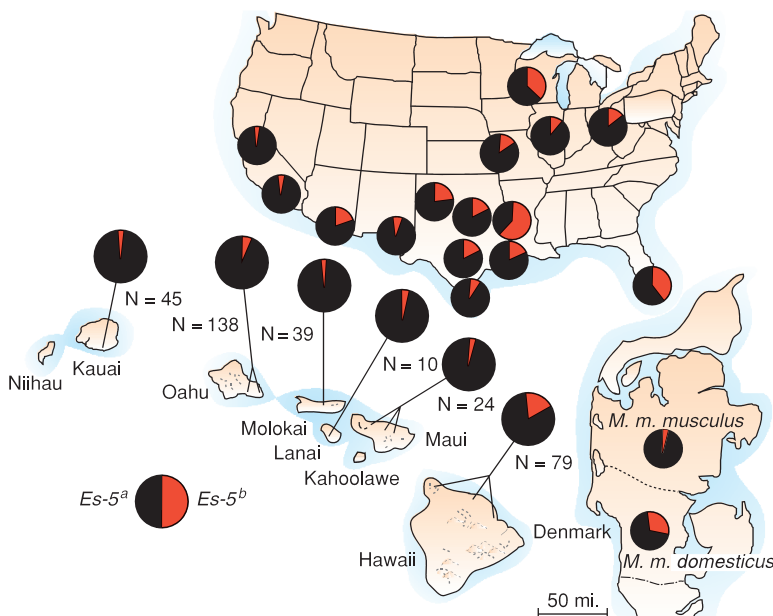


Figure 21.9 Frequency distribution of the *Es-5* alleles of an esterase locus in house mice. Each circle represents allelic frequencies at that geographic location. Note the general tendency for the *Es-5^b* allele to increase from west to east in the continental United States. (From L. Wheeler and R. Selander, "Genetic Variation in Populations of the House Mouse, *Mus musculus*, in the Hawaiian Islands," *Studies in Genetics*, VII, 1972. University of Texas Publication 7213. Reprinted with permission of M. R. Wheeler.)

Knowledge of the changes in amino acid sequences can be used to estimate the rate of evolutionary change. That is, the data show how many amino acid substitutions have occurred between two known groups of organisms. The genetic code dictionary allows us to estimate the minimum number of nucleotide substitutions required for this change. For example, if one protein contains a phenylalanine in position 7 (codons UUU, UUC), and the same protein in a different species has an isoleucine in the same position (AUU, AUC, AUA), we can see that the minimum number of substitutions to convert a phenylalanine codon to an isoleucine codon is one (UUU → AUU). When we know the minimum number of substitutions, we can calculate molecular **evolutionary rates**, nucleotide substitutions per million years. In a sense, these rates provide us with a **molecular evolutionary clock** that measures evolutionary time in nucleotide substitutions.

Many studies of the rate of amino acid and nucleotide substitutions have been done on hemoglobin, on cytochrome *c*, on a class of proteins involved in blood clotting called fibrinopeptides, and on many others. Figure 21.10 shows the way in which an amino acid sequence differs among species. From comparisons of this type, we can calculate the actual number of amino acid differences, as well as percentage differences. Table 21.4 is a compilation of percentage differences between various species based on the cytochrome *c* protein. This type of information can be used two ways.

First, we can construct a **phylogenetic tree** that tells us the evolutionary history of the species under consideration (fig. 21.11). This tree can be compared with phylogenetic trees constructed by more classical means using fossil evidence and evidence from morphology, physiology, and development. From the comparisons, we can look at areas of disagreement in an attempt to find out the best way to create phylogenetic trees. In addition, molecular phylogenies can give us information unattainable in any other way, as, for example, when the fossil record is incomplete or ambiguous.

A second use of DNA or amino acid difference data is to determine average rates of substitution. Once we

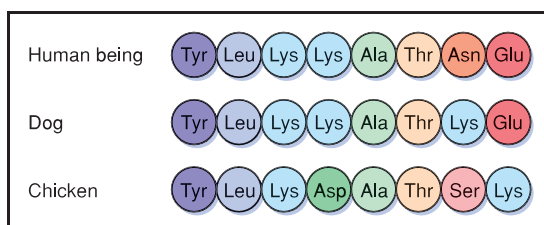


Figure 21.10 The amino acids making up the terminal portion of cytochrome *c* in three species. Note the similarities and differences.

know the current amino acid differences in the proteins of two species, it is possible to estimate the actual number of nucleotide substitutions that have taken place over evolutionary time using the statistical Poisson distribution, which deals with rare events. The index, *K*, is the average number of amino acid substitutions, per site, between two proteins:

$$K = -\ln(1 - p)$$

in which $\ln$ is the natural logarithm (to the base *e*), and $p = d/n$ in which *d* is the number of amino acid differences and *n* is the total number of amino acid sites being compared. For example, in figure 21.10, *n* = 8 and *d* = 3 between the dog and chicken. Thus

$$K = -\ln(1 - 0.375) = 0.47$$

Therefore, the average number of amino acid substitutions, per site, between dog and chicken is 0.47.

We can take this calculation one step further by determining the per-year rate:

$$k = K/2T$$

in which *k* is the amino acid substitution rate per site per year, and *T* is the number of years since the two species diverged from a common ancestor. We divide by *2T* because each side of the tree has evolved independently for *T* years. When *k*'s are calculated for many proteins over many species, they cluster around 10^{-9} (table 21.5). In fact, Kimura has suggested the unit of a *pauling* to be equal to 10^{-9} amino acid substitutions per year per site in

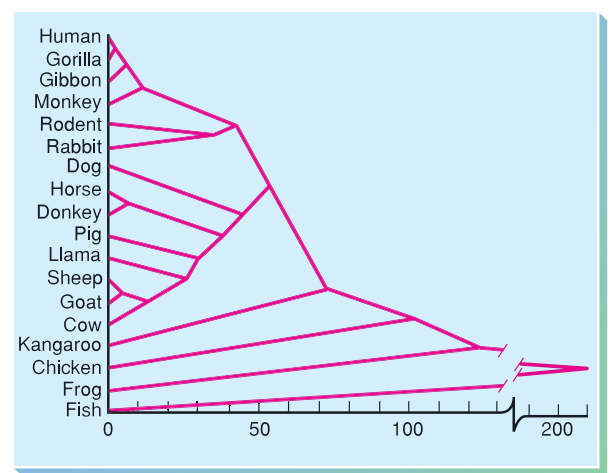


Figure 21.11 Composite evolution of hemoglobin, cytochrome *c*, and fibrinopeptide A. The total number of nucleotide substitutions appears on the horizontal axis. Note how the tree groups similar organisms and generally agrees with classical systematics. (From C. H. Langley and W. M. Fitch, "An examination of the constancy of the rate of molecular evolution," *Journal of Molecular Evolution*, 3:168. Copyright © 1974 Springer-Verlag, Heidelberg. Reprinted by permission.)

Table 21.4 Amino Acid Differences (By Percentage) in Cytochrome *c* Between Different Organisms

	Human being	Pig	Horse	Chicken	Turtle	Bullfrog	Tuna	Carp	Lamprey	Fruit fly	Screwworm	Silkworm	Sesame	Sunflower	Wheat	<i>C. krusei</i>	Yeast	<i>N. crassa</i>	<i>R. rubrum</i>
Human being	0	10	12	13	14	17	20	17	19	27	25	29	35	38	38	46	41	44	65
Pig, bovine, sheep		0	3	9	9	11	16	11	13	22	20	25	38	40	40	45	41	43	64
Horse			0	11	11	13	18	13	15	22	20	27	39	41	41	46	42	43	64
Chicken, turkey				0	8	11	16	14	17	23	21	26	40	41	41	45	41	44	64
Snapping turtle					0	10	17	13	18	22	22	26	38	39	41	47	44	45	64
Bullfrog						0	14	13	20	20	20	27	41	42	43	46	43	45	65
Tuna							0	8	18	23	22	30	42	43	44	43	43	45	65
Carp								0	12	21	20	25	40	41	42	45	42	43	64
Lamprey									0	27	26	30	44	44	46	50	45	47	66
Fruit fly										0	2	14	42	41	42	43	42	38	65
Screwworm fly											0	13	41	40	40	43	42	38	64
Silkworm moth												0	39	40	40	43	44	44	65
Sesame													0	10	13	47	44	48	65
Sunflower														0	13	47	43	49	67
Wheat															0	45	42	48	66
<i>Candida krusei</i>																0	25	39	72
Baker's yeast																	0	38	69
<i>Neurospora crassa</i>																		0	69
<i>Rhodospirillum rubrum</i>																			0

Source: From M. O. Dayhoff, ed., Atlas of Protein Sequence and Structure," National Biomedical Research Foundation, Washington, D.C., 1972. Reprinted with permission.

honor of Linus Pauling, who, along with E. Zuckerkandl, first proposed the concept of a molecular clock in 1963. If the values of k (such as those in table 21.5) form a normal distribution around 10^{-9} , then 10^{-9} would be the rate of "the" molecular evolutionary clock. So far, the data have been too limited to determine the distribution.

Although controversy still exists, the neutralists have interpreted the relative constancy of the molecular evolutionary clock as strong evidence in support of the neutral gene hypothesis. A constant rate of molecular evolution over many groups of organisms over many different time intervals implies that the substitution rate is a stochastic or random process rather than a directed or selectional process. This is not to say that no adapted changes occur in proteins or that there are no constraints. In fact, the evidence suggests that three classes of amino acids can be grouped in terms of substitution rate: invariant, moderately variant, and hypervariant. It seems possible that virtually no substitutions of amino acids will occur in and around the active site of the enzyme since any amino acid change in that area might be deleterious or lethal. For ex-

ample, a segment of cytochrome *c* that runs from amino acids 70 to 80 is invariant in all organisms tested. This area includes a binding site of the protein.

DNA Variation

If the neutralist view of molecular evolution is correct, we should be able to make some predictions about rates of change in DNA. For example, we predict that DNA under greater constraint should amass fewer base changes than DNA under lesser constraint. We could test this by looking at the accumulation of mutations in the three positions of the codon, or we could look at DNA that is not directly translated, such as pseudogenes (see chapter 15) or introns, which are probably under lesser constraint. Let us first look at the three positions of the codon.

A reexamination of the codon dictionary (see table 11.4) shows that the third, or wobble, position of the codon should be under less constraint. Eight amino acids belong to unmixed families; their amino acids are defined by the first and second positions coupled with any of the

Table 21.5 Evolutionary Rates (k) as 10^{-9}
Substitutions Per Amino Acid Site Per
Year for Various Proteins

Protein	k
Fibrinopeptide	8.3
Pancreatic ribonuclease	2.1
Lysozyme	2.0
Hemoglobin alpha	1.2
Myoglobin	0.89
Insulin	0.44
Cytochrome <i>c</i>	0.3
Histone H4	0.01

Source: From M. Kimura, *The Neutral Theory of Molecular Evolution*, Cambridge University Press, 1983. Reprinted with the permission of Cambridge University Press.

four bases in the third position of the codon. The remaining amino acids belong to mixed families; the first two positions and the purine or pyrimidine nature of the third position in their codons is important. Hence, the wobble (third) position of the codon is under the least constraint and should build up the most neutral or near-neutral mutations.

In addition, analysis of changes in the first and second positions indicates that more drastic change takes place by mutation of the second rather than the first position of the codon. Thus, we predict that evolutionary distance, as measured by base substitutions, should be greatest for the third codon position and least for the second position. This turns out to be generally true (table 21.6).

It should be clear that a major problem facing those who study evolutionary clocks is how to calibrate them. Are average changes uniform throughout lineages? Do clocks speed up, slow down, or show other unpredictable changes through time? There is evidence, for example, that both the nuclear and mitochondrial DNA clocks have slowed down in the hominid lineage as com-

Table 21.6 Evolutionary Distance of Codons,
Measured in Base Substitutions Per
Nucleotide Site

	Codon Site		
	2	1	3
Beta globin, human being vs. mouse	0.13	0.17	0.34
Beta globin, chicken vs. rabbit	0.19	0.30	0.64
Rabbit, alpha vs. beta globin	0.44	0.54	0.90

Source: From M. Kimura, *The Neutral Theory of Molecular Evolution*, Cambridge University Press, 1983. Reprinted with the permission of Cambridge University Press.

pared with old world monkeys. If the clocks change speed in different lineages, at different times, and for different parts of the genome, there will be errors in interpreting lineages and errors in using averages to understand the general patterns of change.

At this point, it is probably safe to say that while natural selection acts to create organisms that are adapted to their environments (see box 21.2 on mimicry), many nucleotide and amino acid changes may not have measurable effects on the fitness of the organism, and hence their frequencies may be determined by the stochastic processes of mutation and genetic drift. Adaptation is by natural selection, but neutral variation most certainly also occurs in organisms.

SOCIOBIOLOGY

We close this chapter by looking at a level of evolution only recently addressed. In 1975, E. O. Wilson published a mammoth tome entitled *Sociobiology: The New Synthesis*. This book has been at the center of major controversies that have spread to the fields of sociology, psychology, anthropology, ethology, and political science. The basic premise of the book is that social behavior is under genetic control. Although Wilson's book contains twenty-six chapters concerned with the animal kingdom, controversies have arisen because of the one chapter that applies the theory to human beings.

Altruism

V. C. Wynne-Edwards published a book in 1962 entitled *Animal Dispersion in Relation to Social Behavior*. In it, he suggested that animals regulate their own population density through altruistic behavior. For example, under crowded conditions, many birds cease reproducing. The interpretation of this phenomenon was that these birds were being altruistic: Their failure to breed was for the ultimate good of the species. (**Altruism** means risking loss of fitness in an act that could improve the fitness of another individual.) Wynne-Edwards suggested a mechanism called **group**



Edward O. Wilson (1929–).
(Courtesy of Dr. Edward O. Wilson.
Photo by Pat Hill/OMNI Publications
International, Ltd.)

BOX 21.2

Mimicry is a phenomenon whereby an individual of one species gains an advantage by resembling an individual of a different species. There are at least two types of mimicry.

In **Müllerian mimicry**, named after F. Müller, several groups of organisms gain an advantage by looking like one another. This mimicry occurs among organisms in which all the mimetic species are offensive and obnoxious. The classic example is the general similarity among bees, wasps, and hornets.

In **Batesian mimicry**, named after H. W. Bates, a vulnerable organism (mimic) gains a selective advantage by looking like a dangerous or distasteful organism (model). The classic example of Batesian mimicry was, until 1991, the monarch (*Danaus plexippus*) and viceroy (*Limenitis archippus*) butterflies (fig. 1). Although the viceroy is smaller and, on close examination, looks different from the monarch, the resemblance is striking at first glance. Monarch butterflies feed on milkweed plants, obtaining noxious chemicals called cardiac glycosides, which the monarchs store in their bodies. When a bird tries to eat a monarch, it becomes sick and regurgitates what it has eaten. Thereafter, the bird will not only avoid eating monarchs, but it will also avoid eating any butterflies that look anything like monarchs. Previously it was believed that the mimetic viceroy butterfly gained a selective advantage by looking like the monarch and fooling bird predators into thinking that the viceroy was bad to eat. However, D.

Experimental Methods

Mimicry

Ritland and L. Brower demonstrated a previously unrealized fact: The viceroys taste as bad as the monarchs to birds. This fact changes the mimicry of these two species from Batesian to Müllerian mimicry.

Examples of Batesian mimicry do occur in numerous butterfly species. For example, in West Africa, *Pseudacraea* species mimic species of the genus *Bematistes* (fig. 2). These species are primarily black and white or black and orange, and in some the sexes differ, each having a different mimic. Upwards of twenty species can be involved in these mimicry complexes in one area. Both forms of mimicry depend on the selective pressure generated by predation. Certain requirements must be met for each system to work properly. Batesian mimicry has the following requirements:

1. The model species must be conspicuous and inedible or dangerous.
2. Both model and mimic species must occur in the same area, with the model being very abundant. If the model is rare, predators do not have sufficient opportunity to learn that its pattern is associated with a bad taste. In fact, the reverse can happen; the model can

be at a selective disadvantage if it is rare because the predators will learn from the mimic that the pattern is associated with something good to eat.

3. The mimic should be very similar to the model in the morphological characteristics predators perceive but not necessarily similar in other traits. The mimic is not evolving to *be* the model, only to look like it.

Müllerian mimicry requires that all the species be similar in appearance and distinctly colored. They can, however, be equally numerous. And, as the British geneticist P. M. Shepard pointed out, the resemblance among Müllerian mimics need not be as good as between the mimic and model of a Batesian pair because Müllerian mimics are not trying to deceive a predator, only to remind the predator of the relationship.

Although there have been some critics of mimicry theory, especially critics of the way in which the system could evolve, the general model put forth by population geneticist and mathematician R. A. Fisher is generally accepted. According to Fisher, any new mutation that gave a mimic any slight advantage would be selected for. As time proceeded, other loci that might favorably modify the expression of mimetic genes would also be selected for in order to increase the similarity of mimic and model. This mechanism surmounts the criticism that a single mutation could not produce a mimic that so closely resembled its model.

selection: groups that had altruistic behavior would have a survival advantage over groups that did not.

In 1966, G. Williams, in his book *Adaptation and Natural Selection: A Critique of Some Current Evolutionary Thought*, refuted the altruistic view with the charge that individuals that performed altruistic acts would be selected against. In other words, organisms not performing altruistic acts would have a higher degree of fitness.

Williams held that apparent altruism had to be interpreted on the basis of benefits accruing to the individual performing the altruistic act. After his book, the idea of doing something for the good of the species became passé. How, then, can apparent altruism be accounted for? How can we explain why ground squirrels appear to put themselves at risk to predators by giving alarm calls, and why female workers in ant, wasp, and bee colonies



(a)



(b)

Figure 1 Müllerian mimicry. (a) Monarch butterfly and (b) viceroy butterfly. Both have similar colors (orange and black) and a generally similar color pattern. ([a] © Robert Finke/Photo Researchers, Inc. [b] © Richard Parker/Photo Researchers, Inc.)



Figure 2 Batesian mimicry seen in West African butterflies that live in the same places. Those on the left are species belonging to the genus *Bematistes*. Those on the right that mimic them are different species belonging to the genus *Pseudacraea*. (© J. A. L. Cooke/Oxford Scientific Films/Animals, Animals.)

forsake reproduction in order to work for the colony? **Sociobiology**, the study of the evolution of social behavior, attempts to answer these questions.

Kin Selection and Inclusive Fitness

In 1964, W. D. Hamilton developed concepts that explained altruistic acts without resorting to group

selection. Starting with the known fact that relatives have alleles in common, Hamilton suggested that natural selection would favor an allele that promoted altruistic behavior toward relatives because the result might be an increase in copies of that allele in the next generation. The proportion of alleles shared by two individuals can be defined as a **coefficient of relationship, r** . If an individual has a certain allele, the probability that a particular

relative also has that allele is r . Siblings have an $r = 1/2$. A squirrel is likely to have virtually all its alleles still viable if it sacrifices itself for two or more siblings. In fact, natural selection should definitely favor altruism of an individual toward three siblings because, in a sense, natural selection is weighing 1 copy of an individual's alleles (the individual itself) versus 1.5 copies (three siblings).

This sort of reasoning has been termed the **calculus of the genes**. It does not imply that individuals actually think these things out; rather, natural selection has favored the individuals that behave this way. Hamilton referred to the sum of an individual's fitness plus the fitness effects of alleles that relatives share as **inclusive fitness**. He referred to the way natural selection acts on inclusive fitness as **kin selection**.

Hamilton applied his ideas of inclusive fitness and kin selection to explain sterile castes in the eusocial (truly social) hymenoptera (bees, ants, and wasps). The workers in these colonies are sterile females. Why do they forsake their ability to reproduce in order to help maintain the hive or colony? The answer seems to come from **haplodiploidy**, the unusual sex-determining mechanism of these species. In the eusocial hymenoptera with sterile castes, fertilized eggs produce diploid females, whereas unfertilized eggs produce haploid males (drones). The difference between a reproductive queen and a sterile worker in bees is larval nutrition: larvae fed "royal jelly" can become queens. Hamilton showed that since a worker is more closely related to her sisters than to her own potential offspring, kin selection could favor a worker who helps her sisters at the expense of her own reproduction.

Figure 21.12 shows a queen (female) with alleles A_1 and A_2 at the A locus and a haploid drone (male) with the A_3 allele. A daughter will have either the A_1A_3 or A_2A_3 genotype. If we compare one of these daughters with her sisters, we see that the average $r = 0.75$ —half of the time, $r = 1.0$, and the other half of the time, $r = 0.5$. A queen and her daughters have an $r = 0.5$. Thus, we see that workers (females) are more closely related to their sisters, and hence are at a reproductive advantage by raising them rather than their own young. Wilson has pointed out that sterile caste systems have evolved among insects in only one other group beside the eusocial hymenoptera, the termites. Although eusocial hymenoptera make up only 6% of insects, sterile castes have independently evolved at least eleven times. This is compelling evidence for the validity of Hamilton's analysis. Only one noninsect example of a caste has been discovered: the naked mole rat, a small subterranean rodent living in Africa, has this type of social system.

Many studies concerned with apparently altruistic acts have provided a large body of support for Hamilton's theory of kin selection and inclusive fitness. P. Sherman, working with ground squirrels, for example, has observed that the individuals that make the alarm calls have the most to gain from the standpoint of inclusive fitness; these individuals are resident females surrounded by kin.

One other explanation for altruism is also consistent with benefits to individual fitness. It is that many apparently altruistic acts are in reality selfish—they just look altruistic. To be altruistic, an individual must risk reducing its fitness to potentially benefit the fitness of others. We may, in fact, misinterpret some acts as altruistic that simply are not.

This turnaround in thought, from group selection to individual selection, has been an intellectual revolution in modern evolutionary biology. Before this revolution, many of the behaviors in nature that involved apparent altruism were difficult to explain. Now sociobiological reasoning provides an explanation.

The reason so much controversy has sprung up over the theory of genetic control of social behavior is because of the implications the theory has for human social, political, and legal issues. Human husband-wife, parent-child, and child-child conflicts, for example, may be built into the genes. Altruism, our highest form of nobility, may be mere selfishness. Many critics fear that sociobiological concepts can be used to support sexism and racism. For human beings, the alternative to the theory of sociobiology is the theory that most human behavior, including cultural learning, is determined by the environment. At present, although much evidence remains to be gathered, the sociobiology concept is compelling to many evolutionists.

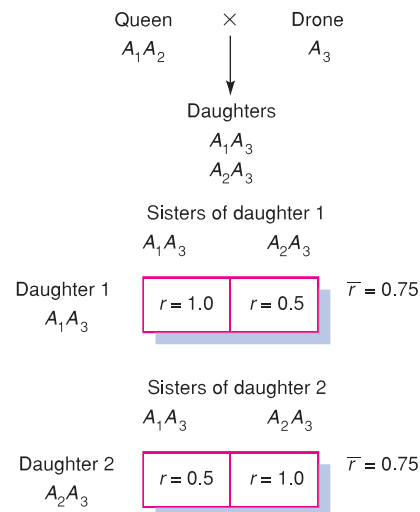


Figure 21.12 Haplodiploidy in eusocial hymenoptera produces sisters with an average r of 0.75. Because drones (males) are haploid, queens produce daughters of only two genotypes at any locus. A given daughter has an r of 1.0 with sisters of identical genotype and an r of 0.5 with sisters of the other genotype, for an average r of 0.75. In other words, females have a 75% genetic similarity with their sisters but only a 50% similarity with their own offspring.

S U M M A R Y

STUDY OBJECTIVE 1: To analyze the mechanisms of evolution and speciation 589–595

The theory of evolution by natural selection was put forward by Charles Darwin, who recognized the natural variation among individuals within a population of similar organisms. He noted also that offspring are overproduced in nature, and this overproduction inevitably leads to competition for scarce resources. Darwin assumed that, when competition occurs, the most fit will survive; through time, then, a population will become better adapted to its environment through the process of natural selection. Applying the algebra of population genetics to this theory leads to the modern concept of evolution, neo-Darwinism.

Cladogenic speciation occurs when reproductive isolating mechanisms arise, usually after gene flow in a population is blocked. Different populations of a species can then evolve independently. When individuals from the isolates can no longer interbreed, speciation has taken place. If the isolates then come in contact again, they will remain as separate species. Speciation may occur gradually or in a punctuated manner; it can be by allopatric, parapatric, or sympatric mechanisms.

STUDY OBJECTIVE 2: To investigate the mechanisms of the maintenance of genetic variation in natural populations, both selective and neutral 596–603

Evolution depends on variation. In 1966, Lewontin and Hubby, using electrophoresis, showed that a tremendous

amount of heterozygosity occurred in natural populations. Attempts to explain this variation have led to two major competing theories: (1) variation is maintained selectively and (2) variation is not under selective pressure, but is instead neutral. Two areas of evidence support the neutralist view.

First, the molecular evolutionary clock (the per-year, per-amino acid, substitution rate) appears to be fairly constant at 10^{-9} . This constancy implies that the majority of amino acid changes are the result of stochastic processes. Second, there have been greater numbers of nucleotide substitutions in DNA under lesser constraint than in DNA under greater constraint. For example, the third, or wobble, position of the codon has accumulated more mutations than the other two positions. We conclude that natural selection creates adapted organisms, but the majority of base and amino acid changes may be neutral.

STUDY OBJECTIVE 3: To discuss sociobiology, the evolution of social behavior 603–606

Sociobiology is another term for evolutionary behavioral ecology. It attempts to provide evolutionary explanations for social behaviors. Apparent altruistic behavior can be explained either as kin selection or as selfishness. Sterile insect castes have come about because of the unusual haplodiploid sex-determining mechanism in the eusocial hymenoptera. There is much controversy about and little information for applying sociobiological principles to human behavior.

S O L V E D P R O B L E M S

PROBLEM 1: What are the roles of reproductive isolating mechanisms in the process of evolution?

Answer: Reproductive isolating mechanisms prevent individuals in two populations from mating with each other or producing viable offspring. These mechanisms can be prezygotic or postzygotic. They usually evolve while populations are isolated from each other, either physically or during parapatric or sympatric speciation. For example, if a species is split by a new river, the populations on either side of the river can evolve in isolation from each other. Reproductive isolating mechanisms usually evolve irrespective of the other facets of evolution taking place. Thus, if, after time, the two populations come into contact (the river dries up), reproductive isolating mechanisms may have evolved to prevent mating.

If weak reproductive isolating mechanisms have evolved, natural selection usually favors strengthening them by selecting against hybrids and against any mating behavior that leads to the formation of hybrids.

PROBLEM 2: What is our modern evolutionary concept of altruism?

Answer: An altruistic act is one in which an individual risks the loss of fitness in order to benefit another individual. Human beings value these “selfless” acts; however, they are not favored in natural animal populations, except under very specific circumstances, because altruistic acts should be selected against. In other words, all other things being equal, an individual that did not do altruistic acts would have a higher fitness than one that did

do these acts. Therefore, fitness is higher for “selfish” individuals. Altruistic acts, however, are expected if the beneficiary of the acts shares genes in common with the

benefactor performing the acts. Generally, altruism can be expected among relatives, following the rules of kin selection.

EXERCISES AND PROBLEMS *

DARWINIAN EVOLUTION

1. Outline the Darwinian mechanism of the process of evolution. What is meant by neo-Darwinism?

EVOLUTION AND SPECIATION

2. Population geneticist Hampton Carson has defined a “population flush” as a period of reduced selection during population increase. Why should there be reduced selection during a flush?
3. Describe how the processes of allopatric, parapatric, and sympatric speciation could take place.
4. Can information on evolutionary rates gained from molecular techniques shed light on the punctuated equilibrium-phyletic gradualism controversy? What additional data are needed to decide this controversy?
5. What is meant by “constraint” in the molecular evolution of DNA and proteins?
6. Recently, a vial of bull semen was stolen from an artificial insemination facility. Your friend is about to undergo artificial insemination and is concerned she may give birth to a Minotaur, or a cow-human hybrid. Provide two explanations for why she should not worry about this possibility.
7. In *Drosophila*, females in populations A and B produce an average of 250 offspring each. When the two populations are crossed, AB females produce only about 100 offspring each. Are populations A and B in the process of becoming different species?
8. A few plants of species *Q* ($2n = 14$) suddenly double their chromosomes ($2n = 28$) and immediately become a new species, *R*. Why are *QR* hybrids sterile?
9. One of the arguments creationists use to refute evolution is the presence of gaps in the fossil record. How can you explain the gaps from an evolutionary standpoint?

GENETIC VARIATION

10. The following electrophoretic data are from a sample of one hundred field mice for their salivary amylase-1 genotypes. The two alleles are *F* and *S*,

for fast and slow migration in an electric field: *FF*, forty-three, *FS*, fifty-four, and *SS*, three. Is selection acting? What would you look for in data to determine whether frequency-dependent selection, heterozygote advantage, or transient polymorphism is at work?

11. What mechanisms permit the maintenance of genetic variability in natural populations? Give examples where possible.
12. Discuss the “neutral gene hypothesis.” What are its alternatives? What data are needed to distinguish among these views?
13. Koehn showed that differently functioning alleles of an esterase system in fish correlated with water temperature. What sorts of selection can you imagine that could affect the same type of alleles in mammals, which are homeothermic (warm-blooded) and hence maintain a relatively constant internal temperature?
14. P. Niemälä and J. Tuomi have suggested that the irregular leaf outlines in some plant species are a form of mimicry. What would the leaves be mimicking? What form of mimicry might this be?
15. From figure 21.10, what is *K* (the average number of amino acid substitutions per site) between human beings and chickens? between dogs and human beings? Do all three possible comparisons support known evolutionary relationships?
16. How does the acceptance of the neutral mutation theory change our basic view of neo-Darwinism?
17. In a given population, the frequencies of *AA*, *Aa*, and *aa* genotypes are 0.36, 0.48, and 0.16, respectively. If the assigned fitnesses are $1.5 - p$, 1.0, and $1.5 - q$, what will the genotypic frequencies be after one generation of selection?
18. If the rate of amino acid substitution per site per year is 2×10^{-9} , and the average number of amino acid substitutions per site is 0.2, how long has it been since the two species diverged?
19. Scientists have examined one thousand amino acids in the proteins of human beings and chimps and have found a difference in twenty-three. Calculate the average number of amino acid substitutions per site.

*Answers to selected Exercises and Problems are on page A-24.

20. Scientists are now using DNA sequences to show phylogenetic relationships between or among species. In many cases, cDNA is made from isolated mRNA and then sequenced. Is the method a reasonable approach to show evolutionary relationships?
21. In which codon position should the greatest abundance of variation occur? Why?
23. If the “calculus of the genes” suggests sacrificing oneself for two siblings, for how many first cousins should one sacrifice oneself?
24. In certain animal populations, infanticide is practiced by one or more males. Do you think this infanticide is random, or would you expect specific individuals to be eliminated?

SOCIOBIOLOGY

22. What are the differences among individual selection, group selection, and kin selection? How could each type of selection explain altruistic acts?

C R I T I C A L T H I N K I N G Q U E S T I O N S

1. The peppered moth (*Biston betularia*) has two phenotypic forms, melanic (dominant) and peppered (recessive). The moths face predation by birds, and the predation is selective against different-colored tree trunks. In an industrialized area, one in which the tree trunks are dark like the melanic form (and thus “hide” the melanic forms from the birds), a sample of moths indicated that the frequency of the allele for peppering was 0.6; the next year, it was 0.5. What is the fitness of the peppered genotype under this circumstance?
2. V. C. Wynne-Edwards suggested that birds form flocks so that they can assess their population numbers. When they assess that their numbers are high, they decide not to breed for the good of the species, so that they do not exhaust their resources. Edwards called this process *group selection*. Why can’t this mechanism work, given that it involves behavior that is for the good of the species?

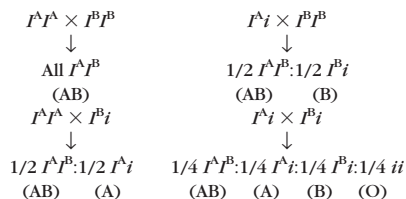
Suggested Readings for chapter 21 are on page B-20.

APPENDIX A

Brief Answers to Selected Exercises, Problems,
and Critical Thinking Questions

Chapter 2 Mendel's Principles

1. Dwarf F_2 ($1/4$ of total F_2), when selfed, produce all dwarf progeny (tt). Tall F_2 ($3/4$ of total F_2), when selfed, fall into two categories: $1/3$ (TT , $1/4$ of total F_2) produce all tall, and $2/3$ (Tt , $1/2$ of total F_2) produce tall and dwarf progeny in a 3:1 ratio. (The 3:1 ratio is from $1/2$ the F_2 , so the tall component is $3/8$ of the total F_3 [$3/4 \times 1/2$], and the dwarf is $1/8$ of the total F_3 [$1/4 \times 1/2$].) Overall, the F_3 are $3/8$ TT (tall), $2/8$ Tt (tall), and $3/8$ tt dwarf (see fig. 2.7).
3. Rule of segregation: adult diploid organisms possess two copies of each gene. Gametes get one copy. Fertilization restores the diploid number to the zygote. Rule of independent assortment: alleles of different genes segregate independently of each other.
5. Black is dominant, white is recessive, and both parents were heterozygous. The progeny are in an approximate 3:1 ratio. Since both parents had the same phenotype, the simplest cross is $Bb \times Bb$.
7. The disease is recessive at the individual level but incompletely dominant at the enzymatic level. Check the glossary for definitions.
9. $Ll \times Ll$. Let L = long ears and l = no ears. We see three phenotypes in an approximate 1:2:1 ratio. One of the phenotypes (short) is intermediate between long ears and no ears. Therefore, we have incomplete dominance.
11. Washed eye mutant, We ; wild-type, We^+ . (W is already the allelic designation for the wrinkled phenotype.)
13. All. Since the child was type A, it must have gotten the I^A allele from its mother. The other allele in the child is either I^A or i . A type A ($I^A I^A$ or $I^A i$), type B ($I^B I^B$ or $I^B i$), type O, or type AB man could have supplied either an I^A or i allele.
15. Universal donor, type O (no red-cell antigens); universal recipient, type AB (no serum antibodies).
17. All AB; or $1/2$ AB, $1/2$ A; or $1/2$ AB, $1/2$ B; or $1/4$ A: $1/4$ AB: $1/4$ B: $1/4$ O. Crosses can be $I^A I^A \times I^B I^B$, $I^A I^A \times I^B i$, $I^A i \times I^B I^B$, or $I^A i \times I^B i$.



19. Steve and his fiancé could be related. Both the dean and Steve's father must be I^i to produce O children, and each could have contributed M to produce M offspring. If the dean and Steve's father each contributed an S allele, the daughter would be SS . Note that if the daughter had B blood, she and Steve could not be related.

21. $RrTt \times$ self yields:

$1/16$ $RRTT$ red, tall
 $2/16$ $RRTt$ red, medium
 $1/16$ $Rrtt$ red, dwarf
 $2/16$ $RrTT$ pink, tall
 $4/16$ $RrTt$ pink, medium
 $2/16$ $Rrtt$ pink, dwarf
 $1/16$ $rrTT$ white, tall
 $2/16$ $rrTt$ white, medium
 $1/16$ $rrtt$ white, dwarf

23. Choice (b) is preferred because although each will give the correct genotype, generally, testcrossing has the greatest probability of exposing the recessive allele in a heterozygote. For example, an Aa genotype, when selfed, produces aa offspring one-fourth of the time; when testcrossed, aa offspring appear one-half of the time; and when crossed with the Aa type (backcross), aa offspring occur one-fourth of the time. Thus, with a limited number of offspring examined per cross, testcrossing most reliably exposes the recessive allele.
25. The F_1 are tetrahybrids ($Aa Bb Cc Dd$). If selfed, an F_1 would form $2^4 = 16$ different types of gametes; 2^4 different phenotypes would appear in the F_2 , which would be made up of $3^4 = 81$ different genotypes; $1/(16)^2 = 1/256$ of the F_2 would be of the $aa bb cc dd$ genotype.
27. a. In the first cross, look at the yellow-to-green ratio, 120:43—almost exactly 3:1. Therefore, yellow is dominant and both parents must be heterozygous. Now look at the tall-to-short ratio, 122:41. Again, we see a 3:1 ratio, which indicates that tall is dominant and each parent is heterozygous. Thus, the cross is probably $YyTt \times YyTt$.
 b. In the second cross, there are no tall progeny. Therefore, either the short phenotypes are homozygous, or short is dominant and at least one parent is homozygous. In the absence of the first cross, we can't determine the mode of inheritance of height. We can, however, conclude that yellow is dominant (we got a 3:1 ratio) and that each parent is heterozygous. Based on the first cross, we can conclude that this cross is $Yytt \times Yytt$.
 c. In the third cross, we see 41 yellow:46 green, and 45 tall:42 short. Both of these ratios are 1:1, and all we can conclude is that these ratios result from matings between a heterozygote and a recessive homozygote. With only this cross, we can't determine dominance. However, we can if we use all three crosses. The cross is $yyTt \times Yytt$.

A-2 Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

29. From the F_1 progeny, we can see that long and tan must be dominant, and the F_2 result confirms this assumption. We see a 3:1 ratio for both tan:dark and long:short. The total number of flies is 80. An ideal 9:3:3:1 ratio would be 45:15:15:5. Our results are very close to this. Therefore, we conclude that tan and long are dominant, and that the F_1 flies were heterozygous.

31. First list all possible genotypes for colored plants:

$AACcRR$
 $AACcRr$
 $AACcRR$
 $AACcRr$
 $AaCCRR$
 $AaCCRr$
 $AaCcRR$
 $AaCcRr$

The first genotype can be eliminated because all progeny should be colored, regardless of the tester strain. $AACcRr$ can be eliminated because the progeny of the first cross would have all been colored. $AACcRR$ can be eliminated because the progeny of the second cross would have all been colored. $AaCCRR$ can also be eliminated because the progeny of the third cross would have all been colored.

We are now left with $AACcRr$, $AaCCRr$, $AaCcRR$, and $AaCcRr$. Try $AACcRr \times aaccRR$ (cross 1), which will give 1/2 colored ($A-C-R$): 1/2 colorless ($A-ccR$) progeny. This could be the genotype, so try it in the second cross: $AACcRr \times aaCCrr$. This too will give 1/2 colored and 1/2 colorless offspring ($A-C-rr$). Since this does not fit the observed result, the unknown genotype is not $AACcRr$. Now try $AaCCRr \times aaccRR$ (cross 1). This fits the results. Try $AaCCRr \times aaccRR$. This will give 3/4 colorless ($aaccRR$, $A-ccRR$, or $aaC-RR$), which does not fit the results; therefore, the genotype is not $AaCCRr$. Now try $AaCcRr \times aaccRR$. This, too, will give 3/4 colorless ($aaccR$, $A-ccR$, or $aaC-R$), which is not seen. Therefore, the genotype must be $AaCCRr$. Confirm this with the other two crosses:

$AaCCRr \times aaCCrr \rightarrow 1/4$ colored : $3/4$ colorless, which fits.

$AaCCRr \times AACcrr \rightarrow 1/2$ colored : $1/2$ colorless, which fits.

33. a. all normal

b. 9 normal: 7 dark

c. 1/2 ebony: 1/2 normal; 1/2 black: 1/2 normal

If we let e = ebony, e^+ = wild-type, b = black, and b^+ = wild-type, the first cross is:

$ee b^+ b^+ \times e^+ e^+ bb$
 $\downarrow$
 $all e^+ e b^+ b$
 $\downarrow$ selfing

9/16 $e^+ - b^+ -$	wild-type
3/16 $ee b^+ -$	ebony
3/16 $e^+ - bb$	black
1/16 $ee bb$	ebony, black

Since it is difficult to distinguish black and ebony, 7/16 will be dark-bodied.

In c, the crosses are:

1. $e^+ e b^+ b \times ee b^+ b^+$
2. $e^+ e b^+ b \times e^+ e^+ bb$

In each case we have a testcross situation for only one gene.

35. 9 wild-type: 3 orange-1: 3 orange-2: 1 pink.

The first two crosses indicate that wild-type is dominant to both oranges, and the fourth indicates that orange-2 is dominant to pink. The fifth cross produces four phenotypes, indicating we are dealing with at least two genes. The presence of two genes is also suggested by orange-1 $\times$ orange-2. If these two traits were allelic, all the progeny should have been orange. The $F_1 \times$ pink produces progeny that resemble those from a double testcross. If $A-B-$ = wild-type, $A-bb$ = orange-1, $aaB-$ = orange-2, $aabb$ is probably pink. The crosses in question are then:

$AAbb$ $\times$ $aaBB$
 (orange-1) (orange-2)
 $\downarrow$
 $AaBb$
 (wild-type)
 $\downarrow$ self

9/16 $A-B-$	wild-type
3/16 $A-bb$	orange-1
3/16 $aaB-$	orange-2
1/16 $aabb$	pink

37. Two loci with epistasis. $AaBb \times$ self yields 9/16 $A-B-$: 6/16 $A-bb$ + $aaB-$: 1/16 $aabb$. Verify by testcrossing the various classes.

39. See for example figure 2.25. Other ratios are 10:3:3; 10:6; 12:3:1; 12:4.

41. 9 nonworkers: 7 hard workers. The F_1 indicates nonworker is dominant; therefore, a worker must be a recessive homozygote. If one gene is involved, the cross of the F_1 female $\times$ worker male ($Ww \times ww$) should produce 1 worker: 1 nonworker, for this is a testcross. This result is not seen, so we must have more than one gene involved. Perhaps a worker can result from more than one gene. Let $A-B-$ = nonworker, and $A-bb$, $aaB-$, or $aabb$ = workers. The original worker is $aabb$, and the cross is:

$aabb$ $\times$ $AABB$
 $\swarrow$
 $AaBb$ $\times$ $aabb$
 (nonworker) (worker)

1/4 $A-B-$	1/4 $A-bb$	1/4 $aaB-$	1/4 $aabb$
nonworker	worker	worker	worker

If $AaBb \times$ $AaBb$
 $\downarrow$

9/16 $A-B-$ nonworkers	
3/16 $aaB-$	workers
3/16 $A-bb$	
1/16 $aabb$	

43. 2 1 3 4
 $? \rightarrow$ indole $\rightarrow$ tryptophan $\rightarrow$ 3-hydroxyanthranilic $\rightarrow$ niacin
 or kynurenine acid

Accumulation: 1, indole; 2, ?; 3, tryptophan or kynurenine; 4, 3-hydroxyanthranilic acid. Without serine, the pathway would be blocked before the point of tryptophan production, after indole.

45. a. Maple sugar urine disease is recessive. If two individuals with the same phenotype produce offspring, some of whom have a different phenotype, both parents must be heterozygous. Since they are heterozygous, their phenotype must be dominant.

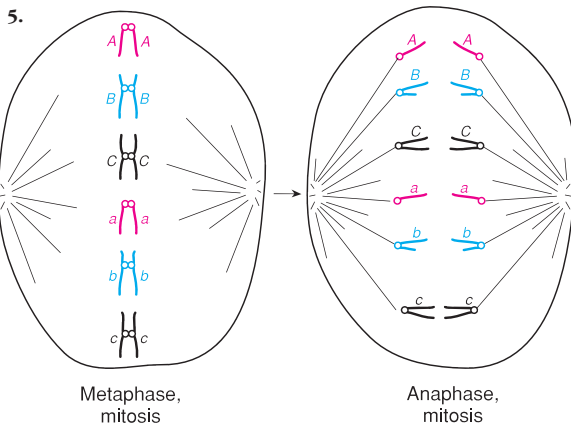
b. 3/4. Let M = normal and m = maple sugar urine alleles. The cross is $Mm \times Mm$. At each conception, the probability is three-fourths that each child is of the $M-$ genotype.

Critical Thinking Question:

1. The 15:1 ratio indicates that all genotypes except the recessive homozygote (*aabb*) produce a triangular capsule. The rounded capsule results from the recessive homozygote. One way to look at this is that the rounded form is a "default" form when neither locus has functional—dominant—alleles. However, a dominant allele at either of two loci is adequate to form the triangular seed capsule. The loci can be considered redundant in the pathway of seed capsule shape since a functional allele at one, the other, or both will provide a dominant (triangular) phenotype. At this point in time, it is impossible to know precisely what the enzymatic function of each dominant allele is.

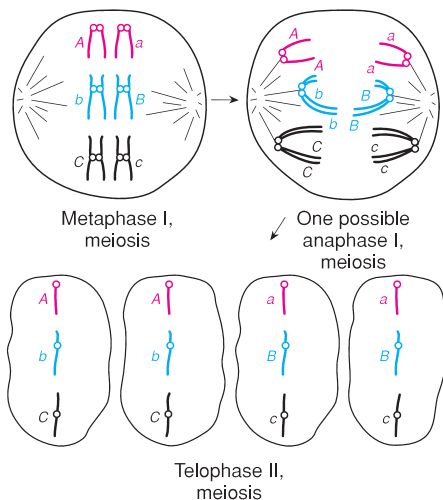
Chapter 3 Mitosis and Meiosis

1. See table 3.1 for a summary answer.
3. When a eukaryotic chromosome replicates during the S phase of the cell cycle, one chromosome becomes two chromatids, attached near the centromere. These are sister chromatids. Chromatids of different chromosomes are nonsister chromatids. Homologous chromosomes are members of a pair of essentially identical chromosomes. In diploid organisms, one member from each pair comes from each parent. Nonhomologous chromosomes do not share this relationship.



7. S phase

9. $2^3 = 8$ different gametes can arise. A crossover between the *A* locus and its centromere does not alter gametic combinations.



11. The intent of this problem is to make you think about the essential steps of meiosis, primarily the necessity to separate members of homologous pairs of chromosomes. Presumably, any method you devise will force you through that process.
13. Meiosis apportions homologous chromosomes the same way Mendel's rules apportion alleles. Each gamete gets one member of a homologous pair of chromosomes. Segregation predicts the same about alleles. The separation of homologues of one chromosome pair at meiosis is independent of the separation of other homologous pairs. Independent assortment makes the same prediction about alleles.
15. A gamete from wheat will have twenty-one chromosomes, and a gamete from rye will have seven chromosomes. Even if the seven rye chromosomes could pair with seven wheat chromosomes, a highly unlikely possibility, the remaining fourteen wheat chromosomes could not pair and would segregate randomly during meiosis. Almost every gamete would get an incomplete set; if fertilization did occur, the zygotes would have extra chromosomes (trisomic) or would be missing some chromosomes (monosomic or nullosomic).
17. 64. The number of combinations is 2^n where n = the number of different chromosomes in the set.

	DNA (Number of Chromatids)	Ploidy
Spermatogonium or Oogonium	2	$2n$
Primary Spermatocyte or Primary Oocyte	4	$2n$
Secondary Spermatocyte or Secondary Oocyte	2	n
Spermatid or Ovum	1	n
Sperm	1	n

21. a. 50 b. 50. The primary oocyte is diploid and will undergo meiosis, but only one functional ovum results from each primary oocyte. The secondary oocyte will divide to produce an ovum and a polar body.
23. Any possible genotype, from *AAABBB* through *aaabbb* can occur in the endosperm. If at a given locus the endosperm is homozygous, so is the embryo. If the locus is heterozygous (e.g., *Aaa* or *Aaa*), so is the embryo. Thus, an *AAabbb* endosperm is associated with an *Aabb* embryo.
25. A greater maternal influence in *Drosophila* and corn than in *Neurospora* means the sexes (mating types) do not show the disparity in size between male and female cells as in *Drosophila* and corn.
27. b. Homologous chromosomes will pair during meiosis. Each gamete gets one of each chromosome, A, B, C, D, and E. Fertilization fuses two cells with the chromosome complement given. Since root cells are somatic tissue, these cells will be diploid.
29. a. 2^{50} or about 1.1×10^{15} b. 2^2 . The number of gametes produced is 2^n , where n = number of independently behaving entities. If the genes are completely independent, we expect 2^{50} , and if they are completely linked, we expect 2^2 . In reality, the number falls between these extremes.

A-4

Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

Critical Thinking Question:

1. Both meiosis and mitosis are processes that initiate under certain circumstances of cell cycle and place. Neither is actually dependent on the chromosomal content of the cell. Thus, meiosis could begin in a haploid cell, but it would not be a successful process because there is no homologue for any chromosome to pair with. Mitosis would, however, be successful because there is no pairing (synapsis) required for successful completion of the process.

Chapter 4 Probability and Statistics

1. a. $(5!/3!2!)(1/2)^3(1/2)^2 = 0.3125$
b. $(1/2)^5 = 0.03125$ (SDSDS, in which S = son, D = daughter)
c. $2(1/2)^5 = 0.0625$ (SDSDS + DSDSD)
d. $(1/2)^5 = 0.03125$
e. $2(1/2)^5 = 0.0625$ (all sons + all daughters)
f. 4 daughters, 1 son + 5 daughters: $(5!/4!1!)(1/2)^4(1/2) + (1/2)^5 = 0.1875$
g. $(1/2)^2 = 0.25$ (DXXXS, in which X is either a daughter or a son, with $p = 1$; $P = [1/2][1][1][1][1/2]$)
3. Remember that albinos have blue eyes. Therefore, 7/16 of the offspring will have blue eyes. If we let B = brown, b = blue, C = normal color expression, and c = albinism, the following genotypes are blue-eyed: $C-bb$ and $cc-$.
a. $(1/4)^5 = 0.0009765$
b. $(1/8)^5 = 0.0000305$
c. $(5!/4!1!)(7/32)^4(9/32) = 0.00322$
d. $(4!/2!2!)(1/8)^2(1/8)^2 = 0.0014648$
5. a. $2(1/2)^4 = 0.125$
b. $(1/2)^4(1/2)^4 = 0.0039063$ (Probability that sperm and egg creating the zygote each had only paternal centromeres.)
7. One-half.
9. a. 81/256 b. 108/256 c. 9/256. In (a), since all children have the same phenotype, each child will have the same probability of having no molars. Therefore, $(3/4)^4 = 81/256$. In (b),

$$P = \frac{4!}{3!1!} (3/4)^3(1/4) = 108/256.$$

When order is given, we multiply the chance of each event, $1/4 \times 1/4 \times 3/4 \times 3/4 = 9/256$.

11. One-eighth. B must be heterozygous (Gg), as must A's father. We assume A's mother is GG , since there is no mention of the disease in her family. Therefore, A has a one-half chance of getting g from his father. If two heterozygotes mate, the chance of a recessive child is one-fourth, so $P = 1/4 \times 1/2$.
13. 1/512. The F_1 progeny are $Aa Bb Cc Dd Ee$. The chance of getting any individual with a particular homozygous genotype is $(1/4)^5$. Since we are looking for two different possibilities, we have $2(1/4)^5 = 2/1024 = 1/512$.
15. 0.049. Since the order is not specified, we use the multinomial formula. We have six mice, so $n = 6$. If p = chance of agouti, $p = 3/4 \times 1/2 = 3/8$; q (black coat color) = $1/4 \times 1/2 = 1/8$, and r (albino) = $1/2$. The equation becomes:

$$p = \frac{6!}{2!2!2!} (3/8)^2(1/8)^2(1/2)^2 =$$

$$\frac{6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 9 \times 1 \times 1}{2 \times 2 \times 2 \times 64 \times 64 \times 4} = 0.049$$

17. Hypothesis: $RrYy \times RrYy$ produces $R-Y:R-yy:rrY:rryy$ in a 9:3:3:1 ratio. The critical chi-square, three degrees of freedom at probability of 0.05, = 7.815.

	<i>R-Y-</i>	<i>R-yy</i>	<i>rrY-</i>	<i>rryy</i>	Sum
Observed	315	108	101	32	556
Expected	9/16	3/16	3/16	1/16	
	312.75	104.25	104.25	34.75	556
$O - E$	2.25	3.75	-3.25	-2.75	
$(O - E)^2$	5.06	14.06	10.56	7.56	
$(O - E)^2/E$	0.016	0.135	0.101	0.218	$0.470 = \chi^2$

Since this chi-square, 0.470, is less than the critical chi-square, we fail to reject our hypothesis of two-locus genetic control with dominant alleles at each locus.

19. We reject the 3:1 ratio as an appropriate null hypothesis. If we calculate the chi-square using 3:1 as the expected ratio, we expect 72 and 24 ($3/4$ and $1/4$ of 96, respectively):

	<i>O</i>	<i>E</i>	<i>O-E</i>	$(O-E)^2$	$(O-E)^2/E$
Curly-winged flies	61	72	-11	121	$121/72 = 1.681$
Straight-winged flies	35	24	11	121	$121/24 = 5.042$

Chi-square = 6.723 ($1.681 + 5.042$). With one degree of freedom, $p > 0.05$ (critical chi-square = 3.841).

Critical Thinking Question:

1. You should change your choice because the box you chose originally has a $1/3$ chance of containing the prize, whereas the remaining box has a $2/3$ chance of containing the prize. The $1/3$ chance of your choice is set by the fact that there were three equally likely choices at the beginning. When your friend eliminated an empty box, she left two choices: your original box and the third box. Since the probability of your original choice has not changed, the probability of the remaining box must be $2/3$ to give a combined probability of 1.0 that a box contains a prize.

Chapter 5 Sex Determination, Sex Linkage,
and Pedigree Analysis

1. The differences are in terminology only, not in shape or size of the chromosomes. In species in which females have a homomorphic pair of sex chromosomes, the members of the pair are called X chromosomes. In species in which males have a homomorphic sex chromosome pair, the members of the pair are called Z chromosomes.
3. $3/8$ males, $3/8$ females, $2/8$ intersexes ($dsx dsx$ homozygotes). The $1/4 dsx dsx$ flies will be intersexes, whereas half of the $3/4$ will be normal males and half will be normal females.
5. The protein is probably a dimer, which, in the heterozygote, can be of fast-fast, fast-slow, or slow-slow subunit combinations. A female heterozygous for a sex-linked gene controlling a dimeric enzyme should show the pattern of lane 3 in whole blood (mixture of slow-slow and fast-fast dimers) and lanes 1 or 2 in individual cells.

7. a. 0; human female, male fly
b. 1; human female, female fly
c. 0; human male, male fly
d. 1; human male, female fly
e. 2; human female, female fly
f. 4; human female, female fly
g. 1/0 mosaic; human male-female mosaic, male-female fly mosaic

9.

	Cross	Reciprocal
P_1 female	X^+X^+	$X^{Lz}X^{Lz}$
male	$X^{Lz}Y$	X^+Y
F_1 female	X^+X^{Lz}	X^+X^{Lz}
male	X^+Y	$X^{Lz}Y$
F_2 females	X^+X^+, X^+X^{Lz}	$X^+X^{Lz}, X^{Lz}X^{Lz}$
males	$X^+Y, X^{Lz}Y$	$X^+Y, X^{Lz}Y$

11. Exemptions should be made minimally for hemophilia in brother, sister's son, mother's brother, mother's sister's son, mother's father, and others more distantly related.

13. P_1 $f_1f_1 X^+X^+$ (female) $\times$ $f_1^+f_1^+ X^{ct}Y$ (male)
 F_1 $f_1^+f_1 X^+X^{ct}$ (female) $\times$ $f_1^+f_1 X^+Y$ (male)

	Male			
	$f_1^+ X^+$	$f_1 X^+$	$f_1^+ Y$	$f_1 Y$
Female				
$f_1^+ X^+$	$f_1^+f_1^+ X^+X^+$	$f_1^+f_1 X^+X^+$	$f_1^+f_1^+ X^+Y$	$f_1^+f_1 X^+Y$
$f_1^+ X^{ct}$	$f_1^+f_1^+ X^+X^{ct}$	$f_1^+f_1 X^+X^{ct}$	$f_1^+f_1^+ X^{ct}Y$	$f_1^+f_1 X^{ct}Y$
$f_1 X^+$	$f_1^+f_1 X^+X^+$	$f_1f_1 X^+X^+$	$f_1^+f_1 X^+Y$	$f_1f_1 X^+Y$
$f_1 X^{ct}$	$f_1^+f_1 X^+X^{ct}$	$f_1f_1 X^+X^{ct}$	$f_1^+f_1 X^{ct}Y$	$f_1f_1 X^{ct}Y$

F_2 : females, 3/4 wild-type, 1/4 fuzzy; males, 3/8 wild-type, 3/8 cut, 1/8 fuzzy, and 1/8 cut and fuzzy.

15. a. X linked

b. gray

c. 1/2 gray:1/2 yellow in both sexes

In both crosses, we see a difference in the phenotypes of the sexes, suggesting sex linkage. The F_1 offspring from the first cross indicate that gray is dominant to yellow. The F_1 females from this cross must be heterozygous, and the two phenotypes in the F_2 males result from each of the X chromosomes in the F_1 female being hemizygous in the F_2 males. The first cross is therefore (calling gray the wild-type):

X^+X^+	$\times$	X^0Y	
	$\downarrow$		
X^+X^0	$\times$	X^+Y	
gray		gray	
	$\downarrow$		
X^+X^+	X^+X^0	X^+Y	X^0Y
gray	gray	gray	yellow

Now diagram the second cross:

X^0X^0	$\times$	X^+Y	
	$\downarrow$		
X^+X^0	$\times$	X^0Y	
gray		yellow	
	$\downarrow$		
X^+X^0	X^0X^0	X^+Y	X^0Y
gray	yellow	gray	yellow

17. Yes. Begin by determining genotypes of the two individuals. The woman must be heterozygous $X^C X^c$. A man with normal vision must be $X^C Y$, and all his daughters must receive his X chromosome and should be normal, either $X^C X^C$ or $X^C X^c$. Since color blindness is recessive, the daughter must have two X^c chromosomes. A *very* rare possibility is that the man is the father and nondisjunction occurred in both parents: at meiosis I in the male and at meiosis II in the female (see chapter 8).

19. a. F_1 : wild-type females, white-eyed males.

b. F_2 : 3 wild-type:3 white-eyed:1 ebony:1 ebony, white-eyed in both sexes.

c. Reciprocal F_2 : 3 wild-type females:1 ebony female:3 wild-type males:3 white-eyed males:1 ebony male:1 ebony, white-eyed male. Let X^+ = red, X^w = white, e^+ = wild-type, e = ebony.

$X^w X^w e^+ e^+$	$\times$	$X^{w+} Y ee$	P_1
$\downarrow$			
$X^+ X^w e^+ e$	$\times$	$X^{w+} Y e^+ e$	F_1
(wild-type)		(white-eyed)	

Use probability for the F_2 generation rather than the Punnett square:

$$\begin{aligned}
 (1/2)X^+ \times (1/2)X^{w+} \times (3/4)e^+ &= 3/16 \text{ wild-type females} \\
 &\times (1/4)ee = 1/16 \text{ ebony females} \\
 \times (1/2)Y \times (3/4)e^+ &= 3/16 \text{ wild-type males} \\
 &\times (1/4)ee = 1/16 \text{ ebony males} \\
 (1/2)X^w \times (1/2)X^{w+} \times (3/4)e^+ &= 3/16 \text{ white-eyed females} \\
 &\times (1/4)ee = 1/16 \text{ white-eyed, ebony females} \\
 \times (1/2)Y \times (3/4)e^+ &= 3/16 \text{ white-eyed males} \\
 &\times (1/4)ee = 1/16 \text{ white-eyed, ebony males}
 \end{aligned}$$

For the reciprocal cross,

$X^+ X^+ ee$	$\times$	$X^{w+} Y e^+ e^+$
$\downarrow$		
$X^+ X^{w+} e^+ e$	$\times$	$X^+ Y e^+ e$
(wild-type)		(wild-type)

All F_2 females will get X^+ ; $e^+ : e$ will be 3:1. The males will be as in the males shown.

21. The female is heterozygous for an X-linked color gene (one of the X chromosomes in the cells of female cats is inactivated, leading to the black and yellow spots). We immediately deduce X-linkage because of the different phenotypes in the sexes. Finding two types of males indicates that the female was heterozygous. The patches of yellow and black come from X-inactivation. The cross is:

X^bX^o		$\times$	X^bY	
		$\downarrow$		
X^bX^b	X^bX^o		X^bY	X^oY
black	black and yellow		black	yellow

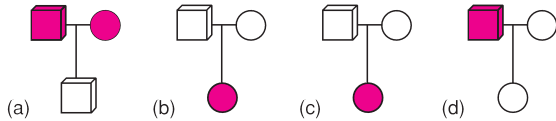
23. Penetrance is the proportion of individuals of a particular genotype that shows the appropriate phenotype; expressivity is the degree to which a trait is expressed.

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Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

25. a. The phenotype is the propensity to have twin offspring. It could be caused by a recessive or dominant, sex-linked or autosomal allele.
b. Autosomal dominant or possibly autosomal recessive inheritance.
c. Autosomal, or sex-linked, recessive inheritance.
d. Autosomal recessive inheritance.

27. Assuming 100% penetrance:



Critical Thinking Question:

1. The immediate effect of a null allele is to make heterozygous genotypes appear to be homozygotes. That is, in the simplest system, we expect one band in a homozygote and two bands in a heterozygote. If we see only one band, we assume that the individual is homozygous for that allele, when in fact that individual might be heterozygous for the allele that produces the particular band and the allele that produces no band (null allele). The null allele can be verified by both the absence of any bands in the null homozygote and the results of breeding experiments when the null allele is suspected.

Chapter 6 Linkage and Mapping in Eukaryotes

1. a.

P ₁ groucho	×	rough
<i>grogro ro⁺ro⁺</i>		<i>gro⁺gro⁺ roro</i>
F ₁ female <i>gro⁺gro⁺ ro⁺ro</i>	×	male <i>grogro roro</i>
F ₂ <i>grogro ro⁺ro</i>		518
<i>gro⁺gro roro</i>		471
<i>grogro roro</i>		6
<i>gro⁺gro ro⁺ro</i>		5
$(6+5)/1,000 = 0.011 = 1.1\%$ recombination		
$= 1.1$ map units apart		

- b. Given the map units, F₁ gametes are produced on the average by females as follows: *gro ro⁺*, 49.45% = 0.4945 (98.9%/2); *gro⁺ ro*, 49.45% = 0.4945; *gro ro*, 0.55% (1.1%/2) = 0.0055; and *gro⁺ ro⁺*, 0.55% = 0.0055. Males, lacking crossing over, produce only two gamete types: *gro ro⁺* and *gro⁺ ro*, each 50% = 0.50. Summing from the Punnett square following, the phenotypes of the offspring would be as follows: wild-type, 50%; groucho, rough, 0%; groucho, 25%; and rough, 25%.

Female	Male	
	<i>gro ro⁺ (0.5)</i>	<i>gro⁺ ro (0.5)</i>
<i>gro ro⁺</i>	groucho	wild-type
(0.4945)	0.24725	0.24725
<i>gro⁺ ro</i>	wild-type	rough
(0.4945)	0.24725	0.24725
<i>gro ro</i>	groucho	rough
(0.0055)	0.00275	0.00275
<i>gro⁺ ro⁺</i>	wild-type	wild-type
(0.0055)	0.00275	0.00275

3. A dihybrid female is testcrossed (with a hemizygous male having both recessive alleles). Each recombinant class will make up about 5% of the offspring. Each parental class will make up about 45% of

the offspring. Phenotypic classes will be equally distributed between the two sexes. The same results will be found for an autosomal locus if the dihybrids are females (no crossing over in males). A reciprocal cross cannot be done for X-linked genes because males cannot be dihybrid. Males dihybrid for an autosomal gene produce only two classes of offspring when testcrossed—parentals.

5. a. The hotfoot locus is in the middle (compare, for example, hotfoot, a double crossover, with the wild-type, a parental); there are 16.0 map units from hotfoot to either end locus: $74 + 66 + 11 + 9$ recombinants between hotfoot and waved and $79 + 61 + 11 + 9$ recombinants between hotfoot and obese.
b. The trihybrid parent was *o b wa/o⁺ b⁺ wa⁺*.
c. The coefficient of coincidence is $20/25.6$ ($20/[0.16 \times 0.16 \times 1,000]$); interference is $1 - (20/25.6) = 0.22$, or 22%.
7. a. Work backward from the 0.61% double recombinants ($0.100 \times 0.061 \times 100$). Thus, there would be 6 of 1,000 double recombinants. In the *an-sple* region, we need the total of single + double recombinants = 100 of 1,000 (10 map units). Thus, $100 - 6 = 94$; divided by 2 (two phenotypes) is 47 each. For the *sple-at* region, the total of single and double recombinants = 61 (6.1 map units). Thus, $61 - 6 = 55$; divided by 2 is 27 and 28. The parentals make up the remainder for a total of 1,000.
b. With a coefficient of coincidence of 0.60, only 0.366% (0.61×0.60) of the expected double recombinants will occur, that is, 4 instead of 6. Thus:

Coefficient of Coincidence

	1.0	0.6
ancon, spiny, arctus oculus	422	421
wild-type	423	422
ancon, spiny	27	28
arctus oculus	28	29
ancon	47	48
spiny, arctus oculus	47	48
ancon, arctus oculus	3	2
spiny	3	2
Total	1,000	1,000

9. a. linked; b. *trans*; c. 28.7%. The cross is a testcross. If the genes were not linked, we would expect a 1:1:1:1 ratio of offspring; we don't see that. The alleles that are linked will appear as the majority classes, which are Trembling, long-haired and normal, Rex. Therefore, Trembling and Rex are in the *trans* position. If we let *T* = Trembling, and *R* = Rex, the cross is

$$\frac{Tr}{tR} \times \frac{tr}{tr}$$

Recombinants are Trembling, Rex and normal, long-haired;

$$\frac{42 + 44}{300} \times 100 = 28.7\%$$

11. a. $\frac{k e^+ cd}{k^+ e cd^+}$ b. $\frac{k}{k^+} \frac{e}{e^+} \frac{cd}{cd^+}$

The initial cross is $\frac{k e^+ cd}{k^+ e cd^+} \times \frac{k^+ e cd^+}{k^+ e cd^+}$

Producing a trihybrid F₁ female: $\frac{k e^+ cd}{k^+ e cd^+}$, in any order.

Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

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The last two classes (3 + 4 offspring in the F_2) are double crossovers and allow us to determine order by comparison with the parents (880 + 887). If the order is $kcd e$, a double crossover in the F_1 females yields $kcd^+ e^+$ and $k^+ cde$. If the order is $cdke$, a double crossover yields $cdk^+ e^+$ and $cd^+ ke$. Therefore, the order must be $kecd$, which gives the correct double recombinants in the F_2 generation. After reconstructing the trihybrid

$$\frac{ke^+cd}{k^+ecd^+}$$

and scoring each of the offspring for crossovers in the $k-e$ and $e-cd$ regions: map units, $k-e = [(64 + 67 + 4 + 3)/2000] \times 100 = 6.9$ and map units, $e-cd = [(49 + 46 + 4 + 3)/2000] \times 100 = 5.1$.

13. 0.0125. This problem requires the manipulation of equations. We know that interference = 1 - coefficient of coincidence, so coefficient of coincidence = 1 - interference = 1 - (-1.5) = 2.5. Since coefficient of coincidence =

$$\frac{\text{observed double crossovers}}{\text{expected double crossovers}}$$

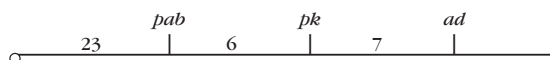
observed double crossovers = (coefficient of coincidence) $\times$ (expected double crossovers). The expected double crossover frequency is (2.5)(0.005) = 0.0125.

15. PD, 1, 2, 4, 6, 8-10; NPD, 3; TT, 5, 7. Map units = $[(\text{NPD} + \{1/2\}\text{TT})/\text{Total}] \times 100 = [(1 + \{1/2\}2)/10] \times 100 = 20$. The loci are 20 map units apart.
17. FDS, 3-5, 8, 10; SDS, 1, 2, 6, 7, 9. The distance between the *arg* locus and its centromere is $(1/2)\% \text{SDS} = (1/2)50\% = 25$ map units.
19. For example, the only variant of the type 2 pattern of table 6.7 is a^+b , a^+b , a^+b , a^+b , ab^+ , ab^+ , ab^+ , ab^+ . Other patterns that are variants of the remaining five categories are derived by inverting the eight spores of a pattern (bottom to top) or by switching spores 1 and 2 with spores 3 and 4 or spores 5 and 6 with spores 7 and 8.
21. $a \times a^+ \rightarrow a/a^+$, which undergoes meiosis. Twelve map units means that the SDS pattern makes up 24% of the asci.
23. For example: 1, 408; 2, 42; 3, 250; 4, 250; 5, 30; 6, 5; 7, 15. To make the numbers work out, classes 4-7 must equal 300, as must classes 3 + 5 - 7, in order to make $(1/2)\% \text{SDS} = 150$. Thus, if classes 3 and 4 are 250 each, 50 must be spread out among classes 5-7. NPD + $(1/2)\text{TT}$ should equal 300 to confirm the arrangement (*a-b* distances). These numbers will give an *a-b* distance of 30.
25. a. yes; PD > > NPD (actually, no NPD). b. a: 2.5 map units; b: 7.5 map units. Classify each ascus—I: PDT, FDS for both; II: TT, FDS for *a*, SDS for *b*; III: TT, FDS for *a*, SDS for *b*; IV: TT, FDS for *a*, SDS for *b*; V: TT, FDS for *a*, SDS for *b*; VI: PDT, SDS for *a* and *b*; VII, PDT, SDS for *a* and *b*; VIII: PDT, SDS for *a* and *b*. We see no NPDs, so genes are linked. For gene to centromere distances, use the formula

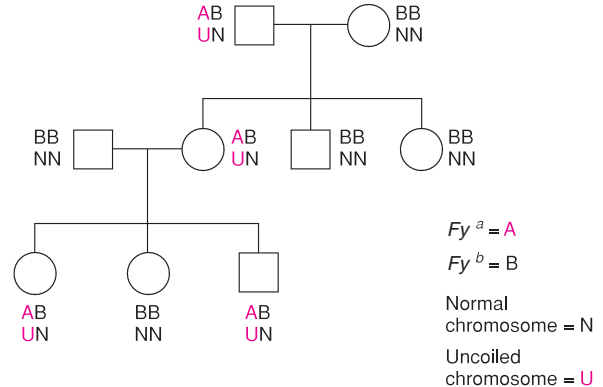
$$\frac{1/2 (\text{number of SDS asci})}{100} \times 100:$$

a to centromere = $(1/2)(3 + 1 + 1)\% = 2.5$ map units; *b* to centromere = $(1/2)(2 + 3 + 2 + 3 + 3 + 1 + 1)\% = 7.5$ map units.

27. 12.5 map units. The only genotype that grows on minimal medium is arg^+ade^+ . If the two genes were unlinked, 1/4 of the progeny should have this genotype; this is not seen. The genes must be linked; wild-type results from recombination between these two genes. The reciprocal class, arg^-ade^- , which has not been selected for, should be equally frequent, so: map units = $(2 \times 25)/400 \times 100 = 12.5$.



29. Construct a pedigree of the Duffy alleles (Fy^a , Fy^b). Arbitrarily assign one allele to a normal chromosome 1 and the other allele to the coiled chromosome. Then, accompanying the pedigree, the alleles and their morphologically proper chromosomes would be associated.



31. Twenty map units apart (two of ten recombinant sons according to the "grandfather method").
33. Thirty-three map units. The woman is heterozygous in *trans* configuration for color blindness and hemophilia: bc^+/b^+c . Recombination between these two markers yields b^+c^+ and bc . We can only detect recombinants in sons, so

$$\frac{\# \text{ normal sons} + \# \text{ double mutant sons}}{\text{total sons}} \times 100 = \text{map distance} = 2/6 = 0.33$$

35. Enzyme A on 11; B on 15; C on 18; D on 3; E on 7. Enzyme A is present in clones X and Y, and chromosome 11 is common to these two clones. Enzyme B is present only in X, and 15 is the only chromosome unique to X. Similar logic allows the assignment of the other genes.

Critical Thinking Question:

1. Three-point crosses capture (allow us to see) double crossovers that have taken place in the two regions defined by the three loci. However, any double crossovers that occur within one region or any crossovers involving more than two events will not be indicated correctly by random-strand analysis.

Chapter 7 Linkage and Mapping in Prokaryotes and Bacterial Viruses

1. The prokaryotic chromosome is a double-stranded DNA circle that is small compared with most eukaryotic chromosomes. Viral chromosomes can be DNA or RNA. Viruses are obligate intracellular parasites. Whether they are alive depends on the definition of the term *alive*.
3. A colony is a visible mass of cells derived usually from a single progenitor. A plaque is the equivalent growth of phages on a bacterial lawn, producing a cleared area lacking intact bacteria.
5. The bacterium could have survived and produced a colony if it was on a λ -free area, it became lysogenic (and thus resistant to further phage attack), or it was genetically resistant to phage λ .
7. 1, $bis^- arg^-$; 2, leu^- ; 3, lys^- ; 4, $bis^- met^-$ or $bis^+ met^-$; 5, arg^- .
9. Where phages cannot grow: *E. coli ton^r*, phage h^+ . Where phage can grow: *E. coli ton^s*, phage b^+ or *h*, or *E. coli ton^r*, phage *b*.

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Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

11. Far. If the selective locus is near, it passes into the F^- cell very early during conjugation. Consequently, there is a great reduction in the recovery of loci distal to the selective marker because both the Hfr and F^- members of a conjugation event can be killed by the selective agent (e.g., an antibiotic such as streptomycin).
13. See figures 7.8, 7.9, 7.15, 7.17, 7.18, and 7.26.
15. 9 min 1 min 8 min 7 min
Origin ——— *az* ——— *ton* ——— *lac* ——— *galB*
17. For example, use an Hfr that is wild-type but *str^s* with the F factor integrated at minute 20. Use an F^- strain that is *pyrD⁻*, *purB⁻*, *man⁻*, *uvrC⁻*, *bis⁻*, and *str^r*. Interrupt mating at one-minute intervals; plate cells on complete medium with streptomycin to kill Hfr cells and grow recombinant and nonrecombinant F^- cells. The next day, after colonies have grown up, replica-plate onto selective media. The following data would be generated:

Colony Growth on Media Selective for

	<i>pyrD⁺</i>	<i>purB⁺</i>	<i>man⁺</i>	<i>uvrC⁺</i>	<i>bis⁺</i>
Minute 0	—	—	—	—	—
1	+	—	—	—	—
5	+	+	—	—	—
16	+	+	+	—	—
22	+	+	+	+	—
24	+	+	+	+	+

19. The order is *a c b*, and *c* is close to *a*. Genes *c* and *a* are cotransformed 76% of the time, suggesting that these two genes are very close and *b* is far away (*a-b* cotransduction is 10%, or 0.1). Two orders are possible: *a-c-b* and *c-a-b*. If the first order is correct, *a⁺ b⁺ c⁻* results from a double crossover; this class should be the least frequent. If the second order is correct, a single exchange between *a* and *c* would yield *a⁺ b⁺ c⁻*, but this frequency should be similar to *a⁺ b⁻ c⁻*, and it is not.
21. *thr leu pro bis*. We see that cells that are *thr⁺* are the most frequent. The chance of interruption in the conjugation increases with the length of time for the mating. Therefore, genes farther from the origin of transfer appear less frequently. We can order the genes based merely upon the frequency of genotypes seen. The order must be *thr leu pro bis*. Since we see no *bis⁺*, and since we stopped the mating at 25 minutes, *bis* must be after minute 25 on the map of this Hfr strain.
23. *a* and *c* are close; *b* is farther away; *c* is probably in the middle. The numbers of the first three transformant classes indicate that each gene, by itself, is readily transformed. We notice that classes with *b* and any other gene are quite rare, a situation that indicates *b* is far from *a* and *c*. We notice that *a* and *c* are cotransformed about 13% of the time, *b* and *c* about 3% of the time, and *a* and *b* about 2% of the time.
25. a. *lys⁺ bis⁺ val⁺*
lys⁺ bis⁺ val⁻
lys⁺ bis⁻ val⁺
lys⁺ bis⁻ val⁻
 Since there is no lysine in the medium, *lys⁺* must be present to allow growth.
 b. *lys⁺ val⁺ bis⁺*
lys⁺ val⁺ bis⁻
 Both *lys⁺* and *val⁺* must be present to allow growth.

- c. *lys⁺ val⁺ bis⁺*
lys⁺ val⁻ bis⁺

Both *lys⁺* and *bis⁺* must be present to allow growth.

- d. *lys⁺ val⁺ bis⁻*
lys⁺ val⁻ bis⁺

We see no *lys⁺ val⁺ bis⁺* cells.

- e. *lys⁺* and *val⁺* are close together; they are cotransformed 75% of the time. Order could be *lys val bis* or *val lys bis*.
 f. *val lys bis*. If the order is *val lys bis*, *val⁺ lys⁻ bis⁺* should be rare, since this genotype results from a double exchange; and indeed, this class is the least frequent.

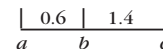
27. Mix the phages together with bacteria with increasing quantities of the two phages. Knowing the numbers of each in a particular case, it is possible to predict the proportion of cells doubly infected (product of probabilities). The recovery of recombinants should increase with that probability. In other words, recombination should occur only in doubly infected cells.

29. 0.6 0.2 1.2
 a *b* *d* *c*

Small recombination frequencies should be approximately additive. Note that recombination distances are twice the value of wild-type plaques since the double mutant recombinants were not counted. Thus, the data table should be:

Cross	Percent Wild-Type Plaques	Percent Recombinants
<i>a × b</i>	0.3	0.6
<i>a × c</i>	1.0	2.0
<i>a × d</i>	0.4	0.8
<i>b × c</i>	0.7	1.4
<i>b × d</i>	0.1	0.2
<i>c × d</i>	0.6	1.2

The largest distance is between *a* and *c*; therefore *a* and *c* must be at opposite ends. Since *a-b* = 0.6, *b* must be 0.6 units to the right of *a*. This position gives *b-c* as 1.4, the observed distance. We now have the following map



If *d* is to the left of *b*, then *d-c* should be greater than 1.4, a result not seen. Therefore, *d* is 0.2 units to the right of *b*.

31. | 0.04 | 0.02 |
 2 1 3

To calculate map distance, you must have the number of recombinations and the total number of progeny. Since all phages grow on strain B, this number must equal the total number of progeny; this is 250×10^7 . Since only wild-type phages grow on K12, and since wild-types result from recombination between two genes,

$$\frac{a \quad b^+}{a^+ \quad b} \xrightarrow{X} \underline{a \quad b} \text{ and } \underline{a^+ \quad b^+},$$

the number that grow on K12 must be recombinants. But this number represents only half of the recombinants, for the double mutant will not grow on K12. Total recombinants are:

Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

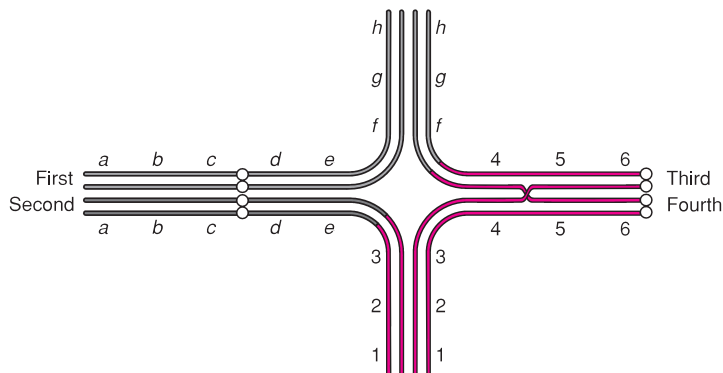
A-9

$$\begin{array}{ll} 1 \times 2 & (2 \times 50)(10^4) = 10^6 \\ 1 \times 3 & (2 \times 25)(10^4) = 5 \times 10^5 \\ 2 \times 3 & (2 \times 75)(10^4) = 1.5 \times 10^6 \end{array}$$

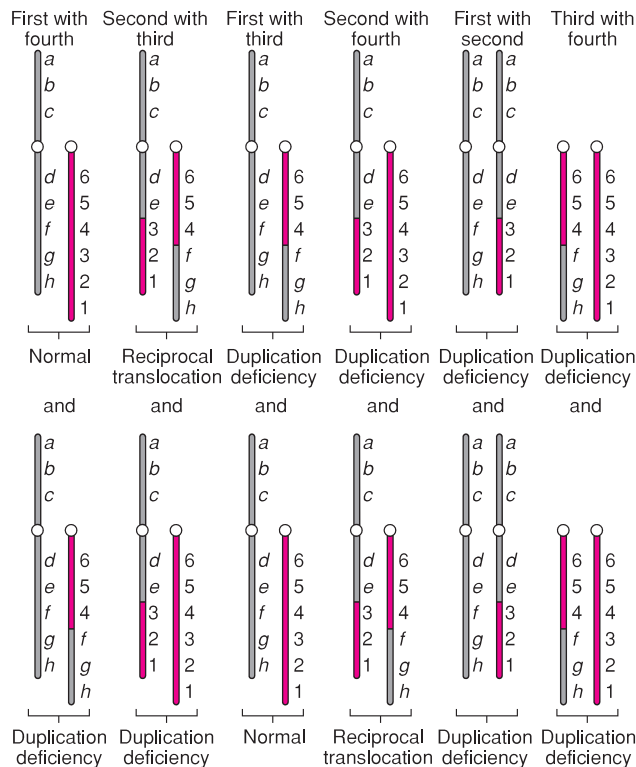
$$\begin{aligned} \text{map distance } 1-2 &= \frac{(100 \times 10^4)}{(250 \times 10^7)} \times 100 \\ &= (0.4 \times 10^{-3})(100) = 4 \times 10^{-2} = 0.04 \end{aligned}$$

$$\begin{aligned} \text{map distance } 1-3 &= \frac{(50 \times 10^4)}{(250 \times 10^7)} \times 100 \\ &= (0.2 \times 10^{-3})(100) = 2 \times 10^{-2} = 0.02 \end{aligned}$$

$$\begin{aligned} \text{map distance } 2-3 &= \frac{(150 \times 10^4)}{(250 \times 10^7)} \times 100 \\ &= (0.6 \times 10^{-3})(100) = 6 \times 10^{-2} = 0.06 \end{aligned}$$



Alternate segregation Adjacent-1 segregation Adjacent-2 segregation



33. Use replica-plating on selective media with arabinose as the sole carbon source, thus selecting for ara^+ cells. Although all three loci can be cotransduced, the rarity of $ara^+ leu^- ilvH^+$ indicates leu is the middle locus ($ara leu ilvH$). Cotransductance frequencies:

$$ara \text{ to } leu = (9 + 340)/(9 + 340 + 32) = 0.92$$

$$ara \text{ to } ilvH = 340/(340 + 32 + 9) = 0.89$$

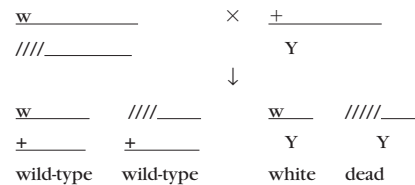
35. $azi^f leu^+ thr^+$. The cotransduction frequency of leu and azi indicated leu is closer to azi than to thr . Thus, two orders are possible: $leu azi thr$ or $azi leu thr$. If the first order is correct, and leu^+ is selected, $leu^+ azi^f > leu^+ thr^+$; this prediction fits. However, with the same gene order, and selecting for thr^+ , $thr^+ azi^f > thr^+ leu^+$; this result is not seen, and the order must be $azi leu thr$. The second order predicts $thr^+ leu^+ > thr^+ azi^f$.

Critical Thinking Question:

- At first, with much smaller data sets, scientists came up with branching models of the bacterial chromosome. However, with the very large data set of table 7.4, it is almost impossible to come up with a reasonable mechanism other than a circular bacterial chromosome.

Chapter 8 Cytogenetics

- All chromosomes form linear bivalents. The cross-shaped figure is seen only in heterozygotes.
- No, there are no inversion loops formed in homozygotes.
- A diagram will show that a crossover between a centromere and the center of the cross can change the consequences of the pattern of centromere separation. For example, in the figure at left, we diagram a crossover between loci 4 and 5, as in figure 8.11.
- Reciprocal translocation (some effects occur only in the heterozygous condition). Look for the cross-shaped figure at meiosis or in salivary gland chromosomes.
- Assume crossovers as shown (following page, top left).
- All females should get a wild-type X chromosome from their father. Irradiation produces chromosomal breaks, so a deletion of part of the X is possible, producing a situation of pseudodominance. (Alternatively, the offspring could have gotten a mutant X^w from the father.)
 - Diagram the crosses (a “/” represents a deleted part of the chromosome and “—” = X chromosome)

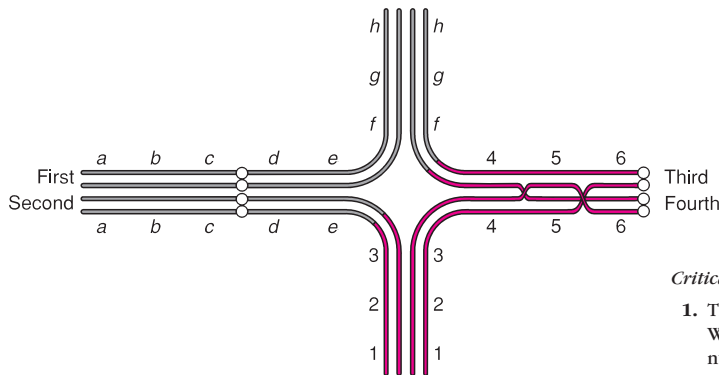


We expect all wild-type females and all white-eyed males, but in a ratio of 2 female:1 male.

- We see that the F_2 offspring from cross A $\times$ B yields fewer progeny than the other crosses. Something unusual must be involved. One explanation is that one of the strains is homozygous for a reciprocal translocation. The translocation, when heterozygous, results in some inviable gametes or progeny, and thus reduces the number of progeny.
- We expect to see about 32% recombination between these two genes, but we see only 2%. The most likely explanation is that an inversion occurred so that these two genes came to lie close to each other; the stocks are homozygous. Since semisterility is not reported, we are probably not dealing with crossover suppression in inversion heterozygotes.

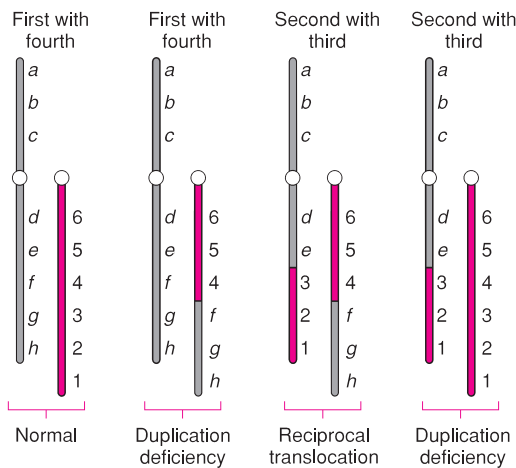
A-10

Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions



Critical Thinking Question:

- The numbers are all multiples of 14 (1, 2, 3, 4, 5, 6, 7, and 8 copies). We can thus hypothesize that the original diploid chromosome number ($2n$) is 14. The other species would be polyploids, multiples of the original 14 (tetraploid, hexaploid, etc.). These are all the even ploids up to 112 chromosomes. As we saw, even ploids have the potential to succeed in meiosis, whereas odd ploids rarely do.

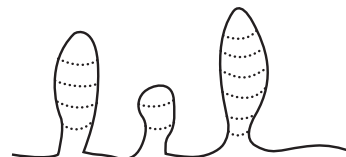


Chapter 9 Chemistry of the Gene

- The genetic code would somehow be read in number of tetranucleotide units, in which each unit consists of one each of the four bases (G, C, T, A). For example, one unit might be the amino acid alanine, two units might be the amino acid arginine, and so on.
- Sugars: DNA has deoxyribose, RNA has ribose; and bases: DNA has thymine in place of uracil, RNA has uracil in place of thymine.
- See figures 9.18 and 9.19.
- 28% G; 28% C; 22% A; 22% T.
 - Same percentages except 22% U, 0% T. Chargaff's rule states that the quantity of A = T and the quantity of G = C. If G = 28%, then C = 28% and G + C = 56%. The sum of all bases must equal 100%. Therefore, (A + T) = 100 - 56 = 44. Since A = T, $1/2(44) = 22\%$. This is the amount of both A and T. For an RNA molecule, proceed the same way, except remember that U replaces T, so we have 22% U.
- There must be regions of complementarity within the single-stranded regions. A melting temperature indicates some regions that are double-stranded. We can envision at least two different possible configurations:
 - Whole molecule complementary



- Fragment complementary



In fact, most single-stranded molecules have some regions that are complementary.

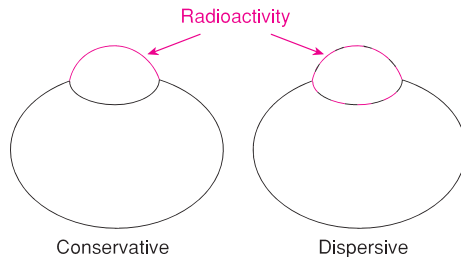
- 19.1 billion base pairs. One base takes up 3.4 Å. Since 1 Å is 1/10,000,000 millimeter, there are $10,000,000/3.4 = 2.94$ million bases per millimeter. Multiply by 1,000 to get to meters, and finally by 6.5 for 6.5 meters: $= 1.91 \times 10^{10}$ bases.

- A translocation from the tip of the normal X in the male to the Y. We expect all males to receive an X chromosome with the white-eye allele from the female. For the male to be wild-type, we still must have part of the wild-type X chromosome. To test, cross this wild-type male with white-eyed females. All the female progeny should be white-eyed and all the male progeny red-eyed. Cytological examination of the chromosomes should reveal the translocation.
- $4n = 92$; $2n - 1 = 45$.
- $n = 8 + n = 6$ equals $14 \times 2 = 28$; $20 + 20 = 40$
- An XO/XXY mosaic can occur by nondisjunction of the Y chromosome in a cell during early cleavage in an XY individual. An XX/XXY mosaic can come about if one of the cells during early cleavage in an XX zygote is fertilized again by a Y-bearing sperm. Trisomy 21 usually comes about from an egg with two copies of the chromosome; the egg had two copies because of meiotic nondisjunction.
- The father. The allele for color blindness can only come from the mother. If meiosis in her is normal, an egg could get the X chromosome carrying the mutant allele. The daughter has only one X chromosome, so the sex chromosomes failed to separate in the man, and a sperm with neither X nor Y fertilized the egg.
- The first meiotic division in the father is normal, producing cells with either two X or two Y chromatids. During the second meiotic division in the cell with the two Y chromatids, both Y chromatids move to the same pole and end up in the same sperm cell.

Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

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13.

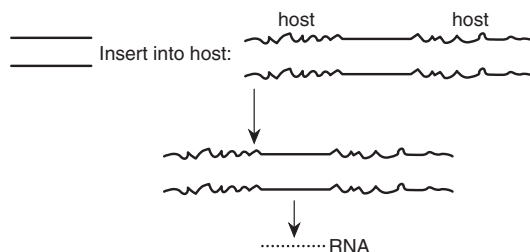
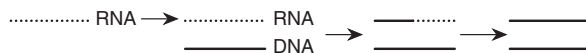


15. A primosome is a helicase plus a primase; it opens the DNA and creates RNA primers on lagging strands and is part of the replisome. A replisome includes a primosome plus two copies of DNA polymerase III; it coordinates replication on both the leading and lagging strands at the Y-junction.

17. See figure 9.28.

19. At one time molecular swivels, presumably protein in nature, located periodically along the DNA, were suggested.

21.



This question really asks how double-stranded DNA can be formed from single-stranded RNA. First, we could synthesize a complementary DNA strand, then begin to make a second DNA strand complementary to the first. While synthesizing the second DNA strand, we begin to degrade the RNA. The double-stranded DNA molecule now inserts into the host DNA. In order to get more viruses, a single-stranded RNA molecule must be made from the DNA. We would need, at a minimum, an enzyme to make a single-stranded DNA to form a hybrid with the RNA; an enzyme to degrade the RNA and to make the second, complementary DNA strand; an enzyme to cut the host DNA to allow the viral DNA to insert itself; an enzyme to ligate the two molecules; and an enzyme to make more viral RNA. The earlier functions are performed by one viral enzyme, reverse transcriptase.

23. Finding small pieces or fragments of DNA suggests the Okazaki pieces are only slowly, if at all, joined, a function of DNA ligase. The fact that not many long DNA molecules are seen also suggests that the DNA is being broken, implicating a nuclease as well.

25. It is unlikely that bases are added faster in developing embryos. So we must look for another mechanism. If there are more replicons, and hence more origins of DNA replication, each replicon will be shorter and be able to duplicate faster. Alternatively, and more likely, the process is regulated to slow down adult division.

Critical Thinking Question:

- One way to study mutations that are generally lethal is by isolating temperature-sensitive mutations. These mutations involve amino acids that disrupt the functioning of the enzymes at some critical temperature but are phenotypically normal at other temperatures. Thus, the mutant organisms can be kept alive by growing them at one temperature (the *permissive* temperature) but their mutant effect can be studied at the temperature in which the protein function is disabled (the *restrictive* temperature). (For additional discussion of these mutations, see chapter 12.)

Chapter 10 Gene Expression: Transcription

- See figure 10.3. Complementarity is achieved between messenger RNA and ribosomal RNA and between messenger RNA and transfer RNA.
- Transcription has higher error rates. Errors of DNA polymerase tend to become permanent, whereas errors of RNA polymerase do not.
- A consensus sequence is made up of the nucleotides that appear in a significant proportion of cases when similar sequences are aligned. A conserved sequence consists of nucleotides found in all cases when similar sequences are aligned. For example, the Pribnow box (fig. 10.6) is the consensus sequence TATAAT.
- See figure 10.8 for a promoter and figure 10.10 for a terminator. The transcript starting from the promoter would be 5'-CUUAUACGGU.... The transcript from the terminator is shown in figure 10.10.
- A stem-loop structure can form when a single strand of DNA or RNA has a double helical section (see fig. 10.10). An inverted repeat is a sequence read outward on both strands of a double helix from a central point (see fig. 10.10). A tandem repeat is a segment of nucleic acid repeated consecutively; that is, the same sequence repeats in the same direction on the same strand:

5'-TCCGGTCCGGTCCGG-3'

3'-AGGCCAGGCCAGGCC-5'

A DNA sequence with a seven-base inverted repeat is

5'-ATTACGCGGTAAT-3'

3'-TAATGGCGCCATTA-5'

- Footprinting is a technique in which DNA in contact with a protein is exposed to nucleases; only DNA protected by the protein is undigested. Promoters could be isolated by protection with RNA polymerase in the absence of ribonucleotides—the polymerase will not move—and then sequenced.
- The superscripts of the sigma factors refer to their molecular weights (e.g., σ^{70} is 70,000 daltons). Different sigma factors usually recognize different prokaryotic promoters.
- 3'-GGTAGTACTGTCTGGGAACGATTGCG-5' ←
 5'-CCATCATGACAGACCCCTTGCTAACGC-3'

Begin by writing the strand that is complementary to the RNA. This will be the transcribed strand. Remember, U in RNA pairs with A in DNA. Since transcription proceeds 5' → 3', the 5' end of the RNA is opposite the 3' end of the DNA.

- The bottom strand is transcribed and the molecule is arranged as



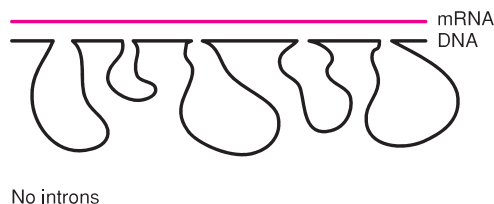
Begin by writing the RNA that could be transcribed from each strand. Since the DNA represents the beginning portion of the gene, the RNA must have an AUG to start protein synthesis. Unfortunately,

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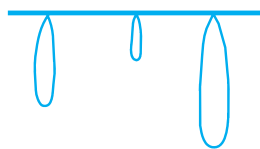
Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

both strands yield RNAs with one or more AUGs. The RNA from the top strand has AUGs in both directions, but in each case the AUG is followed by a termination signal, UAA, or UAG. This RNA could not make a protein. The bottom strand produces an RNA with only one AUG. Since transcription and protein synthesis both proceed $5' \rightarrow 3'$, the left end of the bottom strand must be $3'$.

19. If transcription of the genes is rho-dependent, the RNAs made at 40°C will be longer than those made at 30°C . Since the rho cannot function at the high temperature, RNA polymerase will read past the termination region. If transcription is rho-independent, a rho mutant will have no effect on transcription, and hence, the size of the RNAs.
21. The double helix must unwind in order for transcription to occur. A-T pairs, because they have only two H-bonds, are more easily disrupted than G-C pairs.
23. Removing one base too many or too few would result in a shift in the reading frame during translation (see chapter 11), thus radically altering the protein product.
25. Five introns



27. Group I introns are self-splicing introns that require a guanine-containing nucleotide for splicing. Group II introns are similar but do not require an external nucleotide for splicing. Group I and II introns are released as linear and lariat-shaped molecules, respectively.
29. Spliceosomes are composed of small nuclear ribonucleoproteins (see fig. 10.37).
- 31.



There are three introns, so we expect three single-stranded DNA loops. The coding regions (exons) form RNA-DNA hybrids and appear thicker.

33. Enhancers [E] bind activator proteins that also bind proteins of the polymerase at the promoter [P]. In some eukaryotic genes, there are many enhancers, allowing numerous levels of control and forcing enhancers further and further upstream. For activators bound to enhancers to bind polymerase proteins, the DNA must loop around.

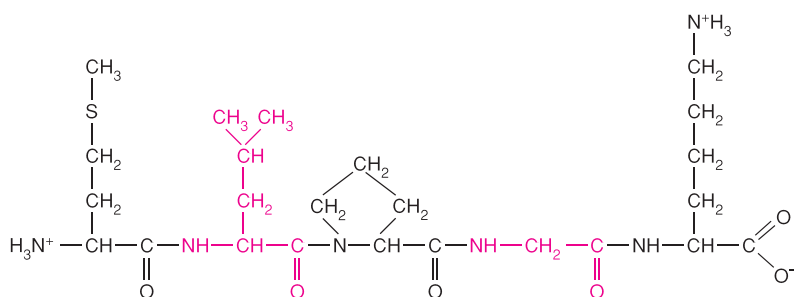


Critical Thinking Question:

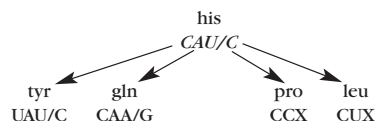
1. Given that a gene controls the production of a protein, there are both realistic and theoretical limits to the size of a gene. From what we know about the functioning of the centromere, a gene would have to occupy no more than the length of a chromosomal arm. However, given that human chromosomes must contain about fifty thousand genes, it is unlikely that any gene is that large. We also know that many functioning proteins are made up of subunits, each controlled by its own gene. Thus, large proteins tend to be conglomerates of smaller ones rather than large functional units. Finally, the larger the protein the more time it takes to transcribe and translate it, making very large size inefficient. As we mentioned before, the average protein is about 300 to 500 amino acids; with introns, and control elements, the gene for an average protein could be quite large. The largest known gene is the human dystrophin gene that codes for a cytoskeletal protein. It is 2,300,000 bases long, has 79 exons, and takes 16 hours to be transcribed.

Chapter 11 Gene Expression: Translation

1. The messenger RNA is $5'\text{-AUGUUACCGGGAAAAUAG-3'}$; the anticodons are $3'\text{-UAC-5'}$, $3'\text{-AAU-5'}$, $3'\text{-GGC-5'}$, $3'\text{-CCU-5'}$, $3'\text{-UUU-5'}$; the amino acids are methionine, leucine, proline, glycine, lysine (see the figure below).
3. See figure 11.16. Use the messenger RNA of problem 1 and be sure to include EF-Tu and EF-Ts.
5. There are approximately twenty aminoacyl-tRNA synthetases in an *E. coli* cell, one for each amino acid. Recognition signals can occur at any point on a given transfer RNA, although the anticodon figures prominently in most.
7. See figure 11.7.
9. Three; see figure 11.29.
11. $5'\text{-UAA-3'}$ $\rightarrow$ $5'\text{-UUA-3'}$ (leucine). The consequence is the failure to terminate the particular protein leading to continued chain elongation to the next nonsense codon or to the end of the messenger RNA. The result is probably a nonfunctioning enzyme or protein.



13. EF-Tu brings a charged transfer RNA to the A site at the ribosome. EF-Ts is involved in recharging EF-Tu (see fig. 11.14). The eukaryotic equivalents are eEF1 α and eEF1 β .
15. A signal peptide is a sequence of amino acids at the amino-terminal end of a protein that signals that the protein should enter a membrane (see fig. 11.25). Although the concept is the same, the situation in eukaryotes is somewhat more complex because there are many different membrane-bound organelles, each having their own membrane-specific requirements. Signal peptides are usually cleaved off the protein after the protein enters or passes through the membrane.
17. NH₂-FGKICABHLNOEDJM-COOH
19. a. 5'-AUG AUU GAA UGC GAG CGG AGU-3'
b. N-met-ile-glu-cys-glu-arg-ser
- First determine the sequence of the RNA complementary to the given DNA strand. Don't forget about polarity; as the strand is written, the 5' end of the RNA will be on the left. Blocking off successive groups of three bases allows the determination of the codons. Use the code to determine the amino acid sequence.
21. 12/27 phenylalanine (UUU, [2/3]³; UUC, [2/3]²[1/3]); 6/27 serine (UCU, [2/3]²[1/3]; UCC, [2/3][1/3]²); 6/27 leucine (CUU, [2/3]²[1/3]; CUC, [2/3][1/3]²); 3/27 proline (CCU, [2/3][1/3]²; CCC, [1/3]³).
23. The table could look the same (see table 11.4) except that the position would be left side = first position (5' end); top = third position (3' end); right side = second position. For example, the codons for valine (currently 5'-GUU-3', 5'-GUC-3', 5'-GUA-3', and 5'-GUG-3') would be 5'-GUU-3', 5'-GCU-3', 5'-GAU-3', and 5'-GGU-3'.
25. We are mixing two RNA strands that are complementary; these strands will form a double-stranded RNA molecule. Since we observed the incorporation of no amino acids, the ribosome must not be able to read a double-stranded molecule.
27. If we write out (GUA)_n as GUA GUA GUA GUA ..., we see that we could use any of three different reading frames: GUA, UAG, or AGU. Since we see only two amino acids incorporated, either two of the possible codons code for the same amino acid, or one of the codons is a stop codon. If you look at the code, you will see that UAG is a stop codon.
29. The stop codon has probably mutated to give a codon for the amino acid leucine. The longer-than-normal protein suggests that the original stop was not read. Numerous possibilities exist. If the second letter of a stop codon were changed to a U, we would have UUA or UUG leucine codons. Alternatively, the insertion of a C before the U would yield CUA, CUG, or CUA as leucine codons. Similarly, an insertion of a U next to the first G yields UUA or GUG as leucine codons. Since the next amino acid is phenylalanine, the next codon must be UUC or UUU. If a base were added, as above, the next codon would have to begin with A or G, and phenylalanine does not begin with A or G. Therefore, the most likely explanation is a change of the second letter from an A to a U.
31. Either CAU or CAC. Write down all possible codons for each amino acid.



For *leu*, note that UUA/G cannot result from a single change in the *bis* codon. Therefore, leucine must be CUX. All of the other amino acids could result from single changes in either the first or second base, and we are left with either codon being the one for *bis*.

Critical Thinking Question:

- In general, transcriptional and translational signals are independent. We could look at this by asking the question, how does changing one of the signals affect the other process? In other words, if we changed a translational signal such as a start or stop codon, would that affect the transcription of that gene? In general, the answer is no.

Chapter 12 DNA: Its Mutation, Repair, and Recombination

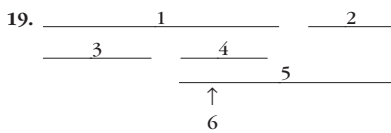
- For example, if the twenty individual cultures of table 12.1 had values of 15, 13, 15, 20, 17, 14, 21, 19, 16, 13, 27, 14, 15, 26, 12, 21, 14, 17, 12, 14, then the mutation theory would not have been supported because the variation between the individual and bulk cultures would not have been different.
 - Reading across each row, we gather more and more information.
Row 1: 1, 6, and 7 are part of one complementation group.
Row 2: 2 and 5 are part of one complementation group.
Row 3: 3 and 4 are part of one complementation group.
Row 4: no new information.
Row 5: no new information.
Row 6: reinforces that 6 and 7 are part of the same complementation group.
Row 7: no new information.
- Thus, we conclude that there are three complementation groups present: 1, 6, and 7 are mutually noncomplementing, as are 2 and 5, and 3 and 4. The half-table missing is a mirror image because a cross of 1 and 3 is the same as a cross of 3 and 1 (reciprocity). The diagonal always contains negative elements because every mutant is a functional allele of itself.
- All mutants should be crossed in pairwise combinations yielding heterozygote daughters. (Presumably, earlier crosses indicated that these are X-linked loci.) All F₁ daughters will be wild-type: The mutations complement and therefore are not alleles. When eosin flies are crossed in a similar fashion, daughters will be wild-type except when the parents were eosin and white. In that case, daughters will be mutant, showing the lack of complementation and hence that the mutations are alleles.

- Prokaryotic and phage genes generally do not have intervening sequences. Benzer and Yanofsky worked at a time when introns were unknown and it was assumed that the length of a gene was transcribed and then translated. If the genes had introns, Benzer and Yanofsky would have been unaware of them; introns would not affect colinearity or mapping. Introns would affect physical measures of the lengths of DNA, with which Benzer and Yanofsky were not involved.
- The *rex* gene of phage λ represses growth of phage T4 *rII* mutants.
- In replicating DNA, a transition mutation can occur by tautomerization of a base in the template strand (template transition) or entering the progeny strand (substrate transition).
- 5-bromouracil (pyrimidine analogue) and 2-aminopurine (purine analogue) are incorporated into DNA as thymine and adenine, respectively. However, each undergoes tautomeric shifts more frequently than the normal base. Both cause transitions. Nitrous acid also promotes transitions by converting cytosine into uracil, which acts like thymine, and adenine into hypoxanthine, which acts like guanine. Proflavin induces insertions and deletions by intercalating and buckling DNA. Ethyl ethane sulfonate removes purine rings and thus promotes transitions and transversions.

- See figure 12.25.

A-14 Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

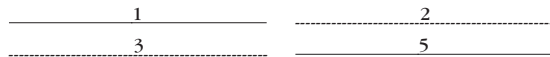
17. Three genes. Gene *A*: mutants 1, 4, 8; gene *B*: mutants 2, 5; gene *C*: mutants 6, 7. Mutant 3 probably contains a deletion that spans genes *A* and *C*. Begin by finding mutants that do not complement. These should have mutations in the same gene. Mutants 2 and 5 are in the same gene. Initially, we suspect that mutants 1, 3, 4, and 8 are in the same gene, which is different from the gene that contains 6 and 7. If mutant 3 is in gene *A*, it should complement 6 and 7, and it does not. One explanation is that 3 is a deletion spanning genes *A* and *C*. Alternatively, mutant 3 could be in gene *A* but be a polar mutation. Either possibility implies that the order is *B A C*.



Begin with deletions that yield mostly “–”s. These must be large deletions that cover most of the other deletions. Mutations 1 and 5 are such mutations. Since they give no wild-type, they must overlap:



Now look at mutant 2. It gives wild-type recombinants with 1 but not 5. Therefore it must overlap the region deleted in 5. Mutant 3, by similar logic, must cover part of deletion 1. We can draw these results as follows. Broken lines indicate we do not know yet how long the deletion is:



Now look at 4. It gives “–” (no wild-type recombinants) with both 1 and 5 and therefore must be in the region that 1 and 5 overlap. If 4 extended to and overlapped either 2 or 3, we expect to see “–” with them. Since this prediction is not met, 4 must be a small deletion spanning at least part of the overlap of 1 and 5. Since mutant 6 gives no wild-type with 1, 4, or 5, it must be within the common region deleted in all three strains.

21. We know that the anticodon pairs with the codon, and we expect nonsense suppressors to contain an altered anticodon. The fact that the nonsense codon can be read by a transfer RNA with a normal anticodon but altered dihydrouridine loop suggests that the way in which this loop interacts with the ribosome causes the anticodon sequence to be misread.
23. $x^+ \rightarrow x$: $AT \rightarrow GC$. $y^+ \rightarrow y$: $GC \rightarrow AT$. This problem requires logic and a knowledge of how mutagens work. For x , the key is the response to HA, which only causes $GC \rightarrow AT$ transitions. Mutant x is reverted by HA, therefore, x must be GC , and the normal x^+ was AT . AP-induced mutations can also be reverted by AP. Since y is not reverted by HA, y must be AT . Therefore, y^+ must be GC .
25. A deletion that spans regions of more than one gene. Consider the following two genes:



By definition, mutations in gene 1 will complement mutations in gene 2 but not other mutations in gene 1. If we have a deletion, x , that covers part of both 1 and 2 (slashes),



presumably gene 1 will be nonfunctional because it is missing the last part of the protein. Gene 2 will be nonfunctional because the

beginning portion of the gene is missing. The following genotypes will give no complementation:

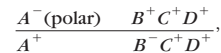


No functional product of gene 1 No functional product of gene 2

We could also get a lack of complementation when mutants are in two genes if we have a bacterial operon in which one of the genes contains a polar mutation creating a transcription stop signal. Such a mutation eliminates all distal functions. Thus, if the operon is



and we construct the partial diploid



we will get no complementation because the top DNA is effectively $A^- B^- C^- D^-$.

27. The auxotroph probably contains a deletion. If a few bases are missing, nothing is available to cause transitions or transversions. It is highly unlikely that the correct number of missing bases could be spontaneously and correctly inserted.
29. Excision repair endonucleases can recognize dimerizations, mismatched bases, and apurinic-apyrimidinic sites.
31. See figure 12.38 for a diagram of recombination. Branch migration is shown in figure 12.39.

Critical Thinking Question:

1. The gene is a linear entity that specifies the linear order of amino acids in a protein in a colinear fashion. Although scientists in the 1960s were convinced of colinearity, there were other alternatives possible. For example, DNA could be a branching structure. Or, transcription of DNA could take place such that the beginning, middle, and end of the gene were not in order. Thus colinearity supported our understanding of the shape and functioning of DNA.

Chapter 13 Genomics, Biotechnology, and Recombinant DNA

1. Type II endonucleases are valuable because they cut DNA at specific points and many leave overlapping or “sticky” ends.
3. In DNA with a random sequence, a four-cutter will find sites approximately once in 4^4 bases ($= 1/256 = 0.0039$). A six-cutter will find sites approximately once in 4^6 bases ($= 1/4096 = 0.0002$). An eight-cutter will find sites approximately once in 4^8 bases ($= 1/65,536 = 0.000015$).
5. DNA can be joined by having compatible ends to begin with or by blunt-end ligation (linkers combine these methods). The appropriateness of a method depends on what DNA is to be cloned and how that DNA can be obtained. Having DNA with “sticky” ends created by the same restriction enzyme would be easiest but sometimes is not available. Adding linkers by blunt-end ligation with a particular restriction site is usually the best compromise.
7. A plasmid is a self-replicating circle of DNA found in many cells. Foreign DNA inserted into a vector forms an expression vector if that foreign DNA produces a protein product. Cosmids are plasmids that contain *cos* sites and are useful for cloning large segments of DNA (up to 50 kb). YACs, yeast artificial chromosomes, have the loci to replicate in yeast (centromere, replication origin, and telomeres). They can be used to clone very large pieces of DNA, upwards of one million bases.

Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions

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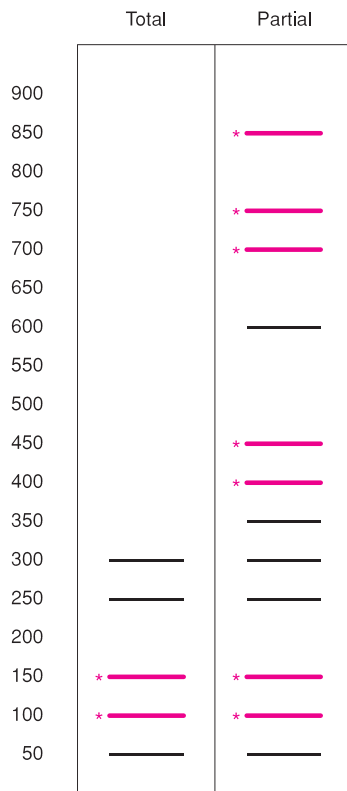
9. Chromosome walking is a technique for cloning overlapping chromosomal regions starting from an arbitrary point (see fig. 13.38). It is useful for determining relative locations of genes in uncharted regions as well as cloning regions too big to fit in a single vector.
11. Southern and northern blotting are gel transfer techniques used to probe for DNA and RNA sequences, respectively. Western blotting is a technique used for locating a protein by antibody recognition. Dot blotting is a probing technique for cloned DNA that eliminates the electrophoretic separation step.
13. Plasmids of *E. coli* origin survive in yeast when a yeast centromere (CEN region) is added, allowing them to replicate within the yeast cell. Inactivated SV40 viruses can function in the presence of intact "helper" viruses that allow them to complete their life cycles. Phage λ has parts of its chromosome that can be removed while still allowing it to complete its life cycle.

15. Partial digestion of molecule 2 leads to the following molecule:

AAAAAAAA
TTTTTTTT

Some of these molecules will form a circle with the single-stranded Ts paired with single-stranded As. The circle eliminates the free 5' phosphate, and the enzyme can no longer work.

17. We must insert DNA that has no introns into bacterial plasmids. This DNA can be obtained by isolating mature, cytoplasmic messenger RNA and then using reverse transcriptase to make double-stranded cDNA. Plasmids with cDNA inserted can then be used to produce human proteins (expression vectors).
19. Electrophoretic bands of the total digest are (* indicates end label) 50, 100*, 150*, 250, 300 bp. Bands of the partial digest are 50, 100*, 150*, 250, 300(×2), 350, 400*(×2), 450*(×2), 600, 700*, 750*, 850* bp.



21. Mutant A: elimination of site between the 300- and 50-bp segments. Mutant B: elimination of site between 100- and 300-bp segments. Mutant C: creation of a new site within the 300-bp segment, dividing it into 75- and 225-bp segments.
23. This problem begins as a trial and error attempt to overlay two restriction maps, made more difficult by the fact that one enzyme, *Bam*HI, has made three cuts that are unordered, leaving many possibilities. However, a bit of thought beforehand makes this problem much easier. If you compare the double digest with the *Bam*HI digest, they share 200-, 250-, and 400-bp segments. The double digest has 50- and 100-bp segments replacing the 150-bp segment in the *Bam*HI digest. The inference is that there is an *Eco*RI cut in the 150-bp segment leading to the 50- and 100-bp segments, with all other segments of the *Bam*HI digestion left uncut. That leaves only two possibilities, as shown below; the data are insufficient to distinguish between the two choices.

<i>Eco</i> RI				<i>Eco</i> RI			
300	↓	700		300	↓	700	
200	↑	150	↑	250	↑	150	↑
[any order]				[any order]			
<i>Bam</i> HI				<i>Bam</i> HI			

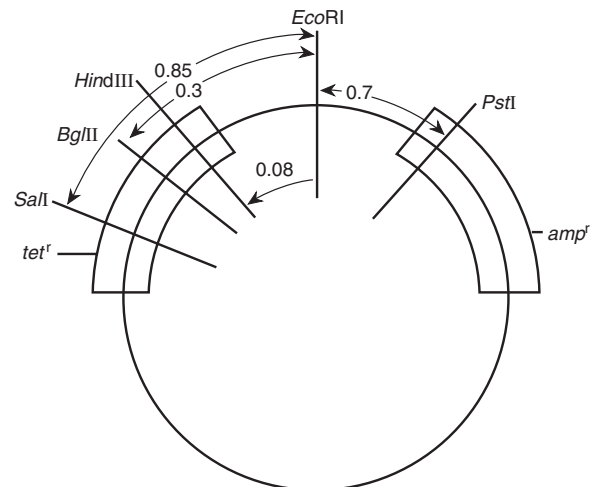
<i>Eco</i> RI	6.2	2.8	4.6	7.4	8.0
<i>Bam</i> HI	10.0		13.0		6.0

We know that the 6.2 and 8.0 kb *Eco*RI fragments are at opposite ends, and that the 10.0 and 6.0 kb *Bam*HI fragments are at the ends. Therefore, the *Bam*HI 13.0 kb fragment must be in the middle. If the 6.2 and 6.0 kb fragments are at the same end, a double digest should produce a fragment of 0.2 kb. This is not seen, so they are at opposite ends:

6.2	(2.8, 4.6, 7.4)	8.0
10.0	13.0	6.0

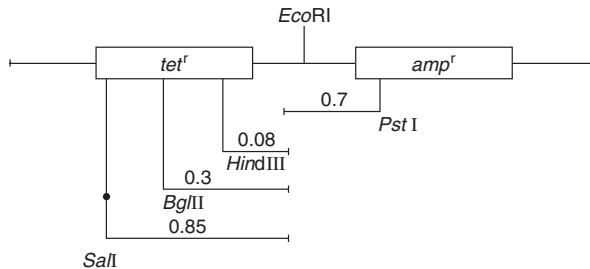
If 7.4 is next to 6.2, we should see a 3.8 fragment. Similarly, if 4.6 is next to 6.2, we should see a 3.8 fragment. We see neither of these fragments, so 2.8 is next to 6.2. The 4.6 fragment must be next.

27. Since *Eco*RI does not eliminate either resistance, its site must be between the *tet*^r and *amp*^r genes. The *Pst*I site must be within the *amp*^r gene, since insertion of DNA into this site eliminates ampicillin resistance. By similar logic, the other sites must be in the *tet*^r gene. In the double digests, the smaller fragment must represent the distance from the *Eco*RI site to the other site. We can draw part of the plasmid as:



A-16

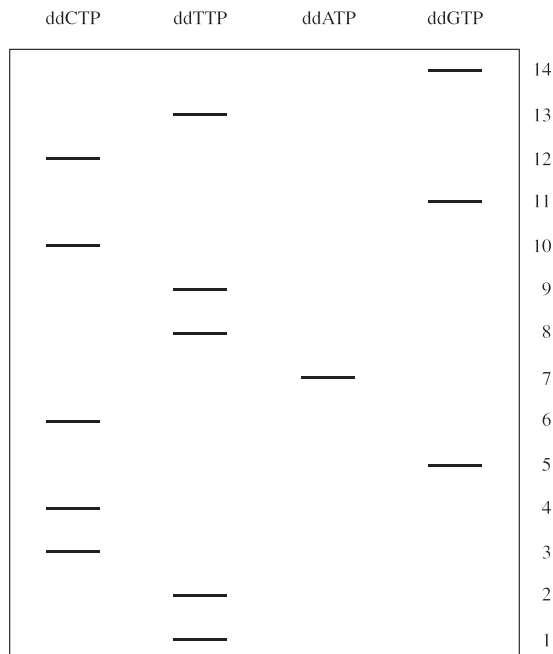
Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions



(b)

If this arrangement is correct, digestion with *Bgl*II, *Eco*RI, and *Pst*I should yield 0.3 kb (*Bgl*II + *Eco*RI), 0.7 kb (*Eco*RI + *Pst*I) + 2.0 kb (*Pst*I + *Bgl*II), and this is, in fact, observed.

29. The second possibility predicts that a 1.5-kb fragment should be seen in the double digestion. This fragment would result from digestion of the 3.5-kb *Eco*RI fragment with *Bgl*II. Since we don't see a 1.5-kb fragment in the double digest, the second possibility does not agree with the observed results.
31. Two of the three number 21 chromosomes present in the child came from the father, not the mother. Since the probe produces different bands in the mother and the father, all of these bands must also be present in the child. The intensity of a band is proportional to the amount of DNA present. The bands that are of paternal origin are more intense than the maternal bands. The father contributed two number 21 chromosomes.
33. For the steps in the dideoxy sequencing method, see figures 13.32 and 13.33. Use of fluorescent dyes has allowed for the automation of the process and the elimination of radioactive tags.
35. The DNA can be inserted into the M13 general sequencing vector.
37. What will appear in the gel are fragments of the newly synthesized strand. Since DNA synthesis proceeds 5' → 3', the 5' base will be T in the new strand. Proceed up the gel by indicating the base complementary to the sequence given.



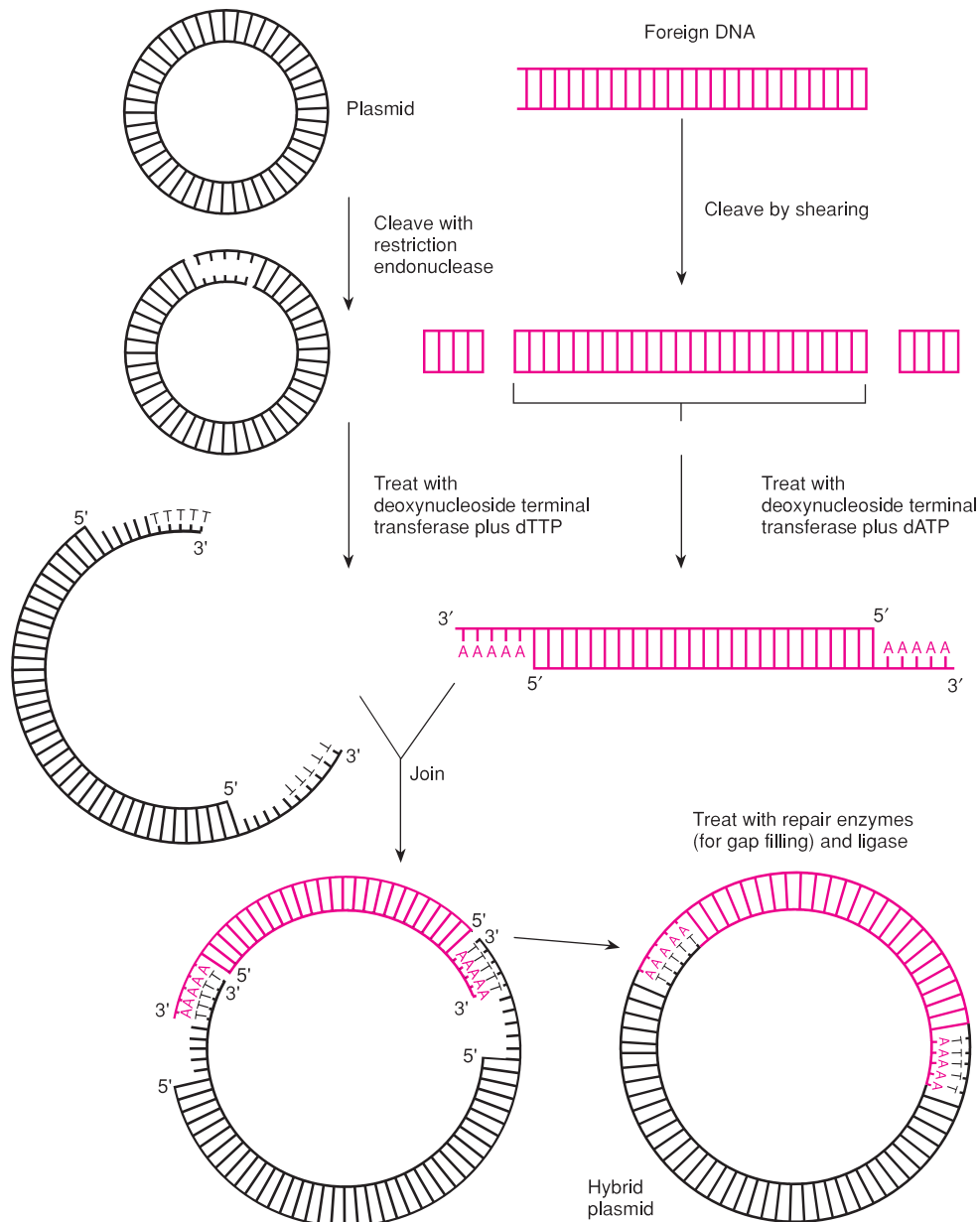
39. Hypervariable DNA is DNA showing a great deal of interindividual variation. A RFLP (restriction fragment length polymorphism) is a polymorphism (variation) that shows up after Southern blotting and probing of restriction digests. A VNTR (variable number of tandem repeats) locus is one that is hypervariable due to unequal crossing over among the tandem repeats. VNTR loci are a hypervariable subset of RFLPs. Sequence-tagged sites are unique sites in the genome that can be amplified with polymerase chain reaction. Microsatellite DNA, repeats of very short segments such as CA, forms VNTR loci that are usually examined by polymerase chain reaction (PCR) if primer sequences are known.

Critical Thinking Question:

1. Sticky ends would exist if the plasmid DNA had a 3' overhang of one nucleotide residue, while the foreign DNA had a 3' overhang of its complement. The exact number of bases would not need to be the same because repair enzymes plus ligase could close the gap. For example, the plasmid could have a 3' tail of thymines added, whereas the foreign DNA could have a 3' tail of adenines added. This method (see figure next page) was called the poly-dA/poly-dT technique.

Chapter 14 Gene Expression: Control in Prokaryotes
and Phages

1. a. inducible (wild-type); b–d. constitutive; e. neither, super-repressed; f. inducible
3. One mutant could fail to bind to operator DNA but could still bind the inducer (*op⁻, in⁺*). This mutant would have constitutive transcription of the operon. The reverse situation could also be true; the repressor could bind the operator but not the inducer (*op⁺, in⁻*). This mutant would be off all the time. A third mutant could fail to bind both (*op⁻, in⁻*), being constitutive.
5. a. Operator and repressor. b. Make a partial diploid with wild-type; the operator mutant will make β -galactosidase constitutively, and the repressor mutant will make it only in the presence of lactose. Four mutations are possible in the *lac* operon: mutations in the *z* gene or the promoter never make the enzyme. Mutations in the repressor always make the enzyme because the repressor cannot bind DNA. In operator mutations, a good repressor can never bind DNA. In a partial diploid, *i⁻ o⁺/i⁺ o⁺*, the wild-type repressor is *trans* acting and can bind to both operators, creating an inducible situation. In *i⁺ o⁻/i⁺ o⁺*, repressor cannot bind to *o⁻*, and this DNA is always on, even though the wild-type DNA is off in the absence of lactose.
7. Cyclic AMP, combined with CAP protein, attaches to CAP sites enhancing transcription of nonglucose, sugar-metabolizing operons in *E. coli*. Glucose inhibits its formation by inhibiting adenylcyclase.
9. We must think about how these operons are controlled. Not only do they need inducer, but they also require the catabolite repression-activation system. These mutants could be unable to make cAMP because the adenylcyclase gene is defective. Alternatively, they could be making a defective catabolite activating protein (CAP).
11. b = tryptophan synthetase gene, a = operator, c = repressor. Look first for the single mutation that never gives enzyme; this genotype will give the letter of the structural gene. The genotype *a⁺ b⁻ c⁺* fits this requirement, so b is the structural gene, and a and c represent control regions. Genotypes 4 and 5 tell us nothing. Look at genotype 6. The right DNA will never make the enzyme. If c is the operator, the left DNA should always make the enzyme, and this is not seen. Therefore, a must be the operator. Check these assignments with genotype 7. The right DNA will never make the enzyme. If a is the operator, the left DNA is always on. If a is the repressor, the right DNA makes a good repressor that will bind to the top DNA and regulate it.



13. The *E. coli trp* operon functions as a normal repressible operon. In addition, attenuation, based on secondary structure and stalling of the ribosome on the leader transcript, can further prevent transcription (see fig. 14.16). Attenuator control can be exerted based on other amino acids if their codons appear in the leader transcript, causing ribosome stalling.
15. Assuming that the mutants produce inactive proteins: *cl*, *cII*, and *cIII*, lytic response; *N*, neither lytic nor lysogenic responses possible; *cro*, no lytic response possible; *att*, no lysogenic response possible; *Q*, no lytic response possible.
17. The λ chromosome has one circular and two linear forms. The circular form is the infective cellular form. A break at one point (*cos* site) takes place during packaging into the phage head, and

a break at another point forms the linear integrative prophage (see fig. 14.18).

19. The prophage region of the Hfr chromosome enters the F^- cell with no repressor present. The situation is thus similar to regular phage infection, which can go either way (lysogenic or lytic cycles).
21. The cells that form colonies do not contain a prophage. The initial heat shock inactivated the repressor and allowed the prophage to excise. The lower temperature activated the repressor, and the repressor bound to the excised phage DNA, preventing gene expression and reintegration of some of the phages. These "cytoplasmic" phages failed to replicate and were lost during cell division. The cells that retained the virus were lysed when the temperature was raised because the phage could now make RNA.

A-18 *Appendix A Brief Answers to Selected Exercises, Problems, and Critical Thinking Questions*

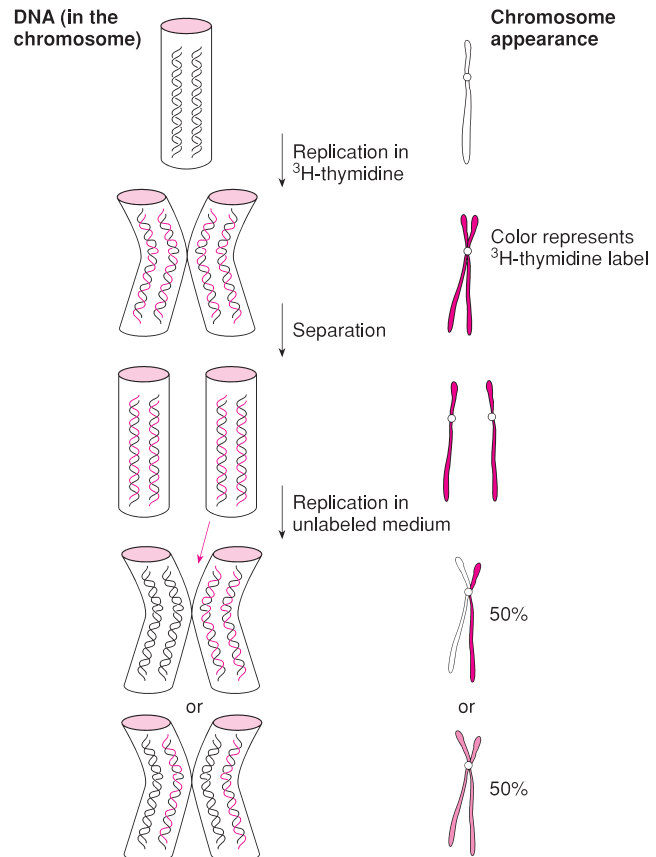
23. An IS element is a simple transposon, which is a segment of DNA that can make a copy of itself to be inserted at another place in the genome. An intron is an intervening sequence, a region excised from messenger RNA before expression. A plasmid is an autonomous, self-replicating genetic particle. A cointegrate is an intermediate structure in transposition.
25. See figure 14.29.
27. See figure 14.35.
29. Transcription is the level at which most control mechanisms work. These include sigma factors, efficiency of promoter recognition, catabolite repression, operon-repressor systems, attenuation, and transposition. Translational control mechanisms include polarity placement, antisense RNA, differences in the efficiency of processes due to nucleotide sequence differences, codon preference, and the stringent response. Posttranslational mechanisms include feedback inhibition and differential rates of protein degradation.
31. Heat shock proteins are normally induced by the presence of a specific sigma factor, which itself is induced by heat shock.
33. The stringent response is the response of a prokaryotic cell to amino acid starvation. The idling reaction of the ribosome results in production of 3'-ppGpp-5', whose appearance is associated with the cessation of transcription, especially of transfer RNAs and ribosomal RNAs, through an unknown mechanism.
35. Feedback inhibition: allosteric enzymes in some synthetic pathways can be inhibited by the end product of that pathway. In addition, many repressor proteins are allosteric.

Critical Thinking Question:

- One might think that a bacterium should metabolize any sugar in its environment. However, glucose is the most efficient sugar to metabolize and if it is present, it should be metabolized first. Even under maximal growth, it takes an *E. coli* cell about 20 minutes to grow and divide. If the cell were to begin to take up and metabolize other sugars, there would be a cost for the manufacture of new enzymes and the inefficiency of the initial steps of the sugar metabolism. In fact, the growth of the cell would slow down under these circumstances and be at an evolutionary disadvantage.

Chapter 15 The Eukaryotic Chromosome

- In general, prokaryotes are small, have a relatively small circular chromosome, and have little internal cellular structure compared to eukaryotes. Most prokaryotic messenger RNAs are polycistronic, under operon control; eukaryotic messenger RNAs are highly processed, monocistronic, and usually not under operon control. Prokaryotes are mostly single-celled organisms, whereas eukaryotes are mostly multicellular. Eukaryotes have repetitive DNA, absent for the most part in prokaryotes. Prokaryotic chromosomes are not complexed with protein to anywhere near the same extent that eukaryotic chromosomes are.
- Assume that each chromosome contained two complete copies of the same DNA. Following the protocol of figure 15.1, the final results would be chromosomes, before separation, that consisted of either two labeled chromatids or only one labeled chromatid, in a 1:1 ratio (barring sister chromatid exchanges). The labeled chromatid in chromosomes with just one chromatid labeled will have twice the label of each chromatid in the chromosomes in which both chromatids are labeled (see the following figure).



- The length of DNA associated with nucleosomes was determined by footprinting, in which free DNA was digested, leaving only those segments protected by nucleosomes. Nucleosome hypersensitive sites are sites not in a nucleosomal state; they seem to be sites involved in the initiation of replication, transcription, and other DNA activities.
- See figures 15.10 and 15.11 for the relationship of the 110, 300, and 2,400 Å chromosome fibers.
- See figure 15.19.
- Polytene chromosomes are chromosomes that underwent endomitosis: They consist of numerous copies of the same chromatid (e.g., in the *Drosophila* salivary glands). Regions of active transcription in polytene chromosomes form diffuse areas called puffs or Balbiani rings (see fig. 15.14). Lampbrush chromosomes occur in amphibian oocytes (see fig. 15.18).
- Satellite DNA differs in its base sequence from the main quantity of DNA and thus forms a satellite band during buoyant density analysis. It is usually centromeric heterochromatin, composed of a highly repetitive DNA.
- Telomeres are repetitive DNA sequences at chromosomal ends. They are repetitions of a five- to eight-base sequence. Most telomeres are G-rich. Telomeres protect the ends of chromosomes and probably provide signals on senescence of cells.
- Highly repetitive DNA usually makes up the centromeric and telomeric regions of the chromosome. Unique DNA, making up the bulk of structural genes, has a large component that is transcribed. Repetitive DNA is composed of dispersed DNA (e.g., short and long interspersed elements—SINES and LINES), multiple copies of